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Morphological evolution of menardiform globorotalids at Western Pacific Warm Pool ODP Hole 806C (Ontong-Java Plateau)

Evolution morphologique du groupe de *Globorotalia menardii* au Site ODP 806C (Ontong-Java Plateau, Pacifique tropical occidental)

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Abstract

The morphological evolution was investigated in the tropical Neogene planktonic foraminiferal lineage *Globorotalia menardii*, *G. limbata* and *G. multicamerata* during the past 8 million years at ODP Hole 806C (Ontong-Java Plateau, western equatorial Pacific). This research is an extension of several similar studies since 2007 from the Caribbean Sea, the tropical Atlantic and the Eastern Equatorial Pacific. The goal is to find empirical and quantitative confirmation for morphological speciation – splitting or phyletic gradualism or accelerated evolution – in planktonic foraminifera with the above lineage as model objects. The present study from ODP Hole 806C serves as a test to discriminate between these evolutionary scenarios.

In the western equatorial Pacific warm and stable environments prevailed back to Pliocene times, and potential influences of Northern Hemisphere Glaciation are thought to bear less severely on shell size evolution than in the Atlantic Ocean. A slow and gradual pattern of shell size increase is therefore expected in the western tropical Pacific, in contrast to the intermittent rapid menardiform shell size increase during periods of intensified formation of warm water eddies in the southern to tropical Atlantic.

For this study a total of 11,101 specimens from 37 stratigraphic levels extending over the past 8 million years were morphometrically investigated thanks to the *AMOR* robot for imaging and microfossil orientation. Of those, 6080 specimens comprise the *G. menardii–limbata-multicamerata* plexus. Special attention was given to trends of spiral height (δX) versus axial length (δY) in keel view, for which bivariate contour- and volume-density diagrams were constructed to visualize speciation and evolutionary trends.

The investigation at Hole 806C showed, that *G. menardii* evolved in a more gradual manner than in the Atlantic. Contour plots of δX versus δY reveal modest bimodality between 3.18 Ma – 2.55 Ma with a dominant branch consisting of smaller *G. menardii* (δX<~300 µm) persisting until the Late Quaternary, and a less dominant branch of larger *G. menardii* (δX>~300 µm) until 2.63 Ma. There is evidence for cladogenesis – splitting with subsequent morphological divergence in the Late Pliocene *G. menardii-limbata-multicamerata* lineage, and which may be linked to changes in the thermocline. Due to the general scarcity of *G. multicamerata* at Site 806, the divergence was less clearly expressed than originally expected. Up-section, bimodality vanished but *G. menardii* populations shifted towards extra large shells between 2.19-1.95 Ma.

The morphological evolution of Pacific menardiform globorotalids contrasts the one observed in the Atlantic. This inter-oceanic asymmetry may indicate possible long-distance dispersal of *G. menardii,* at

least during intermittent phases. For plankton biostratigraphy this implies, that correlating events represent more often than previously thought large scale environmental perturbations with local morphological ecophenotypic adaptations and nuances.

Keywords: *Globorotalia menardii*, morphological evolution, Pacific, Neogene, Volume-Density-Diagrams, AMOR

Résumé

L'évolution morphologique a été étudiée dans la lignée des foraminifères planctoniques du Néogène tropical *Globorotalia menardii, G. limbata et G. multicamerata* au cours des 8 derniers millions d'années au sondage ODP 806C (plateau d'Ontong-Java, Pacifique tropical occidental). Cette recherche est le prolongement de plusieurs études similaires sur la mer des Caraïbes, l'Atlantique tropical et le Pacifique équatorial oriental. Le but est de trouver une confirmation empirique de la spéciation morphologique chez les foraminifères planctoniques avec la lignée ci-dessus comme objets modèles d'étude. Cependant, une évolution accélérée peut également être une explication alternative. La présente étude sert de test pour faire la distinction entre ces scénarios évolutifs.
Dans le Pacifique tropical occidental, des environnements chauds et stables régnaient à l'époque du Pliocène. C'est pourqui l'influence de la glaciation dans l'hémisphère nord dans cette zone fûrent moins fortement sur l'évolution des ménardiformes que dans l'océan Atlantique. Donc, on s'attend à une augmentation lente et progressive de la taille de ces foraminifères au Site 806C, contrairement à l'augmentation intermittente rapide pendant les périodes de formation intensifiée d'anneaux d'eau chaude dans l'Atlantique sud à tropical.

Un total de 11101 spécimens provenant de 37 niveaux stratigraphiques a été étudié morphométriquement à l'aide du robot d'imagerie et d'orientation des microfossiles *AMOR*. Parmi ceux-ci, 6080 spécimens comprennent le plexus *G. menardii – limbata-multicamerata*. Une attention particulière a été portée aux tendances detéctées sur la hauteur de la spirale (δX) par rapport à la longueur axiale (δY) en vue de profil latérale, pour lesquelles des diagrammes de densité bivariés de contour – et de volume – ont été construits pour visualiser la spéciation et les tendances évolutives.

L'étude du sondage 806C a montré que *G. menardii* a évolué de manière plus graduelle que dans l'Atlantique. Les données de δX par rapport à δY révèlent une bimodalité modeste entre 3.18 Ma et 2.55 Ma avec une branche dominante composée de *G. menardii* plus petits (δX <~ 300 μm) persistant jusqu'au Quaternaire supérieur, et une branche moins dominante de *G. menardii* (δX> ~ 300 μm) jusqu'à 2.63 Ma. Il existe une évidence de cladogenèse - séparation avec une divergence morphologique subséquente dans la lignée *G. menardii-limbata-multicamerata* dans le Pliocène supérieur, et qui peut être liée à des changements dans la thermocline. En raison de la rareté générale de *G. multicamerata* au Site 806, la divergence était moins clairement exprimée que ce qu'il était'initialement attendu. En haut de la séction, la bimodalité disparait mais les populations de *G. menardii* se modifient vers des coquilles extra-larges entre 2.19 et 1.95 Ma. L'évolution morphologique des globorotalidés ménardiformes du Pacifique contraste avec celle observée dans l'Atlantique. Cette asymétrie interocéanique peut indiquer une possible dispersion longue distance de *G. menardii*, au moins pendant des phases intermittentes. Pour la biostratigraphie du plancton, cela implique que des événements corrélés représentent plus souvent qu'on ne le pensait des perturbations environnementales à grande échelle avec des adaptations et des nuances écophénotypiques morphologiques locales.

Mots clés: *Globorotalia menardii*, evolution morphologique, Pacifique, Neogène, Diagrammes-Volume-Densités, AMOR

1. Introduction

Planktonic foraminifera – marine, pelagic protists – populated the oceans since the Early Jurassic (Hart et al., 2002; Hart et al., 2003; Gradstein, 2017) about 189 million years ago. Every moment in time, their tiny calcite shells settle through the water column, and during more than 100 million years of radiation they began to form massive piles of calcareous ooze on the bottom of the oceans (Cifelli, 1969; Bolli, 1986; McGowran, 2005; Kucera, 2007; Darling et al., 2009; Schiebel and Hemleben, 2017; Gradstein et al., 2017). During their lifetimes and across myriads of generations these organisms find themselves exposed to environmental perturbances, to which they adapted, subtly but continuously by changing their shell morphologies within a species‘ capability of genetic constraints. Due to the high rate of evolution in foraminifera (De Vargas et al., 1997; Darling et al., 1999; De Vargas et al., 2001; Langer, 1999; Weiner et al., 2016), and over millions of years, this process leaves behind an unprecedented record (e.g., Aze et al., 2011; Fraass et al., 2015), which in academia and industry is exploited for measuring geological time and defining stratigraphic units, (e.g., Bolli and Saunders, 1977, 1985; Kennett and Srinivasan, 1983; Cifelli and Scott, 1986; McGowran, 2005; Kucera, 2007; Wade et al. 2018). Because of high level of completeness of the deep-sea record (Lazarus, 2011), planktonic foraminifera are ideal to map the biogeographic dynamics of evolution, pace and mode of speciation, to reconstruct phylogenetic relationships, and to find speciation loci, which are all important for refining evolutionary theory (e.g., Malmgren et al., 1983; Lazarus et al., 1995; Kucera and Malmgren, 1996; Norris, 1996; Spencer-Cervato and Thierstein, 1997; Feldmann, 2004; Si et al., 2018; Weinkauf et al., 2019). Evolution is a cascade of processes at various dimensions, scales and levels, of which the result becomes visible – though often subtly – in morphological transitions from the ancestor to the descendents. Particularly important in this context is detailed knowledge about shell variability not only through time, but also within the paleobiogeographic ranges of a species, which serves as empirical basis for the construction of phylogenetic schemes (Knappertsbusch, 2011).

Next to classical micropaleontologists, it is nowadays biology that delivers much phylogenetic knowledge with the help of molecular taxonomy, also for planktonic foraminifers (Darling et al., 1997, 1999, 2000, 2004; Morard et al., 2009, 2011, 2019; Weiner et al., 2014 and 2016). This approach is limited to individuals, which contain original organic tissue, and the resulting phylogenetic trees and species concepts are beginning to become increasingly comprehensive in context with classical morphologically based taxonomy (e.g., Kennett and Srinivasan, 1983; De Vargas et al., 1997; Aze et al., 2011; Schiebel and Hemleben, 2017), not only in planktonic foraminifers, but also, for example, coccolithophores (Young et al., 2014). In extinct forms, phylogenetic relationships can only be assessed by morphological comparison of fossils. Ironically, it is the phylogenetic "cartoons" from the 1980s that today are frequently applied to validate phylogenetic trees from molecular approaches and to calibrate molecular clocks (e.g., DeVargas et al., 1997; McGowran, 2005; Aze et al., 2011). A problem relies to the fact that phylogenetic schemes for planktonic foraminifers still notoriously lack sufficiently resolved information about transitional phases, such as the momentum of speciation and divergence of clades to final morphological expressions of separate species, especially when cryptic species are involved. Classical examples that empirically documented cladogenesis in shelled microfossils are those of Kellog and Hays (1975); Lazarus (1983, 1986); Spencer-Cervato and Thierstein (1997), or Hull and Norris (2009). A useful (though not strictly suffice) criterion for

recognizing morphological speciation patterns is the increasing empty space between divergent branches, which can be used for tests in the fossil record (Strotz and Allen, 2013; Fordham et al., 2018; Zehady et al., 2019; Ezard, Pearson, Aze, and Purvis, 2011). The same applies to situations where abrupt speciation has been proposed, e.g., Malmgren et al. (1983); Hull and Norris (2009). Too often, phylogenetic schemes remain oversimplified and do not show the real morphological transition between closely related species (e.g., Kennett and Srinivasan, 1983; Cifelli and Scott, 1986; Pearson, 1998; Stewart, 2003; Aze et al., 2011; Fordham et al., 2018; Zehady et al. 2019, in case of foraminifera, and Young et al., 2014, in case of marine phytoplankton). Still, evolutionary studies on microfossils – foraminifera included – lack precise causal explanations for forces that drive evolutionary trends in detail (Norris, 1999). Strong or long-lasting morphological overlap, which may mask speciation events, make detection of first appearance datums difficult. Morphological overlap can be due to mosaic evolution, repeated ecophenotypic variation and paleoceanographic influences (for example regulation of thermocline ventilation), or due to immigration of close homeomorphs (e.g., Si et al., 2018; Sexton and Norris, 2008, 2011). Such problems not only hamper species delineation but also may lead to false stratigraphic conclusions.

2 scientific context of the present study

2.1 Globorotalia menardii and related forms as study objects

In order to help solving previously cited key questions, studies are carried out at the Natural History Museum Basel (NMB), with the extant Neogene planktonic foraminiferal plexus *Globorotalia menardii* and its close, but extinct relatives *G. limbata* and *G. multicamerata* as a model group (Fig. 1).

What makes this group attractive is their lenticular and low trochospiral shells, which can be more easily quantified than, for example, other planktonic foraminiferal architectures like the globigerinids (in the latter, when viewed in umbilical position, the apertural topography is often highly variable, which makes specimen positioning into homologous position more difficult). Also the high shell variability, the strong resistance against carbonate dissolution (see also Mekik and François, 2006), and the potential for genetic investigations of extant representatives qualify the “menardiforms” ideal candidates for morphometric studies. Menardiform globorotalids radiated during the Late Miocene through Pliocene when several forms appeared for short geological time. Several of them are important stratigraphic markers such as *Globorotalia multicamerata*, which evolved from ancestral *G. menardii* via *G. limbata* (Fig. 1). A special advantage is the explicit tropical nature and abundance of *G. menardii* (Bé et al., 1966; Bé, 1977; Chaisson, 2003; Schiebel and Hemleben, 2017) narrowing efforts for paleobiogeographic mapping of morphotypes, and facilitating search of locus and time of evolutionary origin in deep-sea sediments.

2.2 Previous morphometric investigations about the *G. menardii* group

Extensive morpho-taxonomic studies about *G. menardii* were done by Stewart (2003) illustrating the wide morphological variability of this species and its relatives in the geological past. In Basel, prospection for speciation patterns of model "*menardii*" was continued in a series of investigations,

such as in deep-sea cores DSDP 502 (Caribbean Sea) and 503 (eastern equatorial Pacific) on either side of the Isthmus of Panama (Knapppertsbusch, 2007), at ODP core 925B from Ceara Rise in the western tropical Atlantic (Knappertsbusch, 2015a, 2016a), and two global mapping studies of menardiform morphotypes during the Quaternary (Brown, 2007) and the Mid-Pliocene warmth at 3.2 Ma (Knappertsbusch and Mary, 2012; Mary, 2013; Mary and Knappertsbusch, 2013; 2015). These works confirmed the applicability of the concept of evolutionary prospection developed in Knappertsbusch (2011, 2016a,b) by mapping the axial length versus the spiral height of menardiform shells in side view through time and geography. A key finding was the observation of divergent morphological trends in *G. menardii* on both sides of the Central American Landbridge, showing gradual evolution on the Pacific side and a distinct, step-wise evolution past 2.5 Ma in the two Atlantic sites. Most striking was the discovery that the size increase in menardiform shells in the Atlantic was rapid, with a distinct time-lag of appearance of large morphotypes from the southern ODP Site 925 (2.3–1.95 Ma) to the more northerly DSDP Site 502 (2.58–1.7 Ma) (Knappertsbusch, 2016a).

2.3 Agulhas Leakage Dispersal as alternative hypothesis

When replaced in a paleoceanographic context, the modern Atlantic oceanic circulation and the distribution of Quaternary menardiforms in the South Atlantic (Caley et al., 2012; Fenton et al., 2018) raised the suspicion, that evolution of *G. menardii* cannot be only caused by the closure of the ancient Central American Seaway, which happened around about 3 Ma: After about 2.7 Ma and coeval with a major northern glacial pulse (G10) and a sea-level drop, the Panamanian Isthmus probably fell dry (Molnar, 2008; Woodburne, 2010; Leigh et al., 2013) preventing exchange of thermocline dwellers like large *G. menardii* between the equatorial Pacific and the Caribbean Sea (e.g., Haug and Tiedemann, 1998; Schmidt, 2007). Instead, and considering the western Pacific as a possible evolutionary center for closely related *Globorotalia tumida* (Hull and Norris, 2009), the alternative hypothesis arises that menardiforms were occasionally dispersed by the Agulhas Current and its associated large-scale eddies (so-called rings) into the South-Atlantic Ocean, and then further into tropical belts of the western Atlantic and Caribbean Sea. Such large morphotypes had possibly evolved previously in remote Indian Ocean or Pacific isolation before 2.5 Ma (Knappertsbusch, 2016a).

2.4 Goals of the present study

When considering the gradual size evolution of *G. menardii* observed in the eastern Pacific at DSDP Site 503 (Knappertsbusch 2007), it is hypothesized that, until 2.58-1.95 Ma, super large menardiforms have evolved in Indian Ocean or Pacific isolation. Thereafter, menardiform giants would have passively drifted from the tropical Pacific or Indian Ocean reaching the South Atlantic around Cape of Good Hope during warm climate conditions such as interglacials. In this context, the morphological variation of *Globorotalia menardii* and its descendents *G. limbata* and *G. multicamerata* is here investigated for the past 8 million years from remote tropical Western Pacific Warm Pool (WPWP) ODP Hole 806C (Ontong-Java Plateau). In the WPWP, more stable and warm environments prevailed in Pliocene than Quaternary times (Chaisson and Ravelo, 2000), and the environmental pressure caused by Northern Hemisphere glaciation is expected to bear less severely on shell size growth than

in the Atlantic Ocean. Investigating the situation in the WPWP shall thus serve as an end-member scenario of a series of paleoceanographic settings that have driven evolution. However, paleoceanographic reorganisations in the tropical west Pacific in the course of the northward movement of Australia and the uplift of Indonesia, that both lead to restrictions of the Indonesian Throughflow from the Upper Miocene through Lower Pliocene and around 3.5 Ma, together with cooling events (for example the Mammoth cooling event (M2) at 3.3 Ma), and general cooling in the southern Hemisphere (MIS G20 around 3 Ma (Auer et al., 2019; Lam and Leckie, 2020 a), challenge our understanding of evolutionary trends in menardiforms also in that tropical area.

The present investigation of menardiform evolution at Hole 806C serves thus as a further cornerstone experiment to explore the dispersal idea. If the above reasonings hold true, a more gradual evolution of menardiforms in the quiescence of a climatically stable WPWP is expected. Possibly, extra-large representatives of *G. menardii* would have evolved in the WPWP, that can be drifted first through the Indonesian seaways and then occasionally disperse through an intensifying Agulhas Current system from the Pliocene onwards. The South African "valve" would intermittently permit leakage of transient populations of Indian or Pacific tropical menardiform faunas into the South Atlantic at times of major glacial/interglacial transitions (Peeters et al., 2004; Beal et al., 2011; Scussolini et al., 2013, 2015; Hall et al., 2016). Caused by an intermittent and pulsed nature of Agulhas' intensity, the postulated initial gradual evolutionary trajectory of *G. menardii* would thereby be modulated into a more step-wise size distribution through time for this species in the tropical Atlantic. Towards the more tropical tributaries of Atlantic surface currents, populations of *G. menardii* would stabilize and so cause an intermittent and step-wise pattern of size evolution. This scenario is similar to the one drawn by Norris (1999) and Sexton and Norris (2008), who advocated for free and frequent dispersal of plankton over long distances and subsequent buildup and evolution in the new environment as soon as conditions are favorable.

Sediments from Hole 806C provide the opportunity to investigate speciation in the *Globorotalia menardii – limbata –multicamerata* lineage. In this context attention is given to spiral height (δx) versus axial length (δy) in keel view, for which bivariate frequency diagrams and volume density diagrams are constructed to aid analysis and visualization of morphological trends during the past 8 million years.

3. Materials and Methods

3.1 Core location

ODP Hole 806C (latitude 0° 19.1‘ N, longitude 159° 21.7‘ E, water depth 2521m, Fig. 2) is located on the northeastern margin of the Ontong Java Plateau, almost exactly on the equator, in the center of the western Pacific warm pool inside the boundary of the 29°C isotherm (see Figure 1 in Auer et al., 2019), and outside the range of carbonate dissolution (Kroenke, Berger, Janecek et al., 1991; Hermelin, 1989).

This area is ideal to study the evolution of *G. menardii* in Pacific isolation, because sea-surface temperatures remained relatively stable for at least the past 4-5 Ma (O'Brien et al., 2014; LaRiviere et al., 2012; van der Heydt and Dijkstra, 2011; Ford et al., 2015, Wara et al., 2005; Chaisson and Ravelo,

2000 ; Zhang et al., 2014, see also Figure 2e in Van der Lubbe et al., 2021). In addition, it is a classical paleoceanography site of the ODP legacy, representing a sedimentologically undisturbed setting, with exceptional biostratigraphic and stable isotope records of Neogene pelagic carbonate (e.g., Berger et al., 1993; Chaisson and Leckie, 1993; Jansen et al., 1993; Chaisson and Ravelo, 2000; Nathan and Leckie, 2009).

Until a subbottom depth of 309.6 mbsf (upper Miocene) Hole 806C was continuously cored with the advanced hydraulic piston corer achieving a core recovery of >100%, thus excellently serving the requirements for the present study over the time interval of the last 8 Ma.

3.2 Sample selection

Sampling occurred preferably within selected interglacials identified from $\delta^{18}O$ isotope records from Site 806B, that were published in Berger et al. (1993) and Jansen et al. (1993). This event stratigraphic sampling was primarily done in order to avoid cold intervals, where menardiform globorotalids were frequently reported rare or barren at other locations (Ericson and Wollin, 1968; Norris, 1999). Secondly, this sampling strategy helps to record the upper tail of the size-frequency distribution of the menardiforms, that – according to the author's own observations - occur more frequently in warm intervals. This way, the chance for obtaining menardiform populations with highest morphological diversity and disparity is optimized, which is a requirement to assess evolutionary innovation through geological time (Hopkins and Gerber, 2017).

In the period from 0 to 2 Ma, intervals were chosen during warm peaks identified from the Hole 806B $\delta^{18}O$ record for *Globigerinoides ruber* as given in Table 1 of Berger et al. (1993). From 2 Ma to 5.4 Ma interglacials were also identified via the $\delta^{18}O$ record for *G. sacculifer* at Hole 806B, which are given in Table 1 of Jansen et al. (1993). Interglacials were prerentially selected stratigraphically close to those studied in Knappertsbusch (2007 and 2016a). For the segment older than 5.4 Ma "isotope event sampling" was no longer reliably possible at Site 806 because $\delta^{18}O$ peaks were less pronounced and sample selection was based on biostratigraphic evidence only (e.g., model 806B fnrdIM loc95 MK_final, see further below). In total, samples from 37 stratigraphic levels were taken from Hole 806 C, covering a time span for the past 8 Ma (see Table 1).

Because Hole 806B was already heavily sampled for isotope stratigraphy, Hole 806C was chosen. This means that isotope events identified at 806B had to be projected onto Hole 806C, and that depths scales had to be converted between the two holes. For correlation of equivalent depths (mbsf) between Holes B and C the GRAPE conversion scheme shown in Table 1 of Mayer et al. (1993) was used for the first 102 mbsf at Hole 806C (see Fig. 3 and Table 2).

For samples below 102 mbsf equivalent sample depths at Hole 806C were determined by projection of event ages from Hole 806B (age model 806B fnrdIM loc95 MK_final) onto Hole 806C (age model 806C loc95). For details about the construction of numerical age models read the section about time-

scale and age control further below. The control points for the age models at Holes 806B and 806C are given in Table 2, and both models are plotted in Fig. 3.

3.3 Time-scale and age control

For construction of lines of correlation (locs), and dating of samples from Holes 806B and 806C the Age Depth Plot (ADP) program and AgeMaker (AM) programs from Lazarus (1992) were applied. ADP and AM were created in context of the NEPTUNE database and age modeling efforts of the former micropaleontology group at ETH in Zürich, Switzerland (Lazarus, 1994; Cervato, 1999, 2005; Renaudie et al., 2020). Depths to individual cores at both holes were read from the coring summaries for Site 806 given in the Leg 130 Initial Reports (Kroenke et al., 1991). Depth intervals of all available first and last appearance datums of planktonic foraminifera, calcareous nannofossils, radiolaria and diatoms were taken from the Initial and Scientific Reports of the Proceedings of the ODP to Leg 130 (e.g., Chaisson and Leckie, 1993; the range chart in the back-pocket folder to ODP 130, SR, for planktonic foraminifera; Kroenke et al. 1991; Shipboard Scientific Party, Site 806, for calcareous nannofossils, radiolaria and diatoms; and Takayama (1993) for calcareous nannofossils). Own biostratigraphic observations on planktonic foraminifera were also considered (e.g., Appendix 1). The numerical ages of microfossil datums used in the present study follow the Integrated Magnetobiochronologic Scale (IMBS) of Berggren et al. (1995), which is similar to the Geological Time Scale of Gradstein et al. (2004). The author is aware, that since then geological time scales have been revised and refined multiple times, even recently at nearby Site 807A (Kaushik et al., 2020). However, primarily in order to keep the present study one-to-one comparable with earlier morphometric studies about *G. menardii* (e.g. Knappertsbusch, 2007, 2016), and secondly to maintain best compatibility to original ship-board chronologies applied during Leg 130, it was preferred to adopt the geochronology of Berggren et al. (1995). This strategy is sufficient for the present study and more advantageous than converting sample ages and biostratigraphic datums to ever changing newer chronologies. Should there be any need for conversion sample ages to more recent chronologies, the Timescale Creator software package (https://timescalecreator.org/index/index.php) to the Geological Time Scale 2016 of Ogg et al (2016) can be applied, for example (see also Fordham et al., 2018; Zehady et al., 2019). Comments on the interpretation of the $\delta^{18}O$ record at Hole 806C, inclusion of the Brunhes/Matuyama boundary and the Mid-Pleistocene Revolution for Hole 806C are given in the supplementary information.

3.4 Sample preparation

3.4.1 Preparation of bulk sediments

Between 3.90-13.22 grams of bulk sediment were received for this study from IODP Gulf Coast Core Repository (IODP sample request #20957z). Bulk sediments were first weighted, oven-dried at 40°C, reweighted, and dry weight and the water content determined. Thereafter, samples were gently disintegrated in boiling water without any additives and then carefully wet sieved with a minimum amount of tap water for rinsing through a 63μm mesh. The suspension containing the smaller than 63μm fraction was collected in beaker-glasses and coccoliths and fine particles were allowed to settle

during about 3 days. Thereafter, the clear supernatant water was decanted, the fine fraction was oven-dried at 40°C and then stored in plastic cases. The foraminiferal residues larger than 63μm was immediately dried at 40°C. Both, the <63 μm and the >63 μm foraminiferal fractions were weighted to ±0.01 grams using a microbalance from Sartorius.

3.4.2 Preparation of morphometry slides

For morphometric analyses slides were prepared as described in Knappertsbusch (2016a). The foraminiferal fraction >63μm was split with a microsplitter into aliquots of 1/2, 1/4, 1/8 or 1/16 splits depending on the amount of residue and the abundance of menardiform globorotalids. For every sample binary splitting was carefully protocolled in flow-diagrams. From a particular split, all intact menardiform globorotalids were handpicked under the binocular, including juveniles. During microscopic scanning of sample splits, stratigraphically important markers and menardiform species were classified, and the coiling direction and number of chambers in the final whorl of menardiform globorotalids were recorded. It was attempted to collect at least 200 menardiform globorotalid specimens per sample. The amount of 200 specimens was judged to be a reasonable minimum for morphometric and statistical analyses, without exceeding the efforts for picking, mounting and digital image analysis. In the case of rare occurrence of *G. menardii* the entire sample was picked. If necessary, several of the above aliquots were combined. The unused splits and the remains of the picked aliquots were preserved separately in glass vials.

The term "menardiform globorotalids" includes here all representatives of the *Menardella* lineage of Kennett and Srinivasan (1983). Because of frequent observations of morphological transitions of *G. menardii* to *Globorotalia tumida* at Site 806, members of the *Globorotalia* lineage of Kennett and Srinivasan (1983) including *G. merotumida*, *G. plesiotumida*, *G. tumida tumida*, *G. ungulata* and *G. tumida flexuosa* were picked as well (see taxonomic remarks in section 4). Using water soluble wallpaper glue picked specimens were manually mounted into standard Plummer micropaleontological slides from P.A.S.I. Srl (Torino) with 60 fields in upright position as precise as possible. Before mounting the slide, bottom was painted and soaked with paper glue (Pritt from Henkel), so that specimens could be more easily arranged in keel view. Each specimen was placed in the center of a field. In this position, with the aperture facing upwards, and with the spiral side always facing to the left, the final chamber of a sinistrally coiled specimen becomes oriented downwards on the monitor, while in a dextrally coiled specimen the final chamber is directed upwards on the monitor. Prior to imaging and extraction of morphometric data this "keel-view" position was automatically fine-adjusted with AMOR (see sections below). Next to preparation of morphometry slides, a collection of foraminiferal zonal or subzonal slides was prepared with stratigraphic index species and/or representatives characterizing foraminiferal zones. See range chart for the recorded species in Appendix 1 of the electronic version to this article.

3.4.3 Code for sample designation

Each morphometry slide received a unique 15 characters long code, with the first 9 digits indicating the sample designation in DSDP/ODP notation, e.g., site (3 digits), core (2 digits), section (1 digit), cm

interval (3 digits indicating the deeper level of the interval). The next following 2 characters indicate the slide number (a, b, c, d etc) and a "K" for Keel view. The last 4 digits indicate the field number (2 digits) in the slide (01, 02, . . ., 60) and the number of specimens per field (2 digits, always 01 because only one specimen was mounted per field). Example: 806011015aK0201 means the specimen in field no. 2, in morphometry slide a, Keel view, in ODP sample 806C-01-1, 015cm (the label "C" designating Hole C was omitted).

3.4.4 Automated orientation and imaging of specimens

The robot called *AMOR* (Automated Measurement system for shell mORphology) and its successor *System AMOR 2* were both especially developed at the Natural History Museum Basel for automated specimen orientation and imaging (Knappertsbusch et al, 2009, Mary, 2013; Mary and Knappertsbusch, 2013, 2015a; Knappertsbusch, 2015b; Knappertsbusch, 2016a, 2019). In order to avoid possible variation of results between instruments, the old *AMOR* was used solely for the present study. Positioning and imaging was realized in the single-specimen mode of AMOR using software version AMOR 3.28 (Schorpp, 2013) in combination with AutoIt Script AMOR_Manual_Drive_v10.au3.

AMOR is a system consisting of a binocular light microscope for reflected light (MZ6 from Leica) equipped with a DXC-390P video camera from Sony, and a fully motorized universal 4-axis tilting stage (Fig. 4). The microscope was customized for autofocussing and automatic zooming. This system allows automated orientation and imaging of series microfossils mounted in the above mentioned multicellular Plummer-slides (Knappertsbusch et al., 2009, 2019; Schorpp et al., 2009). Images of specimens are captured at a resolution of 640x480 pixels and 8 bit (256 grey levels) in Tiff format. The microscope objective is a 1x achromatic lense, and the connector between camera and microscope consists of an 1 x Cmount piece. The optimum magnification of a specimen is automatically determined by the system when the foraminiferal test fitted just inside of a frame less than 10 pixels of the images‘ width and height. During operation, magnifications (0.63x through 4x of the Leica zoom body) are automatically recorded for every specimen allowing conversion of pixels into micrometers during later outline extraction. In order to avoid disturbing reflections the tips of swan-neck illuminators are capped with turnable linear polarizing filters. The microscope main objective is equipped with a rotatable polarizing filter-set for ring-illuminations from Volpi, which allows imaging of specimens under cross polarized light. This setting enhances the contrast between the bright calcareous shell of the foraminifera against the black cardboard of the Plummer cell, and simultaneously removes reflecting light spots from the cardboard. In this way, processing of images and extraction of morphometric shell parameters is greatly simplified and reproductible.

3.4.5 Processing of images

After collection with AMOR, images were semi-automatically processed from Tiff to binary black-and-white images in Raw format using ImageJ 1.48v (http://imagej.nih.gov/ij). Details about processing images with ImageJ can be read in Appendix 2 and in Knappertsbusch (2021). Thereafter, Raw images were transferred onto an iMac computer for extraction of particle outline coordinates and morphometric parameters. For documentation purposes, the intermediate black and white Tiff images

after cleaning were also stored, next to the original grey-level Tiff images from AMOR and the final images in Raw format.

3.4.6 Derivation of measurements

Extraction of outline coordinates and morphometric parameters from the raw images was done with a suite of Fortran programs (Knappertsbusch, 2015b, 2016a). For outline extraction software, the improved version Trace_AMOR2_batch.out was applied. The sequence of programs in the present study was as follows: MagCorr2.out (performing a correction of the magnifications returned by AMOR 3.28 and AMOR_Manual_Drive_v10.au3); Trace_AMOR2_batch.out (outline extraction from the raw images); Sprep53.out (interpolation of traced outline coordinates at 250 equiangular distances, centering, polar coordinates, basic measurements); KeelWidth13.out (set Q=0.05 and choose output option 1 for determination of derived parameters); and DexFlip (Correction for dextral coiling). For information about MagCorr2.out and imaging resolution see Appendix 2. All above programs were run in batch mode on an iMac computer. Quality check for outline tracing and best fit of osculating circles was performed using auxiliary programs XYPlot2.out and XYPlot3.out, respectively (see documentation of MorphCol in Knappertsbusch, 2015b).

Morphometric parameters generated in this way include the spiral height (δX), axial length (δY), the area of the outline in profile view (Ar), the upper and lower keel angles Φ_1 and Φ_2, apical angle Φ_3, measurements for the spiral and umbilical convexities (A and B, respectively), and the radii of the osculating circles in the upper (R_{up}) and lower (R_{lo}) keel regions in keel view as shown in Fig. 5.

3.4.7 Plane contour plots from individual samples

In the present study measurements of the spiral height (δX) versus axial length (δY) are central for morpho-paleobiogeographic exploration of menardiform globorotalids. Frequency distributions of δX versus δY are therefore presented in form of plane contour diagrams to identify clusters. The basic tenet is that in a bivariate morphospace spanned by a pair of morphological traits, species would cluster, with sparsely populated or empty space between them (Ezard, et al. 2010), and with clusters being represented by a unimodal distribution or a clinal variation along a morphological continuum at a given time. Within short time-intervals and under stable environmental conditions, these distributions would vary little from a mean value or only to some minor degree within limits that are controlled either by the species ecophenotypic plasticity or its genetic makeup. This means, that over short time periods, bivariate frequency distributions of species should stay more or less constant, centering about a stable point in the morphospace and fluctuate only in abundance. Seen over longer time intervals and still assuming environmental stability, morphological variability may accumulate and increase due to evolutionary change and frequency distributions may change accordingly. Vicariant development may happen when, for example, water stratification sets in and permanently separates vertical populations from each other. A similar (allopatric) effect may occur when populations become separated geographically. Over long enough time, the effect will be that unimodal populations may begin to split up and diverge, thereby developing bi- or multimodal clusters in the morphospace under consideration. Such morphological clustering can ideally be studied by comparison of contoured

frequency diagrams from different time levels and/or from one location to another. Observing spiral height (δX) versus axial width (δY) in contour diagrams is hence intuitive and promising, because these are comprehensible parameters that can easily be measured from shells. For construction of such plots a bivariate frequency matrix was first determined for each sample by counting points in cells of a grid overlain to the scatter plot for a sample of δX versus δY. There are theoretical rules for selection of the optimal bin-width, in our case $\Delta\delta X$ or $\Delta\delta Y$ (Hyndman, 1995; Keating and Scott, 1999), but quite often application of these rules results in poor representation of important details of a distribution (Wand, 1997). In the present case, the best resolution was found empirically by experimentation with different bin-widths at a grid-cell size of of $\Delta\delta X$=100 μm x $\Delta\delta Y$=50 μm (Knappertsbusch and Mary, 2012; Knappertsbusch, 2015b). From the bivariate frequency matrix ($\Delta\delta X$, $\Delta\delta Y$, F) with F being the number of specimens per grid-cell contour diagrams were drawn for *G. menardii*, *G. limbata* and *G. multicamerata* for every sample. This was done using Surface III 2.6 plus Software from Kansas Geological Survey on a iMac computer without any further weighting of gridded frequencies (see contour diagrams in section 4.2).

3.4.8 Calibrated Volume Density Diagrams (VDDs)

Next to the analysis of contour diagrams, so called volume density diagrams (VDDs) are an interesting way to visualize morphological divergence of δX and δY through time. This is a more intuitive way than applying simple descriptive statistics or step-by-step comparison of contour plots from one time level to the next. VDDs help to study the variability of stacked contour diagrams (Knappertsbusch and Mary, 2012; Knappertsbusch, 2016a). By visualization of frequency centers and the empty space in-between them, VDDs allow for detection of possible cladogenesis (e.g., Knappertsbusch 2000). Especially the empty space between density clouds of measurements are an important criteria for recognition of budding or divergence of branches in phenetic schemes. Such an approach becomes especially useful when vertical sections are combined with phylogenetic reconstructions from molecular analysis, as was exemplified in Bendif et al. (2019).

VDDs are surfaces of equal abundance that emerge in a stack of contoured frequency representations for a given species, and if equal contour lines are connected from one time level to the next. They can be generated for any combination of bivariate measurements. In the present study, focus is given to VDDs in the morphospace of δX versus δY. Fig. 1c shows an example for 3 nested VDD surfaces obtained for the *G. menardii-limbata-multicamerata* lineage from ODP Hole 925B (Knappertsbusch, 2016a; Knappertsbusch and Mary, 2012), and a brief summary and deviations from those reports are given below.

VDDs are generated by first composing, reformatting and merging the stack of bivariate δY versus δY frequency (F_s) into a 4-dimensional data set of volume elements. These "voxels" are defined by unit vectors of X_i,Y_j,Z and component $Fs_{i,j}$, which in our case is the local absolute frequency F_s of menardiform specimens at grid-cell X_i,Y_j, and from time level Z. X_i and Y_j denote the centers of bins $\Delta\delta$Xi, and $\Delta\delta$Yj along the δX or δY axes, respectively. For the input matrices the bin-width $\Delta\delta X$ is selected at 100μm and 50μm for $\Delta\delta Y$. In order to obtain scales ranging from 0 to 1, the mid-center ranges of $\Delta\delta X_i$ and $\Delta\delta Y_j$ were normalized by division through 675 and 1550, respectively. Z represents

the normalized age (division by -8 million years to scale ages from -1 through 0). The subscript "s" in F_s denotes that only spacial scaling was applied along X,Y and Z axes, but none in the component F_s itself. These transformations were done using auxiliary program Grid_to_Vox3.out from the author. Thereafter, the normalized X,Y,Z,F_s values were imported to Voxler 2 from Golden Software for 3-D display of isosurfaces of local frequencies F_s (as an alternative, it would also be possible to represent F_s as rendered densities, similar to bone-density representations in a X-Ray computed tomographic image). In contrast to Knappertsbusch (2016a), where VDDs for *G. menardii* from ODP Hole 925B were illustrated from *normalized frequencies* F_{sn} at arbitrary isovalues, it is attempted here to construct *calibrated* VDDs for *G. menardii* from Hole 806C. Calibrated VDDs are created using absolute local frequencies F_s instead of relative F_{sn}, and visually modelling the isosurfaces of VDDs until contour representations in Voxler from sampled levels match best with the plane contour plots generated with Grid_to_Vox3.out and Surface III plus from the previous section. This calibrated visualization of specimen densities through time proved to represent the evolutionary pattern of *G. menardii* more realisticly than in the previous studies. The unscaled raw input files for the VDDs developed herein for Hole 806C are provided electronically in Files RawData_9 through 12 of the archive RawData.zip.

In Voxler, and for calibration, several parameters needed to be adjusted in respective property windows for the gridding and graphics output modules: first, the upper limit of component F_s and its age from the entire data set need to be identified. This is found by comparison of the bivariate frequency matrices of F_s that were created with program Grid2.out with the plots created with Surface III plus. The upper limit of F_s is also returned in the input properties window of Voxler (input data limits option). In the present example (input data file called ALL_XYZFs_menardi_S3_txt.txt) the absolute local frequency maximum of F_s amounts to 60 specimens per grid-cell. It occurs in the sample at -6.63 Ma (normalized age Z = -0.8288; $\text{Offset}_{\text{contour slice}}$ = -0.3271, see equation 1 below). Next, the age of this sample is translated into Voxler‘s internal Z-coordinate system by determination of the corresponding offset value for horizontal contour-slices using the following linear relationship:

$$\text{Offset}_{\text{contour slice}} = 0.1244 * \text{Age} + 0.4975 \quad \text{(Equation 1)}$$

In this equation, age is negative and varies from -8 Ma through 0 Ma. The equation was derived by varying the offset slider in Voxler for horizontal slices showing contour plots. In the above example, the numeric age of -6.63 Ma translates into an offset value of -0.3271. This value was then set for $\text{Offset}_{\text{contour slice}}$ for the contour slice in Voxler at -6.63 Ma, where the data set’s highest value of F_s occurs. This scaling is possible because the input data limits of a 3D lattice apply for the global lattice and not just for the two-dimensional cutting plane through the lattice (see page 269 ff. in Golden Software, LLC (2015). Voxler Full User’s Guide), peaks and contour lines can now be directly compared between Voxler output and those obtained with Surface III. In the -6.63 Ma slice, Voxler returns an interpolated local frequency maximum of F_{int}=45.3554, which corresponds to the mentioned absolute peak of F_s=60 specimens of *G. menardii* per grid-cell (see bivariate frequency matrix from -6.63 Ma). This translates into a conversion ratio of

$$F_{int} / F_s = 0.7559 \text{ (specimens per grid-cell)}^{-1} \qquad \text{(Equation 2)}$$

In other words, 1 specimen per grid-cell in the gridded frequency matrix corresponds to an internal frequency value of F_{int} = 0.7559.

For generating contour diagrams in Voxler, the following parameters were then adjusted in the associated contour module for each horizontal slice: The level method was set to "Min,Max,Levels", with Min=0, and Max=46 (e.g., the next higher integer value to the above interpolated maximum F_{int}=45.3554). For contour lines a level interval of F_{int}=2.2678 was found to be ideal for best visualization of a real contour interval of F_s=3 specimens per grid-cell using the ratio in Equation 2. The same contouring parameters were applied for drawing a vertical contour-slice through the center of data set at 45° (offset from center=0). The effect of this operation is that the peripheral contour-lines from the horizontal slices become tangential to the peripheral contours of the vertical 45° section. In this manner, they define the three-dimensional periphery of the VDD (see later section 5.1 and figures therin). In a final step, toggling into the iso-surface module an outer isosurface was generated. This was performed by visually adjusting the isovalue glider, until the isosurface matches best with the peripheral contour line in the vertical 45° plane. This was reached for *G. menardii* at an isovalue at 2.499, which also represents the absolute frequency F_s of 3 specimens per grid-cell (note that this value is not equal but in the vicinity of the above value of F_{int}=2.2678 for contour-levels).

The same procedure was applied for construction of VDDs for *G. limbata* and *G. multicamerata*.

The relationship between the absolute local frequencies F_s from the gridded data and Voxler's internally interpolated node values for F_{int} is not a straight analytical function, but varies and follows a positive linear regression line. This behaviour depends on the complexity of Voxler's internal data model, e.g., the selection of the interpolation (gridding) method and other parameters set in Voxler. In the present study, inverse distance was selected for the gridding method because it produces the best results in most cases (see recommendation in Golden Software, LLC, 2015:. Voxler Full User's Guide, page 175 ff.). Variation and deviation of F_{int} from F_s is caused by interpolation after data import into Voxler at the selected nodal resolution of the grid. In the present case, the data geometry in Voxler's gridder module was set to the default resolution of 50 x 50 x 50 nodes.

3.4.9 Statistical analysis

Statistical analysis for morphometric parameters (e.g., δX, δY, Ar, Φ_1, Φ_2, Φ_3, A, B, R_{up} and R_{lo}, see section 3.4.6 and Fig. 5) was done with Statistica 12 from Statsoft. Per sample and for every parameter basic univariate statistics such as the means, 95% confidence intervals, the range from lower to upper quartiles, medians, 10% and 90% percentiles, absolute minima and maxima were calculated (Figs. 6-14). Per sample and parameter, summary statistics are reproduced in form of Excel sheets for *G. menardii*, *G. limbata*, *G. multicamerata*, and transitional forms between *G. menardii* and *G. tumida* in Appendices 3, 4, 5 and 6 in the electronic version to this article.

3.4.10 Deposition of materials and data

All materials including morphometry and stratigraphic slides, remaining residues, splits, processing protocols and laboratory notes are deposited in the micropaleontological collections of the Natural History Museum Basel (NMB, Collection Knappertsbusch, 2021). Digital images, intermediate results from processing images, the numerical age model used for ODP Hole 806B, morphometric data, Fortran software written by the author, LabView code for AMOR, and AutoIt scripts are stored on the NMB's internal Media server. The technical documentation, notes, drawings, electric schemes and unpublished reports, that have led to the realization of AMOR, are deposited in the geological archive of the NMB. Primary raw data are provided in the supplied electronic archive RawData.zip to the present contribution. Protocols for imaging and post imaging processing steps per sample slide are logged in files RawData_13.xlsx and RawData_14.xlsx, respectively, also in RawData.zip. A mirror containing the full set of images, intermediate data, morphometric and Voxler data sets is available on the PANGAEA data repository (Knappertsbusch, 2021).

4. Taxonomic remarks

Summaries about taxonomic and phylogenetic positions, and nomenclatural solutions are given in Knappertsbusch (2016a) and references therein, and are not repeated here. In the present study, the nomenclatural concept of Knappertsbusch (2007) and Knappertsbusch (2016a) is applied: The taxonomy of Bolli and Saunders (1985) is broadly followed with modifications and descriptions about menardiform globorotalids given in Chaisson and Leckie (1993) and Kennett and Srinivasan (1983). For species concepts, the collection «*Cenozoic planktic foraminifera zonal and subzonal assemblages*» prepared by H.M. Bolli in 1985-1987, that is deposited at the Natural History Museum Basel was consulted, as well as the pforams@mikrotax website (Young, Wade, and Huber, 2020).

Menardiform globorotalids include mainly *G. menardii*, *G. limbata*, and *G. multicamerata*. The first is abundant throughout the studied section, while *G. limbata* is present mostly in lower abundances, and *G. multicamerata* occurred sporadically (see section 5.3., Appendix 7, and Fig. 19 further below about menardiform census counts and coiling behaviour throughout the core).

Samples and diagrams entitled with "*G. menardii*-lin" include the following formal and informal menardiform taxa: *G. menardii cultrata*; *G. menardii menardii*; *G. menardii* (=juvenile forms, undifferentiated forms, and Bolli's (1970) *G. menardii* "A" and "B"); *G. menardii flexuosa* (=forms with a flexuose final chamber); *G. menardii gibberula* (=menardiforms with a distinct conical spiral side, a flat umbilical side, and usually without a flexed final chamber); and rare forms of *G. menardii gibberula*, that bear a flexed final chamber, that were informally abbreviated as "*FlexGib*".

Because the feature of limbation of chamber sutures on the spiral side cannot be reliably applied in practice, *Globorotalia limbata* is separated as a *G. menardii* with strictly seven chambers in the final whorl. This is an artificial criterion, but here applied in an exploratory manner in order to empirically check for any divergence trends from *G. menardii* or descendent *G. multicamerata*. A global survey of surface sediment samples demonstrated the absence of more than six chambers in the final whorl in Late Quaternary to Holocene *G. menardii*, which supports the distinctiveness of *G. limbata* from *G.*

menardii on the basis of this criteria. In most cases *G. limbata* was found to coil dextrally, but seven chambered sinistral forms were found in lower abundances, too.

Adopting the concept of Bolli and Saunders (1985), *Globorotalia multicamerata* is considered as a flat, biconvex menardiform globorotalid with eight chambers or more in the final whorl. *G. multicamerata* shows almost exclusively dextral coiling with few exceptions. In the study of Chaisson and Leckie (1993) from Site 806, *G. multicamerata* is recorded in only a single sample and at low numbers, which is due to their concept of at least nine instead of eight chambers in the final whorl of *G. multicamerata*.

Often, *Globorotalia tumida* is co-occurring with *G. menardii*, and in episodes dominated over *G. menardii* at Hole 806C (see census counts in Appendix 7, and relative abundances in Figure 19). Also *G. tumida* is morphologically highly variable showing strong intergradation to representatives of the above *G. menardii* group, which made differential diagnosis often a challenge. Similar observations from ODP Site 806 were made by Hull and Norris (2009), who reported flat morphotypes in the *Globorotalia plesiotumida – G. tumida* lineage between 6.225 Ma – 5.819 Ma. Transitional forms between *G. menardii* and *G. tumida* are informally recorded as "mentum" but were not included in the present analysis. In the typical adult habitus, *G. tumida* has a distinctly elongate shell in spiral or umbilical view, more than any *G. menardii*, and with a tapering end: One end has an acute keel at the final chamber, and on the opposite side, it has a broaded end with a round keel. In keel view, *G. tumida* is highly inflated at one end, biconvex, and strongly asymmetric. Shells of *G. tumida* have almost always a thick encrusted surface with sugar-like appearance. Rarely, non-encrusted *G. tumida* were encountered, then the surface was smooth and somewhat shiny. *Globorotalia tumida* coils preferentially in sinistral direction, except during an extended Late Zanclean/Early Piacenzian interval, when *G. tumida* showed dextral and sinistral coiling in about equal proportions. In the present study, members addressed with *G. tumida*-lin include the following formal taxa: *G. tumida tumida*, *G. tumida flexuosa* (informally abbreviated as "FlexTum", e.g., tumid forms with a flexuose final chamber), and the as yet not described informal form "newtum".

Other menardiform or tumid morphovariants were only rarely encountered in the samples from Hole 806C and were grouped in a category "others". These include *Globorotalia ungulata*, („ungulat“ in Appendix 1), *G. pseudomiocenica* ("pmiocen"), and *G. margaritae*. *G. margaritae* is a keeled globorotalid within the size range of juvenile *G. menardii*, but typically very flat in side view and with a large, petaloid-shaped final chamber, that exceeds its relative dimension in comparison to the latter species.

5. Results

A total of over 11,101 specimens of menardiform and tumiform specimens from 37 stratigraphical levels were investigated. Of of these, 5658 specimens comprise *G. menardii*, 367 specimens were *G. limbata*, and 55 specimens were *G. multicamerata*, totalling to 6080 measured specimens from the *G. menardii-limbata-multicamerata* plexus. In the following sub-sections the morphological variability of the *G. menardii–limbata–multicamerata* plexus from Hole 806C is presented in the temporal evolution of selected parameters and simple descriptive statistics, stacks of contoured δX vs δY frequency diagrams, and in VDDs for illustration of the mode of morphological speciation.

5.1 Analysis of shell variation through time from individual parameters

Figures 6 to 14 illustrate trends of keel view area (Ar), spiral height (δX), axial length (δY), radii of upper and lower osculating circles in the keel region (R_{up}, R_{lo}), upper and lower keel angle (Φ_1, Φ_2), spiral convexity (A), and the convexity ratio A/B for each of the menardiform groups during the past 8 million years at ODP Hole 806C. For recognition of main variation through time the means, medians, ranges from the lower to upper quartiles, 10% and 90% quartiles and the extremes (minima, maxima) per sample are plotted separately (see panels A, B and C in Figs. 6-14). The raw data to measurements from individual specimens are listed in text files RawData_01 to 04 in the supplied zip archive RawData.

From 8 Ma to Recent *G. menardii* shows an increase in the means of keel view area, spiral height and axial length, spiral convexity and of the convexity ratio. In contrast, upper and lower keel angles (Φ1) and (Φ2) decrease during that period, whereas radii of upper and lower osculating circles (R_{up}, R_{lo}) show little evolutionary change. Most meaningful measures of evolutionary change are the trends of upper 90% percentiles and maxima of shell size parameters for Ar, δX, and δY, which show a strong reduction in the sample at 2.63 M (see also contour diagrams and VDDs in the following sections). Considered over long time the shell size trend of *G. menardii* at Hole 806C shows a gradual evolutionary size increase.

In *G. limbata* most shell parameters (Ar, δX, δY, R_{up}, Φ_2, A and A/B) co-vary through time, showing first gradual increasing trends from 7.96 Ma through 3.36 Ma, and thereafter a significant decrease of measurements from 3.36 Ma to 2.63 Ma (see panels B in Figs. 6-9, 11, and 13-14). In contrast, R_{lo} and Φ_2 show reverse trends and are not of gradual nature.The youngest sample in the present investigation with more than 2 *G. limbata* specimens occurs at 2.19 Ma (sample 806C-6-3, 83-85cm). Thereafter, sporadic occurrence of *G. limbata* was recorded only in single specimens (n=1) until 0.42 Ma in sample 806C-2-2, 99-100cm, see Appendices 4 and 7, and contour plots in Fig. 16 further below.

In *G. multicamerata,* analysis of evolutionary trends is more difficult because of rare occurrence or absence in many samples. Considering the levels with more than 2 *G. multicamerata* specimens (except the sample at 7.26 Ma, see electronic Appendix 5) all morphometric parameters except Φ_1 and convexity ratio A/B are in phase showing increasing trends from 4.35 Ma – 3.36 Ma, and declining trends thereafter, e.g., from 3.36 Ma – 2.63 Ma.

5.2 Most significant parameters for divergence in the *G. menardii* – limbata – *multicamerata* lineage

For recognition of divergence trends between *G. menardii, G. limbata* and *G. multicamerata* the means and 95% confidence intervals for shell parameters per sample and group are re-plotted in each panel D of Figs. 6 to 14. This separate consideration is most interesting and helps to filter out the most distinctive morphometric parameters for recognition of possible cladogenetic trends. Despite of lower

abundance of *G. limbata* and *G. multicamerata* assemlages in comparison to *G. menardii*, separate consideration highlights slow, phyletic divergence towards larger forms. Most significant parameters include the keel view area (Ar), spiral height (δX), axial length (δY), radii of upper osculating circles (R_{up}), and spiral convexity (A), with *G. limbata* being always intermediate between ancestral *G. menardii* and descendent *G. multicamerata*. In mean upper keel angles Φ_1 the phyletic trend is reverse, but still with *G. limbata* in intermediate position. No clear pattern is evident in the lower keel angle (Φ_2) and mean radii of the lower osculating circles (R_{lo}).

5.3 Plane contour frequency diagrams in the morphospace δX versus δY

Contoured frequency diagrams of spiral height (δX) versus axial length (δY) in profile-view have proven to be an intuitive prospection tool for tracking evolution and divergence in the present study. For sample-to-sample comparison, such plane contour diagrams were therefore prepared from bivariate measurements of δX and δY for *G. menardii*, *G. limbata*, and *G. multicamerata* and illustrated in Figs. 15, 16 and 17, respectively. Gridded files to contour frequency diagrams are provided in files RawData_5 through 8 in the electronic archive RawData.zip.

In *G. menardii*, distributions of δX versus δY define a single, almost linear, uni-directional and continous diagonal morphocline in all samples with varying modal centers and broadening as size increases (see Figure 15). From 7.96 Ma until about 1.95 - 1.84 Ma, there is a tendency of steady increase in δX (and corresponding values of δY) with quite large forms appearing at 1.95 Ma and at 1.84 Ma. Until 4 Ma, assemblages are unimodal and most specimens show a spiral height <300 μm (exception: the sample at 4.64 Ma). In levels younger than 4 Ma, assemblages get significantly larger in δX with a weak tendency to develop a bimodality in δX versus δY from 3.03 Ma to 2.63-2.55 Ma. In bimodal samples, the main mode remains in the δX<300 μm domain, while the subordinate mode advances towards values of δX>400 μm. In samples post 2.55 Ma, bimodality vanishes and populations remain unimodal. However, modal centers tend again to increase in δX and δY: From 1.84 Ma and younger, most modal centers have peaks with δX ≥ 300 μm. The sample at 1.95 Ma is special because it extends over a very wide range in δX and δY despite comparably low abundance of *G. menardii*. In that particular sample *Globorotalia tumida* strongly dominated the globorotalid fauna. Generally speaking, trends in δX versus δY in *G. menardii* at Hole 806C display a rather gradual pattern and there is no indication of sudden size reduction. Note, that with a few exceptions (7.96 Ma, 0.42 Ma), modal centers or crests from Hole 806C distribute below a diagonal line connecting coordinates 50/0 μm and 550/1600 μm in the δX versus δY morphospace (Fig. 15).

Globorotalia limbata specimens occupy a very similar band in the δX versus δY morphospace. Apart from being artificially confined to have 7 chambers in the final whorl, *G. limbata* is difficult to distinguish in single specimens from ancestral *G. menardii*, and divergence appears only in mean δX versus mean δY (Fig. 16, compare also panels D in Figs. 7 and 8) and few other parameters, such as mean Ar, mean Rup, and mean A (see panels D in Figs. 6, 9, and 13). For direct comparison the δX versus δY contour diagrams of *G. limbata* were superimposed to those of *G. menardii* (*G. menardii* are grey

contours in Fig. 16). Starting with small tests and at low frequency at 7.96 Ma at Hole 806C, *G. limbata* expanded along the *G. menardii* morphocline showing almost complete overlap, however with modal centers generally propagating faster to larger δX and δY values than in *G. menardii*. From 7.96 Ma onwards up-core, frequency (and variance) of *G. limbata* increased until 3.18 Ma, when the center of the main mode of *G. limbata* stabilized at about 400 μm in δX and 950 μm in δY. Thereafter, until 2.19 Ma, and with a barren interval in between at 2.87 Ma, *G. limbata* returned to smaller sizes with main modes close to or slightly below 300 μm in δX. Post 2.19 Ma, *G. limbata* occurred only sporadicly in single specimens and vanished completely in sediments younger than 0.71 Ma.

Globorotalia multicamerata occurred almost always in low numbers. Only at intermittent levels in Hole 806C it was more frequent. *G. multicamerata* was first observed at 7.26 Ma (Fig. 17). At 4.35 Ma the highest numbers occurred (n=92 in Figure 17), which can be converted to 43.8 specimens per gram dry bulk sediment (as calculated from the abundances and split fractions given in Appendix 7 and the sediment weights given in Raw_Data_14). From 4.35-3.03 Ma *G. multicamerata* was encountered in abundances up to 6.4 specimens per gram dry bulk sediment. The youngest sample containing *G. multicamerata* is at 2.19 Ma (e.g., 1 specimen observed in ¼ split, which converts to ~1 specimen per gram dry bulk sediment). Divergence of *G. multicamerata* from *G. limbata* is therefore rather difficult to document at Site 806. Between 4.35-3.36 Ma, increasing values of δX and δY can be observed, which also populate inside the range of the *G. menardii* morphocline, and which ends in a low frequency mode at 3.36 Ma centered at δX=500 μm/δy=850 μm (Fig. 17). After 3.36 Ma this trend reversed and at 3.03 Ma the modal center moved to coordinates δX=275 μm/δy=700 μm (Fig. 17). Thereafter, *G. multicamerata* was observed only in singular specimens.

6. Discussion

The following sub-sections explore VDDs as an aid to document evolutionary trends in *G. menardii*, *G. limbata* and *G. multicamerata* at Hole 806C.

6.1 VDDs of *G. menardii*, *G. limbata* and *G. multicamerata*

Figure 18 illustrates calibrated VDDs in the morphospace of δX versus δY versus time for *G. menardii*, *G. limbata* and *G. multicamerata* from Hole 806C. As explained in the methods section, the interpolated outer isosurfaces were modeled in such a way, that they best match plane contour diagrams and frequency matrices from individual samples. Because of unequal frequencies between the three species in a sample, the same levels from plane contour diagrams between *G. menardii*, *G. limbata*, and *G. multicamerata* result in different isovalues of their isosurfaces. In *G. menardii*, the density of 3 specimens per grid-cell is represented by an isovalue of 2.499. In *G. limbata* and *G. multicamerata,* the same density was found at isovalues of 1.452 and 1.105, respectively. For comparison with the planar trends of δX and δY through time (Figs. 7D and 8D), the mean δX versus δY for all three groups are also projected into the 3-dimension spaces of each VDD in Fig. 18.

The VDD in Fig. 18A documents the rather gradual evolutionary trend in δX versus δY for *G. menardii*. There is indication of size reduction in the internal structure of δX and δY around 2.6 Ma (e.g., at a normalized time of -0.32875 in Fig. 18A), but – as will be pointed out later in section 5.5 - this reduction is less strong in comparison to the size jumps at Atlantic ODP Site 925 (between 2.3–1.95 Ma) or at Caribbean DSDP Site 502 (between 2.58–1.7 Ma).

The VDDs for *G. limbata* and *G. multicamerata* are patchier, because of their shorter stratigraphic ranges and of lower abundances of both species. *G. limbata* begins to form more dense clusters in samples younger than 5 Ma (normalized time of -0.625 in Figure 18D), roughly coinciding with the larger mode center in *G. menardii* at 4.10 Ma (normalized time of -0.5125 in Figure 18D) and further connecting with the main cluster of *G. multicamerata* in Figure 18D. This pattern is interpreted as a cladogenetic event of *G. limbata-G. multicamerata* from ancestral *G. menardii* that is perhaps more prominent elsewhere in the early Pliocene (Zanclean) ocean (compare also with plane contour lines in Figs. 15, 16 and 17).

6.2 Vertical slices through VDDs

For closer inspection of intraspecific δX versus δY developments in the *G. menardii-limbata-multicamerata* lineage, parallel series of diagonal contoured time-sections through VDDs were created. For brevity, only the central diagonal slice at zero offset is illustrated in Fig. 19. It represents an enlargement of the diagonal (45°) contour plane shown in Fig. 18D, with blue contour lines indicating densities of *G. menardii* at zero offset from center through the VDD and normal to the viewer. In that plane δX and δY approach maximal covariation between each other, and thus represent averaged first principal component values for δX and δY through time. Because the slope of bivariate contour-diagrams remains more or less constant, projections of δX versus δY in the 45° plane offer trend analysis of internal cluster development through time. In such longitudinal sections, higher density ridges to the left may form a main branch of *G. menardii*, while areas of lower density to the right may indicate side branches from the main branch. The clouds of *G. limbata* (7 chambers in the final whorl) and *G. multicamerata* (≥ 8 chambers in the final whorl) strongly overlap in this projection. However, the divergence of mean δX versus mean δY between *G. menardii, limbata* and *multicamerata* becomes maximal in that perspective.

6.2.1 Ranges and indications for divergence of *G. limbata* and *G. multicamerata* from *G. menardii*

According to Chaisson and Leckie (1993), first rare *G. limbata* were encountered at Hole 806B in sample 34-2,30-32cm, (312.31 mbsf) at an age of 8.72 Ma (for age model see Table 2). At Hole 806C (this study, see Appendix 7 in the electronic version of this article) *G. limbata* was observed in sample 806C-29-7, 31-32cm (271.41 mbsf) at an age of 7.96 Ma. Until 5.78 Ma at Hole 806C (sample 806C-20-2, 68-69cm) *G. limbata* was present but remained rare. From 5.78 Ma core-upwards divergence in mean δX versus mean δY can be seen (Fig. 19 and Figs. 7-8). From 4.35 Ma (Z=-0.54) onwards,

morphological divergence of *G. limbata* from *G. menardii* at Hole 806C became more accelerated. At Site 806, *G. limbata* became extinct between 0.42 Ma (sample 806C-2-2, 99-100cm, n=1) and 0.34 Ma (sample 806C-2-1, 31-32cm, n=1, see Appendix 7). This is about 0.65 Ma younger than calculated from the range chart for Hole 806B in Chaisson et al. (1993), where the last occurrence of *G. limbata* was recorded between sample 806B-3-CC and sample 806B-3-5, 22-24cm.

In case of *G. multicamerata* Chaisson and Leckie (1993) observed their rare specimens with 9 chambers only in a single sample (806B-23-5, 30-32cm) at 6.51 Ma (see Appendix 7). In the present study, first *G. multicamerata* (≥ 8 chambers) were encountered already at 7.26 Ma (sample 806C-26-5, 27-28cm, Fig. 19). Further up-core, *G. multicamerata* suddenly reappears quite frequently in sample 806C-14-1, 22-23cm at 4.35 Ma (Fig. 19). From this level onwards it began to develop larger shells. Above 2.63 Ma, only a single specimen of *G. multicamerata* was observed in one sample at 2.19 Ma. The next sample higher up in the core, where *G. multicamerata* became permanently absent is at 1.95 Ma (Appendix 7). The extinction interval of *G. multicamerat*a at Site 806 is hence interpreted between 2.63 Ma and 1.95 Ma.

6.3 Relative abundances of menardiform groups and coiling changes

In the context of the morphological trends, relative abundance changes and coiling behavior within the *G. menardii-limbata-multicamerata* lineage, and the *G. tumida* group are assessed in the following. The left side of Fig. 19 illustrates some co-variation between δX versus δY divergence in the *G. menardii-limbata-multicamerata* lineage and a predominantly dextral coiling phase of *G. menardii* in the interval from 5 Ma to 3 Ma. This coincidence occurs also with other test parameters like the keel view area (Ar), radius of upper osculating circle (R_{up}), and spiral convexity (A) - see the respective plane line-diagrams in Figs. 6D, 9D, and 13D. Also, in that time interval, falls the rise of dextrally coiled transitional forms between *G. menardii* and *G. tumida* (abbreviated as mentum, see the section about taxonomic remarks), which otherwise were mostly sinistral. Noticeable is the increasing proportion of the *G. tumida* group (including *G. tumida* and *G. tumida flexuosa*) in comparison to the *G. menardii* since 5.78 Ma. Since then *G. tumida* raised to a common faunal component of planktonic foraminifera at Site 806, giving the western tropical Pacific area a typical "*Tumida*-look".

G. *menardii* is known to dwell in the middle to upper thermocline (Chaisson and Leckie, 1993; Keller, 1985; Ravelo and Fairbanks, 1992; Chaisson, 1995, 2003; Chaisson and Ravelo, 1997; Schiebel and Hemleben, 2017), i.e. at depths where water density discontinuity leads to maximum chlorophyll contents and where *G. menardii* seems to track moderately low oxygen concentrations in its upper thermocline habitat (Sexton and Norris, 2011). *Globorotalia tumida* populates deeper, more at the base of the thermocline (Schweitzer, 1989; Ravelo and Fairbanks, 1992; Chaisson and Ravelo, 1997, Wara et al., 2005). The abundance and coiling changes of members of the *G. menardii-* and *G. tumida* groups, and the concurrent divergence trends in the *G. menardii-limbata-multicamerata* lineage may thus provide hints to changes of the upper water column structure and stratification over longer time.

6.4 Evolutionary history of menardiforms in light of Late Neogene Pacific paleoceanography

During the Late Neogene, the WPWP was subjected to complex changes in ocean gateways, paleoceanographic reorganisations, and climatic perturbations. Major influence during the stratigraphic extend of the present study were caused by the constriction of the Indonesian seaway during the middle Miocene-lower Pliocene (see Nathan and Leckie, 2009 for age constraints of the gradual constriction between ~ 17 to 3-4 Ma, and van der Lubbe et al., 2021 for a summary of Pacific climate transitions during the past 7 million years). Today, the Indonesian Seaway is still very much open with an estimated total throughflow of water of 15 Sv (Tillinger, 2011; Feng et al., 2018), and from which 11.6 Sv and 1.1 Sv pass from the Pacific Ocean through the Makassar and Limafatola straits, respectively (Wikipedia, 2021; Leckie, written comm.). During the Late Miocene, the thermocline dwelling *G. menardii-G. tumida* group evolved and increased in abundance at nearby DSDP Site 289 (Keller, 1985) and expanded in the tropical Pacific (Kennett et al., 1985), a phenomenon these authors interpreted to reflect strengthening of the Equatorial Countercurrent system and development of a shallower thermocline after constriction of the Indonesian Seaway. These oceanographic reorganisations were superimposed by influences through West Antarctic Ice Sheet expansion between 6.5-5 Ma (foraminiferal zone N17b, Chaisson and Leckie, 1993), leading to increased equator-pole temperature gradients, increased trade wind strength, altogether enforcing a shallower western equatorial thermocline in the Pacific (Chaisson and Leckie 1993, p. 152). The rapid evolution to larger *Globorotalia tumida* reported by Malmgren et al. (1983) between about 5.6 and 5.0 Ma in the Indian Ocean may have been caused by the Southern Hemisphere cooling and was perhaps associated with coeval thermocline strengthening in the eastern part of the tropical Indian Ocean. The gradual constriction of the Indonesian seaway has led to the formation of a permanent Equatorial Pacific undercurrent (Chaisson and Leckie, 1993; Nathan and Leckie, 2009) and thermocline shoaling in the equatorial Pacific (LaRiviere et al., 2012). Niche space expanded for upper intermediate water dwellers and favoured the *G. menardii-G. limbata* complex and *G. tumida* (Kennett et al., 1985; Chaisson and Leckie, 1993). Blocking of westward flow of tropical waters from the Pacific to Indian Ocean at 5.2 Ma may have resulted in extinction of *Pulleniatina spectabilis* in the Indian Ocean (Srinivasan and Sinha, 1998). The same authors report on the onset of tropical circulation in the Pacific due to emergence of the Indonesian archipelago. Further restriction of Indonesian Throughflow culminated around 3.5 Ma due to northward movement of the Australian continent (Auer et al., 2019; Kaushik et al., 2020), reinforcing stratification of upper watermasses and thermocline shoaling in the WPWP. According to Karas et al. (2016), tectonic reorganization of the Indonesian region caused a change in Throughflow from warm and salt South Pacific to fresher, cooler North Pacific subsurface waters during 3.5-2.95 Ma, which also affected the fresher and cooler Agulhas Current system. After 2.2 – 2.0 Ma, the intensification of the Pacific Walker cell led to development of similar zonal sea-surface temperature gradients in the equatorial Pacific than today (van der Lübbe et al., 2021), and to a massive warm mixed layer that is characteristic for the modern WPWP (Chaisson, 1995). Since the constriction of the Indonesian seaway, increasing upwellings of cooler waters result in a thermocline of regionally varying thickness and development of the Pacific cold tongue (e.g., La Riviere et al., 2012).

The closure of the Central American Seaway (CAS) also played a key role on tropical ocean circulation and watermass structure, again influencing the WPWP: shoaling of the CAS began already during the Middle Miocene (Lutz, 2010). Until 5.1 Ma, stable El-Niño-like conditions persisted in the eastern tropical Pacific, which vanished therafter and recurring in cycles in modern-like east-west contrasting El-Niño-Southern Oscillations (e.g., von der Heydt and Dijkstra, 2011; Ham, 2018; Caj et al., 2018; Ford et al., 2015). Increased restriction in exchange of tropical waters between the Caribbean Sea and Pacific Ocean occurred from about 4.8-4 Ma onwards (Haug and Tiedemann, 1998; Norris, 1999; Lutz, 2010; Karas et al. 2016; Chaisson and Leckie, 1993; Lam and Leckie, 2020), leading to tropical Atlantic-Pacific bioprovincialism between normally perforated and finely perforated menardiforms (Chaisson, 2003). Also the development of a stronger keel can be in some instances associated with the invasion of new depth habitats (Ezard, Aze, Pearson and Purvis, 2011). As a consequence, a zonal Sea Surface Temperature (SST) gradient established between the western equatorial Pacific Site 806 and the eastern equatorial Pacific Site 846 (Karas et al., 2016; Wara et al. 2005; Chaisson and Ravelo, 2000; Schmidt, 2007; Nathan and Leckie, 2009), all embedded in a continuous trend of shoaling thermocline since about 13 Ma (La Rivière et al., 2012). Final closure of the CAS occurred around 3 Ma or shortly thereafter (Cronin and Dowsett, 1996; Schmidt, 2007; Sexton and Norris, 2008; Knappertsbusch, 2016a). It further influenced equatorial Pacific surface circulation patterns and contributed to regionally differential thermocline structures in the east (e.g., Ford, 2015), and also reinforced general cooling of the Northern Hemisphere. After 3 Ma, *G. menardii* and *G. limbata* became intermittently restricted to the Indo-Pacific basins (Norris, 1999). Between 2.7 Ma-2.5 Ma (Chapman, 2000; Bailey et al., 2013; Hodell and Channell, 2016; Ögretmen et al., 2020) the Northern Hemisphere Glaciation (NHG) started: Escalating phases of glacial-interglacial oscillations developed from the late Pliocene (e.g., Sexton and Norris, 2008), each influencing sea-level height, restriction or amplification of currents through the Indonesian Archipelago, inducing cyclic reinforcement/weakening of the upper water stratificartion also in the WPWP area. Past 2.4 Ma – and notably not directly after the closing of the Panama seaway (Norris, 1999) – the modern oceanographic and productivity asymmetry between the eastern and western equatorial Pacific established. As large scale climatic transitions are generally reflected in the spaciotemporal pattern of the microfossil record (Fraass et al., 2015), the above described paleoclimatic history may also be mirrored in the size evolution of menardiform shells.

During the Pleistocene, *G. menardii* populations repeatedly re-invaded the tropical Atlantic during interglacial times and became widely reduced during glacials (Ericson and Wollin, 1968; Norris, 1999; Broecker and Pena, 2014). They may have entered the South Atlantic at times, when especially favorable conditions allowed far dispersal through the Indonesian Seaway and intensified leakage of warm Indian Ocean Waters by the Agulhas Current System. Such long distance inter-oceanic exchange is relatively well known for the Agulhas Current system (Ericson and Wollin, 1968; Gordon, 1985; Lutjeharms and Gordon, 1987; Berger and Vincent, 1986; Berger and Wefer, 1996; Flores et al., 1999; Darling et al., 1999; Giraudeau et al., 2002; Peeters et al., 2004; Rau et al., 2002, 2006; Sexton and Norris, 2008; Caley et al., 2012; Zahn et al., 2012; Marino and Zahn, 2015; Hall et al., 2015). It has however, only rarely been taken into account for *G. menardii* (but see Broecker and Pena, 2014;

Iwai et al., 2014), and was discussed controversially in favour to the deglacial proliferation model (Sexton and Norris, 2011). Repeated seeding of tropical plankton faunas from the Indian Ocean into the South-Atlantic during glacial-interglacial transitions is known since at least the Pleistocene (Schott, 1935; Ericson and Wollin, 1968). The sudden and prominent menardiform shell size increase at ODP Hole 925A since about 1.95 Ma (Late Pliocene), i.e. after a period of strong size reduction during Northern Hemisphere cooling and glaciation around 2.6 Ma (Knappertsbusch 2016a), is now seen as a possible leakage of menardiform globorotalids into the South Atlantic and further extending into the tropical Atlantic Ocean. It is obvious, that the morphological development of *G. menardii-limbata-multicamerata* lineage and extant *G. menardii* must be sought in this complex paleoceanographic scenery (Fig. 19).

The evolution of the *G. menardii-limbata-multicamerata* lineage and *G. menardii* past NHG occurred possibly in combination with a decreased upper thermocline ventilation strength and associated low oxygen saturation levels similar to the deglaciation proliferation model for late Quaternary menardiforms of Sexton and Norris (2011). In this context, the $\delta^{18}O$ isotope records for surface dweller *Globigerinoides sacculifer* from Hole 806B from Berger et al. (1993) and Jansen et al. (1993) are juxtaposed to the shell size trends of *G. menardii*, *G. limbata* and *G. multicamerata* on Figure 19. The curves of Berger et al. (1993) and Jansen et al. (1993) were preferred here over the benthic $\delta^{18}O$ stack of Lisiecki et al. (2005) because the former two were directly from Site 806. Also indicated in Figure 19, is the record of the difference in $\delta^{18}O$ between subsurface dwellers *G. menardii* (data from Nathan and Leckie, 2009) and *G. tumida*, and surface dwellers at Site 806 from LaRiviere et al. (2012). This $\Delta\delta^{18}O$ is a proxy for the prolonged phase of declining mixed layer thickness during the Pliocene.

Figure 19 shows that, during the Zanclean-early Piacenzian (e.g., between 5 Ma until the end of the Mid Pliocene Warmth Period, MPW), the $\delta^{18}O$ and $\Delta\delta^{18}O$ curves roughly parallel the trend of the VDDs. This suggests that the shell-size evolution of *G. limbata* and *G. multicamerata* may be caused by the warming trend seen in the isotope record during that period, and that it perhaps traces occupation of a shallower habitat of menardiforms when the mixed layer gradually expanded but simultaneously became less ventilated during warming.

During NHG, and accentuated at MIS 100 cold event, *G. limbata* and *G. multicamerata* became rarer and simultaneously smaller in shell size, until both became extinct after 2.19 Ma (*G. multicamerata*) or much reduced in numbers (*G. limbata*). Also *G. menardii* became reduced in overall size variability at 2.63 Ma even under the tropical conditions of the WPWP. However, at Site 806 the effect on shell size is by far less severe than during the size reduction at 2.58 Ma in the Atlantic (e.g., Knappertsbusch, 2016a). Interestingly, and in line with the observed constriction in δX and δY of the younger *G. menardii* group, there was a period of reduced SST by up to 2°C from ~3.2 Ma until about 1.8 Ma at Site 806 (see Fig. 2B in Wara et al, 2005). During such times, with $\delta^{18}O$ signature indicating extreme cold conditions, shells developed spiral heights δX predominantly smaller than 300 μm.

Thereafter, and notably from the Mid Pleistocene Revolution (MPR) onwards, distributions of δX versus δY of *G. menardii* tended to develop alternate modal variations between neighbouring samples, with separation of main modes around δX=300 μm (compare Fig. 19 with 15). In the literature,

variations in abundance and size of *G. menardii* were associated with glacial/interglacial cycles, as reported in cores from the northeastern Indian Ocean back to isotope stage MIS 5 (Bhonsale and Saraswat, 2012). In the present study, however, sampling is too coarse and samples were selected only within interglacials. One can therefore only suspect, that modal variations may reflect the establishment of two vertically separated populations of *G. menardii* in the water column with varying abundances. The larger sizes perhaps preferred shallower water levels, where a stronger chlorophyll maximum developed. This idea needs more future investigation at glacial-interglacial resolution. The idea would fit with the observation, that, during the Neogene and at low latitudes, growth to larger foraminiferal sizes is often associated with surface water stratification intensity (Schmidt et al., 2003, 2006; Schmidt, Thierstein, Bollmann and Schiebel, 2004; Schmidt, Thierstein and Bollmann, 2004) and/or warming (Todd et al., 2020). When considering average values of δX vs δY at Hole 806C (see Figs. 8, 9, 15, and line diagrams in Figs. 18-20), *G. menardii* shows an offset to generally larger sizes in comparison to pre-NHG times, but with a slight tendency to decrease from the Calabrian to the Holocene.

6.5 Asymmetry in morphological evolution between Atlantic and Pacific

The morphological trends of *G. menardii* at Ontong-Java Plateau most probably reflect changes in upper watermass structure, that were caused by complex oceanographic reorganisations of the Indonesian Throughflow, closure of the Central American Seaway (Lam and Leckie, 2020), and water mass oscillations in the course of the subsequent NHG. Seen over the Pliocene-Pleistocene time span, the pattern of size increase of menardiform globorotalids at Site 806 is rather gradual, and similar to the pattern observed by Knappertsbusch (2007) at eastern equatorial Pacific Site 503 (see Figure 20). These observations from the equatorial Pacific are different to the morphological evolution of *G. menardii* in the tropical Atlantic, where suddenly much larger forms can be observed past 2.6 Ma in the Caribbean Sea (Knappertsbusch, 2007) and the Western Atlantic (Knappertsbusch, 2016a, compare VDDs in Figure 20 around the normalized time Z = 0.325). This asymmetry in the morphological evolution of menardiforms between the two ocean basins is most striking and it is interpreted here by fundamental differences in hydrography between Atlantic and Pacific with respect to circulation, thermocline structure, seasonality, and nutrient regime (Norris, 1999).

If the above reasonings about menardiform globorotalids were generalized to other microzooplankton this means, that large- and smaller- scale hydrological and nutritional conditions, seasonality, and possibly differences in the thermocline ventilation state (Sexton and Norris, 2011), fostered a richness of niches in the upper watercolumn, where these organisms could narrowly adapt (Norris, 1999; Schmidt et al., 2006). Each species follows thereby its own ecological preferences, and reacts subtly upon environmental stress. Ecological pressure and selective forces over generations, that are primarily driven by external, abiotic influences, are the causes for a continuously escalating morphological diversity and size changes (Schmidt et al., 2004). As the watermass structure modified, so did the planktic foraminifera, by adaptation of their shells. Yet on the individual level, the precise physiological causes for shell growth and change remain still largely unknown. Therefore, and in order

to better understand evolution and adaptive processes, morphological changes under undisturbed, stable conditions need to be distinguished from morphological changes caused by dispersal of species when ocean circulations change due to opening and closing oceanic gateways.

The existence of graduated micro-niches in the upper water column has important consequences for biostratigraphers: Straight biostratigraphic correlatability of species from one place to the next is not necessarily guaranteed. Instead, evolutionary synchrony or diachrony of first and last appearance datums, divergence patterns, biogeographic preferences and dispersal of stratigraphic markers must be carefully be proven from one place to the next, ranging from local to global level, before correlation with planktonic index forms can be practized at high resolution. This remains a major task for future micropaleontology.

7. Conclusions

The most important outcome from the present study is that *Globorotalia menardii* evolved in a more gradual manner at western equatorial Pacific Hole 806C than in the Atlantic across 2.6 Ma, and shows no evidence of abrupt size increase between 2.3 Ma - 1.7 Ma. This observation is similar to the Eastern Equatorial Pacific DSDP Site 503, where also a rather gradual pattern of morphological evolution was found in menardiform globorotalids by Knappertsbusch (2007). These observations document a distinct asymmetry in morphological evolution of *G. menardii* between the tropical Atlantic and Pacific Ocean, which possibly applies to other lineages of planktonic foraminifera too, and which may be of general nature. In more detail, the δX versus δY structure of *G. menardii* from Hole 806C shows development of some internal bi-modality between 3.18 Ma – 2.63 Ma, which is expressed in a higher frequency mode of specimens smaller than about 300 μm in δx until modern times. A subordinate mode of *G. menardii* with δX>400 μm tailored to very large specimens appeared from 3.18 Ma until 2.63 Ma. Thereafter, this slight bimodality vanished but the *G. menardii* assemblage shifted towards larger specimens. The gradual morphological evolution of tropical Pacific *G. menardii* stands in strong contrast to the pattern of rapid morphological change in the tropical Atlantic. This contrast may perhaps be related to an episodically waxing and waning Agulhas Current system in the past, which modulates the morphological pattern of menardiforms through time in the Atlantic Ocean.

At Site 806 the First Appearance Datum (FAD) of *G. limbata* occurred in the interval between 8.71 Ma and 8.81 Ma Ma and this form showed morphological divergence from *G. menardii* between 5.78 Ma until 3.18 Ma. *G. multicamerata* occurred first at Site 806 between 7.58 Ma and 7.26 Ma, about 1.25 Ma earlier than found by Chaisson and Leckie (1993). It was rarer than *G. limbata* but signs of morphological divergence can be seen too in several parameters. Divergence of *G. limbata* and *G. multicamerata* appeared to accentuate in parallel to a general warming trend. In the course of the NHG both forms exhibited terminal dwarfism, which is manifested in shell-size reduction prior to extinction of *G. multicamerata* and the strong decline in abundance of *G. limbata* past 2.19 Ma (*G. limbata* existed in singular specimens until 0.42 Ma).

Globorotalia limbata and *G. multicamerata* show evidence for cladogenetic splitting from ancestral *G. menardii*, although both species are subordinate in frequency at Site 806C. Cladogenesis is seen in shifts of mean δX and mean δY of *G. limbata* and *G. multicamerata* away from those of *G. menardii*.

8. Disclosure of interest

The author declares that he has no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Figure captions

Fig. 1. A. *Globorotalia menardii* in umbilical view. B. Phylogenetic scheme for menardiform globorotalids after Kennett & Srinivasan (1983). C. Spiral height (δX) versus axial length (δY) measurements for the *G. menardii-limbata-multicamerata* lineage at ODP Site 925B for the past 8 Ma illustrated in form of a volume density diagram (from Knappertsbusch, 2016a). *G. menardii* (yellow) is ancestral, from which intermediate *G. limbata* (pink) and terminal *G. multicamerata* (red) separated. For more information about volume density diagrams see sections further below.

Fig. 2. Position of ODP Hole 806C (0° 19.1'N / 159° 21.7'E, 2521m).
Map from https://maps.ngdc.noaa.gov/index.html

Fig. 3. Comparison of the numerical age models from Holes 806B and 806C. The plotted data are tabulated in Table 2.

Fig. 4. Motorized 4-axis tilting stage of AMOR. The prepared morphometry slide is placed in the metal plate sample holder and can be translated in X- (left-right) and Y (forward-backward) direction, and tilted left-right („Roll") and forward-backward („Pitch").

Fig. 5. Plane morphological parameters collected during the present study. A. Test of *Globorotalia menardii* (X-ray CT image) in standard keel view position with the apex facing to the left side and the apertural area facing upwards. In the illustrated dextral specimen the final chamber is directed to the top, while in a sinistral specimen the final chamber would be directed to the bottom. B. δX spiral height, δY axial length, Φ_1 upper keel angle, Φ_2 lower keel angle, Φ_3 apical angle, R_{up} radius of osculating circle in upper keel region, R_{lo} radius of osculating circle in lower keel region, A spiral convexity, B umbilical convexity. The area Ar is the area enclosed by the outline in keel view. For comparison of measurements from sinistrally and dextrally coiled specimens *non* coiling-invariant parameters like Φ_1, Φ_2, R_{up}, and R_{lo} were mirrored in the data matrix using program DexFlip.out (see text).

Fig. 6. Variation of keel-view area (Ar, in mm^2) during the past 8 million years for *G. menardii* (A), *G. limbata* (B), and *G. multicamerata* (C) at ODP Hole 806C. Thick solid line through open symbols indicate sample means. Small solid squares represent medians. Horizontal bars indicate the range from the lower to the upper quartile in each sample. Dashed lines indicate the 10% and 90% sample percentiles. Grey lines are the sample minima and maxima. The straight line is the linear regression line through sample means described by the equation in each respective graph, r^2 is the linear regression coefficient. In Fig. D the sample means and ±95% confidence intervals about the means are plotted again for direct comparison of trends between *G. menardii* (squares), *G. limbata* (circles), and *G. multicamerata* (triangles). Single symbols without confidence intervals are from samples with only 2 or less specimens of the respective menardiform group. Note change of horizontal scale for

better visibility in panel D. See summary statistics for Ar per sample in the electronic Appendices 3, 4, and 5 to this article for *G. menardii*, *G. limbata*, and *G. multicamerata*, respectively. Measurements from individual specimens for *G. menardii*, *G. limbata*, and *G. multicamerata*, are provided electronically in files RawData_1, 2, and 3, respectively in the supplied archive RawData.zip. The group „*G. menardii*-lin“ in (panel A) includes *G. menardii cultrata*, *G. menardii menardii*, undifferentiated and juvenile *G. menardii*, *G. menardii flexuosa*, and *G. menardii gibberula* (see section with taxonomic notes).

Fig. 7. Variation of spiral height (δX, in μm) during the past 8 million years for *G. menardii* (A), *G. limbata* (B), and *G. multicamerata* (C) at ODP Hole 806C. For explanation of symbols see Fig. 6. Note change of horizontal scale in panel D. See summary statistics for δX per sample in the electronic Appendices 3, 4, and 5 for *G. menardii*, *G. limbata*, and *G. multicamerata*, respectively. Measurements from individual specimens for *G. menardii*, *G. limbata*, and *G. multicamerata*, are provided electronically in files RawData_1, 2, and 3, respectively in the supplied archive RawData.zip.

Fig. 8. Variation of axial length (δY, in μm) during the past 8 million years for *G. menardii* (panel A), *G. limbata* (B), and *G. multicamerata* (C) at ODP Hole 806C. For explanation of symbols see Fig. 6. Note change of horizontal scale for better visibility in panel D. See summary statistics for δY per sample provided in the electronic Appendices 3, 4, and 5 for *G. menardii*, *G. limbata*, and *G. multicamerata*, respectively. Measurements from individual specimens for *G. menardii*, *G. limbata*, and *G. multicamerata*, are provided electronically in files RawData_1, 2, and 3, respectively in the supplied archive RawData.zip.

Fig. 9. Trends of the radii of upper osculating circles (R_{up}, in μm) during the past 8 million years for *G. menardii* (panel A), *G. limbata* (B), and *G. multicamerata* (C) at ODP Hole 806C. For explanation of symbols see Fig. 6. See summary statistics for R_{up} per sample in the electronic Appendices 3, 4, and 5 for *G. menardii*, *G. limbata*, and *G. multicamerata*, respectively. Measurements from individual specimens for *G. menardii*, *G. limbata*, and *G. multicamerata*, are provided electronically in files RawData_1, 2, and 3, respectively in the supplied archive RawData.zip. Note that in panel A the range of R_{up} is from 0 to 400 μm, while in panels B and C R_{up} ranges from 0 to 200 μm, and in panel D it ranges from 0 to 150 μm for better visibility.

Fig. 10. Trends of the radii of lower osculating circles (R_{lo}, in μm) during the past 8 million years for *G. menardii* (panel A), *G. limbata* (B), and *G. multicamerata* (C) at ODP Hole 806C. For explanation of symbols see Fig. 6. Note change of horizontal scale from panel A to panels B,C and D. See summary statistics for R_{lo} per sample provided in the electronic Appendices 3, 4, and 5 for *G. menardii*, *G. limbata*, and *G. multicamerata*, respectively. Measurements from individual specimens for *G. menardii*, *G. limbata*, and *G. multicamerata*, are provided electronically in files RawData_1, 2, and 3, respectively in the supplied archive RawData.zip.

Fig. 11. Variation of upper keel angle (Φ_1, in degrees) during the past 8 million years for *G. menardii* (panel A), *G. limbata* (B), and *G. multicamerata* (C) at ODP Hole 806C. For explanation of symbols see Fig. 6. Note change of horizontal scale from panels A, B and C to panel D. See summary statistics for Φ_1 per sample in the electronic Appendices 3, 4, and 5 for *G. menardii*, *G. limbata*, and *G. multicamerata*, respectively. Measurements from individual specimens for *G. menardii*, *G. limbata*, and *G. multicamerata*, are provided electronically in files RawData_1, 2, and 3, respectively in the supplied archive RawData.zip.

Fig. 12. Variation of lower keel angle (Φ_2, in degrees) during the past 8 million years for *G. menardii* (panel A), *G. limbata* (B), and *G. multicamerata* (C) at ODP Hole 806C. For explanation of symbols see Fig. 6. Note change of horizontal scale from panels A, B, C to panel D. See summary statistics for Φ_2 per sample in the electronic Appendices 3, 4, and 5 for *G. menardii*, *G. limbata*, and *G. multicamerata*, respectively. Measurements from individual specimens for *G. menardii*, *G. limbata*, and *G. multicamerata*, are provided electronically in files RawData_1, 2, and 3, respectively in the supplied archive RawData.zip.

Fig. 13. Spiral convexity (A, in μm) versus time during the past 8 million years for *G. menardii* (panel A), *G. limbata* (B), and *G. multicamerata* (C) at ODP Hole 806C. For explanation of symbols see Fig. 6. Note change of horizontal scale from panels A, B and C to panel D. See summary statistics for A per sample in the electronic Appendices 3, 4, and 5 for *G. menardii*, *G. limbata*, and *G. multicamerata*, respectively. Measurements from individual specimens for *G. menardii*, *G. limbata*, and *G. multicamerata*, are provided electronically in files RawData_1, 2, and 3, respectively in the supplied archive RawData.zip.

Fig. 14. Convexity ratio (A/B) versus time during the past 8 million years for *G. menardii* (panel A), *G. limbata* (B), and *G. multicamerata* (C) at ODP Hole 806C. For explanation of symbols see Fig. 6. Note change of scale from panels A, B and C to panel D. See summary statistics for A/B ratios per sample provided in the electronic Appendices 3, 4, and 5 for *G. menardii*, *G. limbata*, and *G. multicamerata*, respectively. Measurements from individual specimens for *G. menardii*, *G. limbata*, and *G. multicamerata*, are provided electronically in files RawData_1, 2, and 3, respectively in the supplied archive RawData.zip.

Fig. 15. Contoured bivariate plane frequency plots of spiral height (δX) versus axial length (δy) of *G. menardii* in keel view from Hole 806C and from 7.96 Ma through Recent. Sample ages are indicated by the numeral in the upper left corner. Contour intervals (int, in specimens per grid-cell size of 50μm δX x 100μm δY) are indicated in the upper right corner. The base contour line in every diagram is at 0 specimens per grid-cell. Peak heights can be estimated by counting up int values per contour line from base to top. Example: The maximum number of specimens per grid-cell in the diagram at 3.36 Ma is 23. Therefore, with a contour interval of 5 the top contour line in that diagram is at 20 specimens per grid-cell). The value n indicates the number of *G. menardii* specimens included per sample. Grid-cell

size is $\Delta\delta X$ = 50μm, $\Delta\delta Y$ = 100μm. The oblique line from δX = 50 μm / δY = 0 μm to δX = 550 μm / δY = 1600μm separates *G. menardii menardii* (below line) from *G. menardii cultrata* (above line) and is the same as described in Knappertsbusch (2007, 2016a). The vertical line separating specimens <300 μm from larger forms helps to better interprete modal shifts from one sample to the next. Open squares indicate the sample mean for δX and δY, together with 95% confidence intervals about the mean (small, intermediate bars) and 95% prediction intervals for observation (large, thin bars). Closed squares indicate sample medians for δX and δY. Quartile ranges (25% to 75%) about the medians are indicated by thick bars. Statistical parameters shown in Fig. 15 are tabulated in Appendix 8. The diagram 0.00 - 7.96 Ma (at the end of the continuation of Fig. 15) represents the summary frequency diagram for Hole 806C including all *G. menardii* specimens (n = 5652, 6 specimens of *FlexGib* excluded) from Recent to 7.96 Ma. Taxa included are *G. menardii cultrata*, *G. menardii menardii*, undifferentiated and juvenile *G. menardii*, *G. menardii flexuosa*, and *G. menardii gibberula* („*G. menardii*-lin“, see section taxonomic notes).

Fig. 15, continuation. The sample at 2.63 Ma was used for calibration of VDDs.

Fig. 15, continuation.

Fig. 15, continuation.

Fig. 16. Contoured bivariate plane frequency diagrams of spiral height (δX) versus axial length (δy) of *G. limbata* in keel view from Hole 806C (black contour lines). Grey-shaded diagrams are those from Fig. 15 (*G. menardii*) for comparison. See captions to Fig. 15 for explanations. Statistical parameters illustrated in Fig. 16 are tabulated in Appendix 9. No statistical parameters were calculated for samples with n ≤ 2.

Fig. 16, continuation.

Fig. 16, continuation.

Fig. 16, continuation.

Fig. 17. Contoured bivariate plane frequency diagrams of spiral height (δX) versus axial length (δy) of *G. multicamerata* in keel view from Hole 806C (black contour lines). Grey-shaded diagrams are those from Fig. 15 for *G. menardii* for comparison. See captions to Fig. 15 for further explanations. Statistical parameters shown in Fig. 17 are summarized in Appendix 10. No statistical parameters are indicated for samples where n ≤ 2.

Fig. 17, continuation.

Fig. 17, continuation.

Fig. 18. Oblique orthographic views of VDDs for the group „*G. menardi-lin*“, *G. limbata* and *G. multicamerata* for spiral height δX (red axis) versus axial length δY (black axis) from Hole 806C during the past 8 Ma. The vertical blue axis is normalized time. A. Iso-surface for *G. menardii* at an isovalue of 2.499 (this corresponds to 3 specimens per grid-cell). Horizontal contour slices from top to down are at 0 Ma [3], 1.95 Ma [3], 2.63 Ma [3], and 6.63 Ma [3]. The contour slice at 2.63 Ma (third from top, normalized time of -0.32875) represents the sample with the strongest size reduction at Hole 806C. The stippled 45° plane shows the frequency distribution of *G. menardii* through time in the central part of the VDD (zero offset from the stippled diagonal plane), also at contour intervals of 3 specimens per grid-cell. An interactive version of this diagram was prepared using software PDF3D ReportGen version 2.17.4 from VisualTechnology Services Ltd and can be inspected in the electronic Appendix 11.

B. Iso-surface for *G. limbata* at an isovalue of 1.452 (corresponding to 3 specimens per grid-cell. Horizontal contour slices from top to down are set at 2.19 Ma, 3.18 Ma, 4.35 Ma, 6.63 Ma, and 7.26 Ma, again each with contour intervals of 3 specimens per grid-cell.

C. Iso-surface for *G. multicamerata* at an isovalue of 1.105, which corresponds to 3 specimens per grid-cell as above. The shown horizontal contour slice is at 4.35 Ma (contour interval of 3 specimens per grid-cell). Contour-slices from the other investigated levels mostly disappear at 3 specimens per grid-cell because of the scarcity of *G. multicamerata*.

D. All three VDDs nested into each other. The VDD for *G. menardii* (cyan, only back half is shown) is opened in order to exhibit the vertical contoured section of densities in the stippled diagonal plane at 45°. Move to Fig. 19 for an enlarged view normal to the 45° plane at offset=0. VDDs for *G. limbata* (back half) and *G. multicamerata* (front and back) shown in orange and red, respectively. The projected line diagrams are the means of δX versus δY per sample for *G. menardii* (black), *G. limbata* (orange), and *G. multicamerata* (red), lines are thickeneded for the respective group in each diagram. Samples with ≤ 2 specimens were omitted for calculation of means and are not shown in these graphs. Compare oblique line diagrams from Fig. 18 with plane diagrams of δX versus age and δY versus age in Fig. 7 D and Fig. 8 D, respectively. The group „*G. menardii*-lin“ includes *G. menardii cultrata*, *G. menardii menardii*, undifferentiated and juvenile *G. menardii*, *G. menardii flexuosa*, and *G. menardii gibberula* (see section taxonomic notes).

Fig. 19. Enlarged view of Fig. 18D showing the 45° plane at diagonal offset=0 (stippled) containing a vertical contour section through the VDD for *G. menardii* seen in normal direction to the viewer. In this orientation axes of spiral height (δX) and axial length (δY) superimpose each other completely and lengths of axes are seen at 45° and -45° angles, respectively. The vertical axis is time in Ma (back-translated from the normalized time Z). The epochs and ages on the right side are indicated following the Geological Time Scale v. 5.0 of Walker et al., 2018). Blue contour intervals (blue numerals next to

contour lines) indicate 3 specimens per grid-cell of size $\Delta\delta X$=100μm x $\Delta\delta Y$=50μm. The lowermost contour line shown is at 3 specimens per grid-cell. Superimposed are the VDDs for *G. limbata* and *G. multicamerata*. Horizontal grey lines mark some selected sample levels. The grey line at 2.63 Ma indicates the sample used for calibration of the VDD and contour representations for *G. menardii*. Grey arrows indicate interpreted evolutionary trend. Inside VDD: Line diagrams for mean δX versus δY in *G. menardii* (black squares), *G. limbata* (orange circles) and *G. multicamerata* (red triangles) show maximal divergence in this perspective (samples with ≤ 2 specimens omitted, compare with Figs. 7D and 8D). The morphological separation of the *G. limbata-multicamerata* branch from ancestral *G. menardii* substanciated around 4.35 Ma (Z=-0.54), but became visible already at 5.78 (Z=-7.2) Ma in *G. limbata*. The $\delta^{18}O$ isotope record of *Globigerinoides sacculifer* (black curve) is from Hole 806B and was taken from Berger et al. (1993) [0-2.115 Ma] and Jansen et al. (1993) [2.1 Ma-5.4 Ma, using age model 806B fnrdIM loc95, see Table 2]. The $\Delta\delta^{18}O_{subsurface-surface}$ curve (in orange) reflects the thickness of the mixed layer at Site 806 and is redrawn from Fig. 1c in La Rivière et al. (2012). Small italic numerals and letters are selected oxygen isotope events following the nomenclature of Prell et al. (1986), obliquity cycles indicated in Berger et al, 1993). 100 indicates cold spell of MIS 100 (see Fig. 1 in Jansen et al. 1993). MPW marks the Mid Pliocene Warmth Period between 3.05 Ma-3.3 Ma, MPR the Mid Pleistocene Revolution. Blue boxes show intervals of absence of *G. menardii* at DSDP Site 502 and ODP Site 925B (see Knappertsbusch, 2016a). The double arrow above the blue boxes indicate entry of large *G. menardii* at these western Atlantic/Caribbean locations. The red box indicates the occurrence of extra large *G. menardii* at Hole 806C from 2.19 Ma to 1.95 Ma (see also plane contour diagrams in Fig. 15). On the left side indicated are trends of coiling direction and relative abundances of menardiform and tumid globorotalids. Labels in abundance diagram: 1 (black), „*G. menardii*-lin“ including *G. menardii cultrata*, *G. menardii menardii*, undifferentiated and juvenile *G. menardii*, *G. menardii flexuosa*, and *G. menardii gibberula*; and flexed forms of *G. menardii gibberula*. 2 (grey), transitional forms between *G. menardii* and *G. tumida*; 3 (hatched), *G. limbata*; 4 (hatched), *G. multicamerata*; 5 (white), „*Tumida*-group“ including formal taxa like *G. tumida tumida*, *G. tumida flexuosa*, *G. plesiotumida*, *G. merotumida*, and an undescribed new tumid form. 6 (stippled), others including *G. ungulata*, *G. pseudomiocenica* and *G. miocenica*, see section taxonomic notes. Census counts for calculation of relative abundances are given in Appendix 7.

Fig. 20. Comparison of VDDs (all vertical sections at 45°, at zero offset from the stippled diagonal plane) for *G. menardii* between tropical Atlantic sites and tropical Pacific sites. The vertical axis shows the normalized time Z (Z=- Age/8). A. DSDP Site 502 (Caribbean Sea, replotted from Knappertsbusch and Mary, 2012. Settings in Voxler: Isovalue=1.28, contour method = Min,Max,Count, min level=1, max level=10, number of levels=5). B. DSDP Site 503 (Eastern Equatorial Pacific, replotted from Knappertsbusch and Mary, 2012. Isovalue=1.28, contour method = Min,Max,Count, min level=1, max level=10, number of levels=5). C. ODP Site 925B (Céara Rise, western Atlantic. Data replotted from Knappertsbusch, 2016a. Isovalue=0.4665, Contour method = Min,Max,Count, min level=0.5, max level=10, number of levels=5). D. ODP Site 806C (Ontong-Java Plateau, this sudy. Isovalue=2.499 [calibrated], Contour method = Min, Max, Count, min level=0, max level=26, level interval=2.268 corresponding to 3 specimens per grid-cell, see text). Note that VDDs from DSDP Sites 502 and 503,

and from ODP Hole 925B were generated on normalized (F_{sn}) local frequencies, while the one from Hole 806C is a calibrated VDD derived from absolute (F_s) local frequencies (see section 2.4.8).

Figure_01.jpg

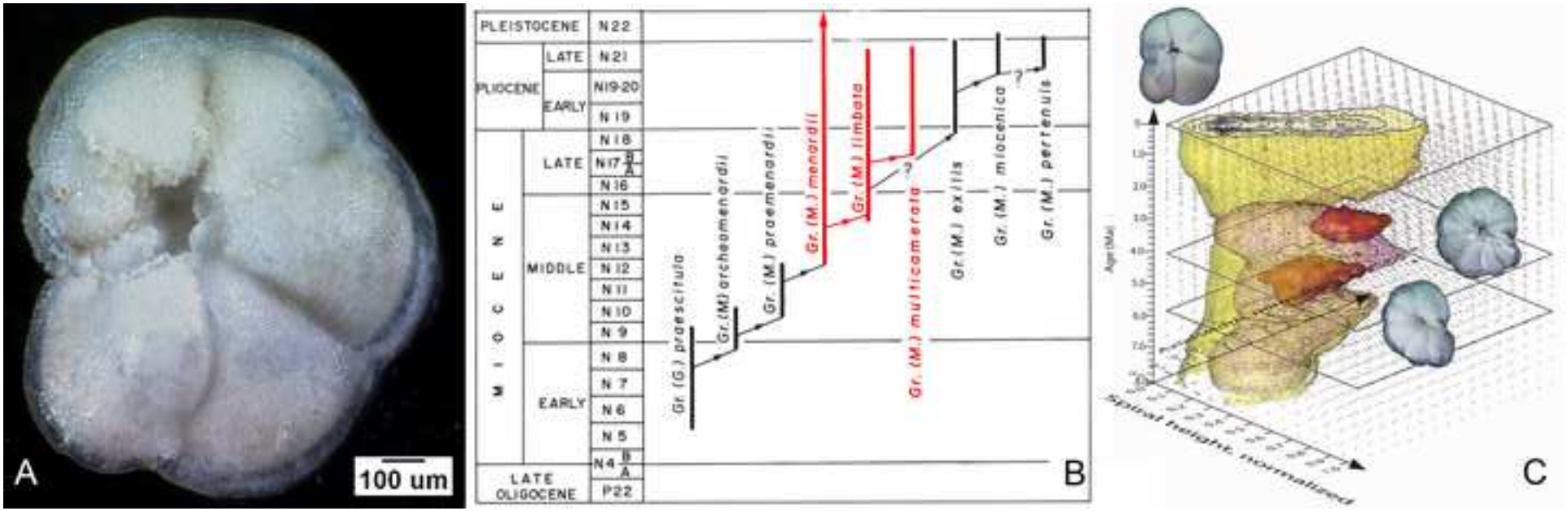
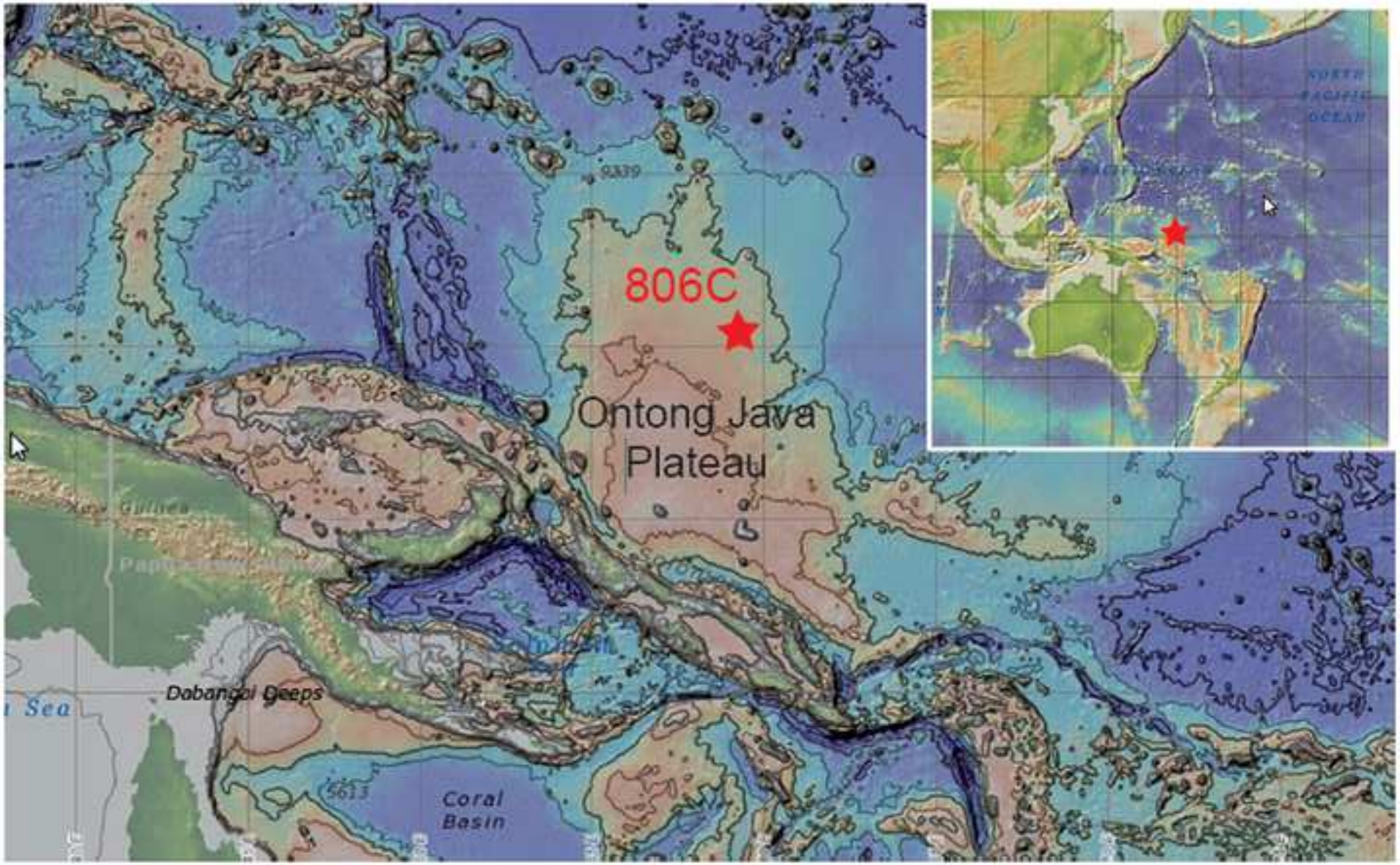


Figure 02, jpg



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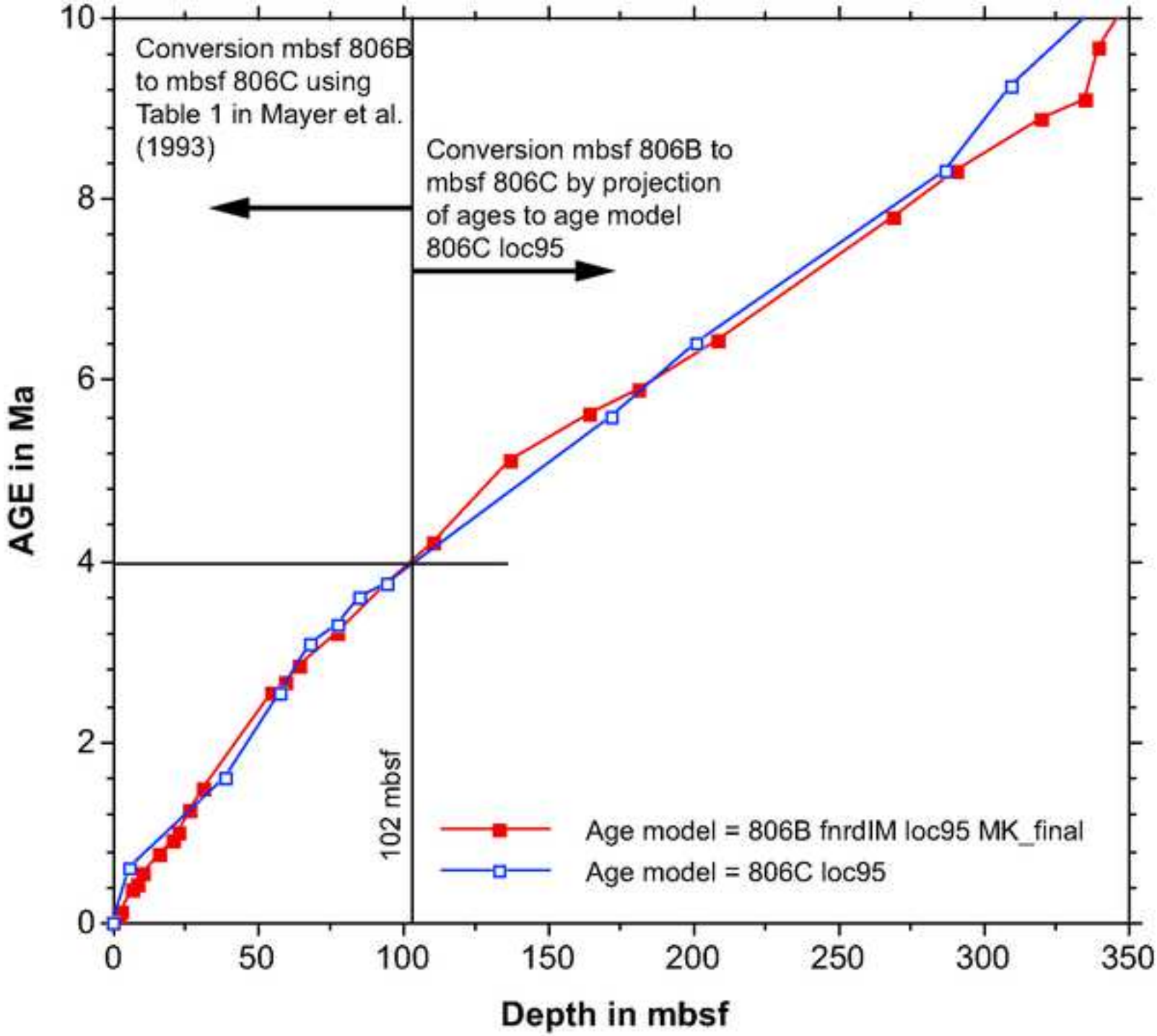
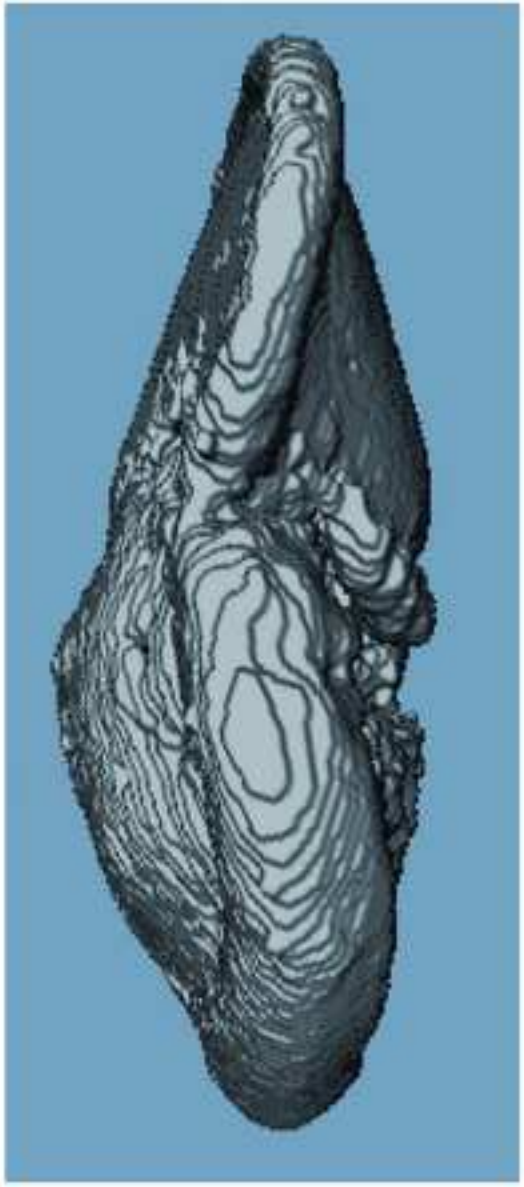


Figure 04, jpg

Figure 05, jpg



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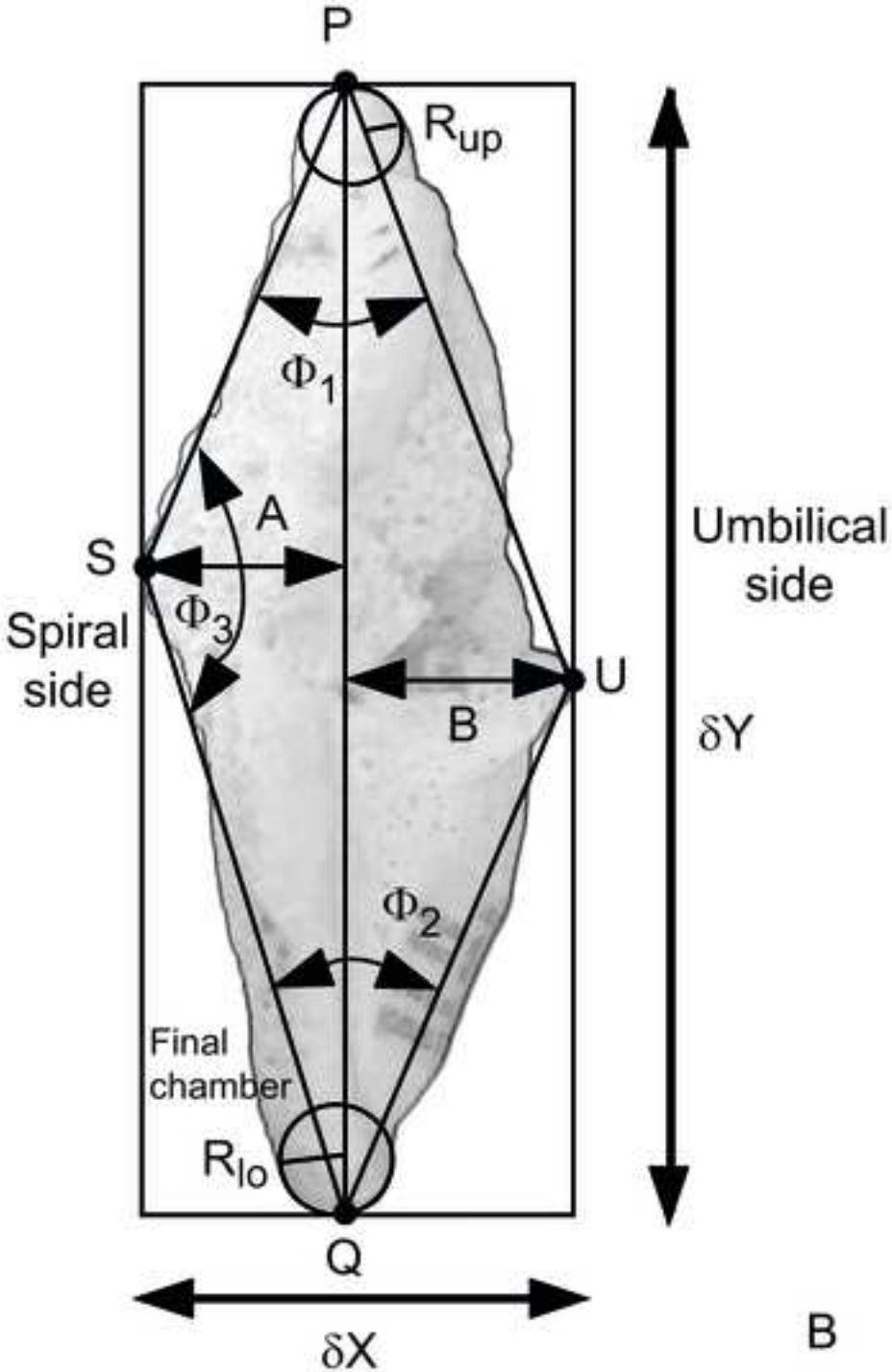

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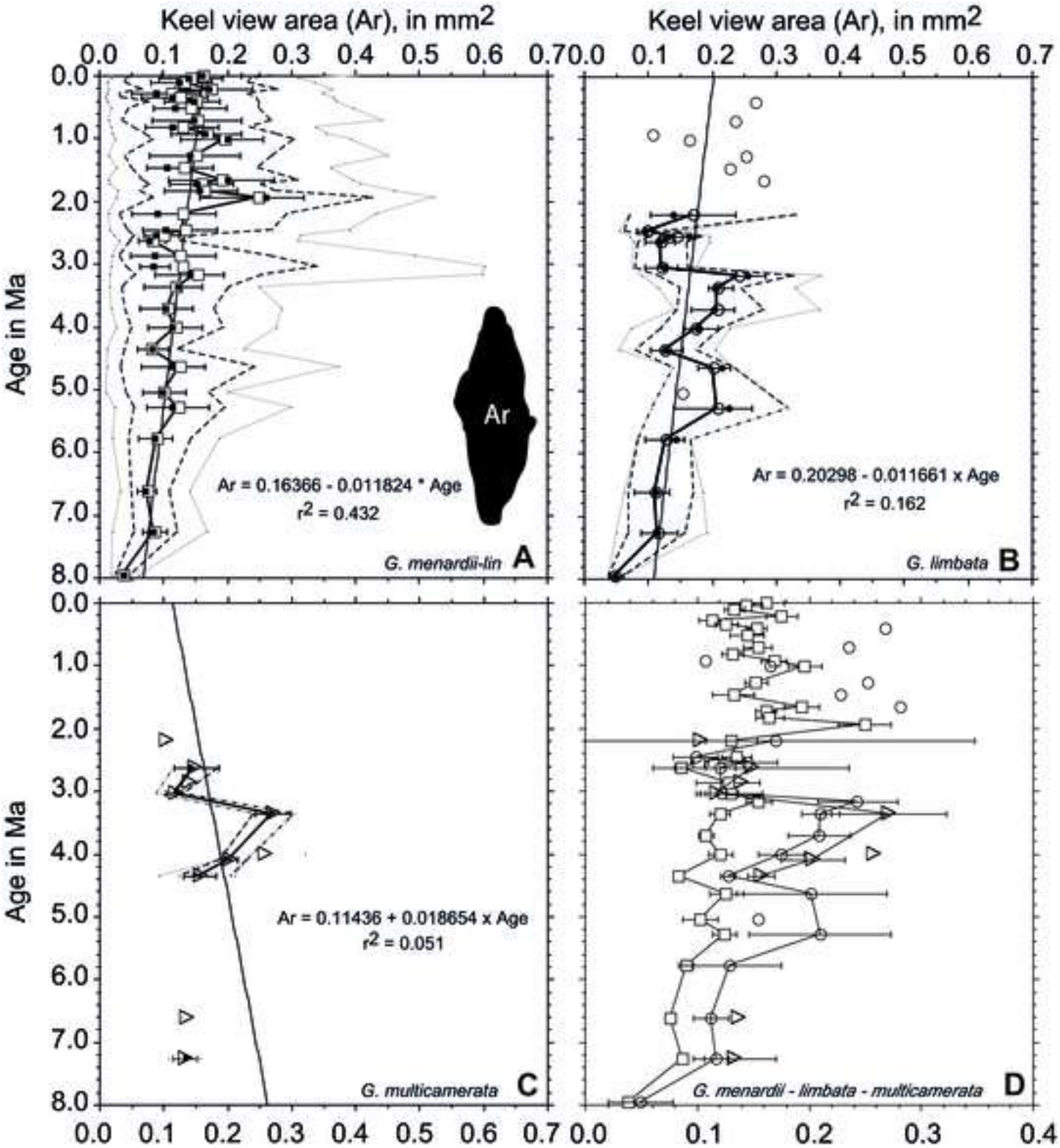

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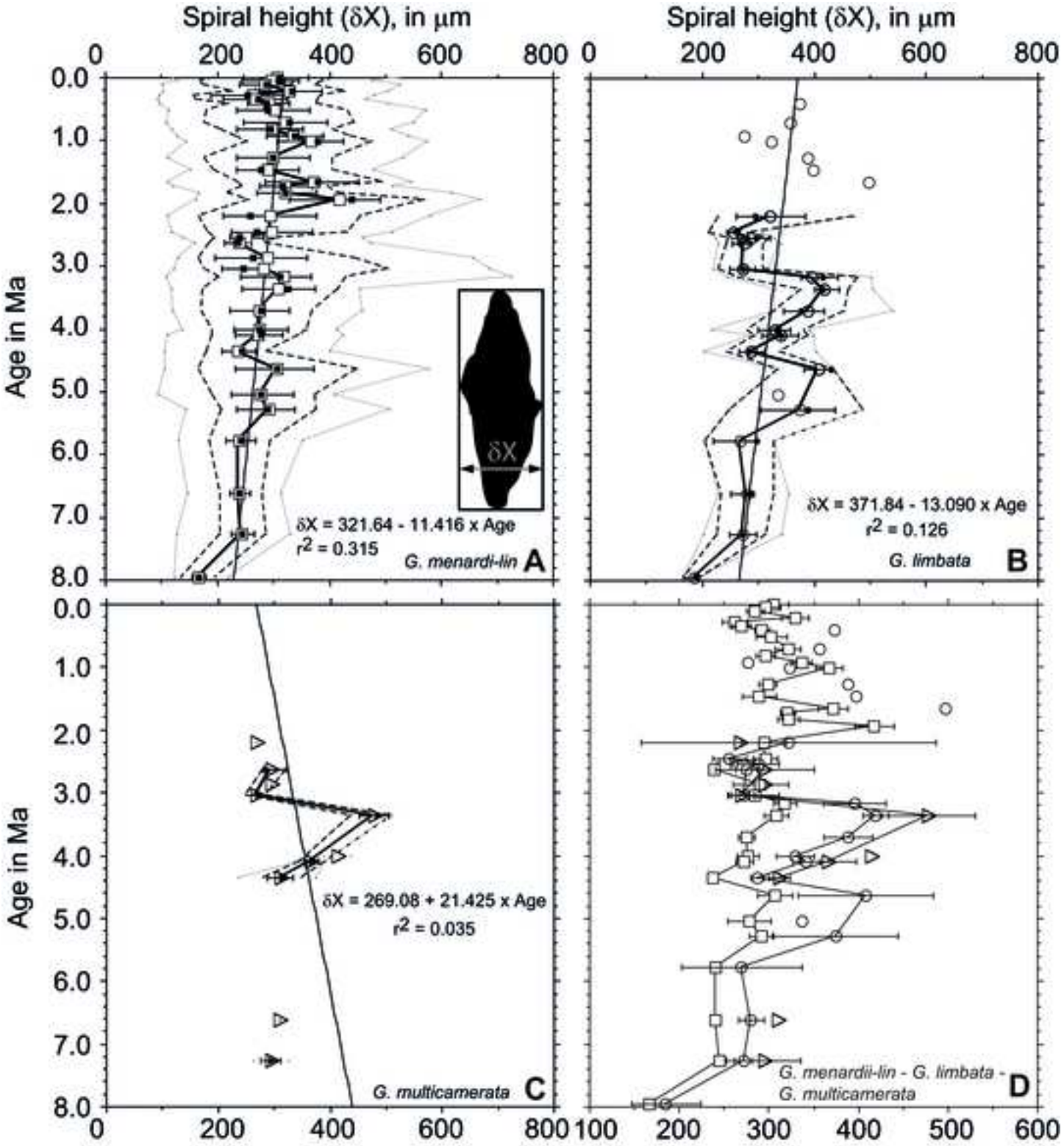

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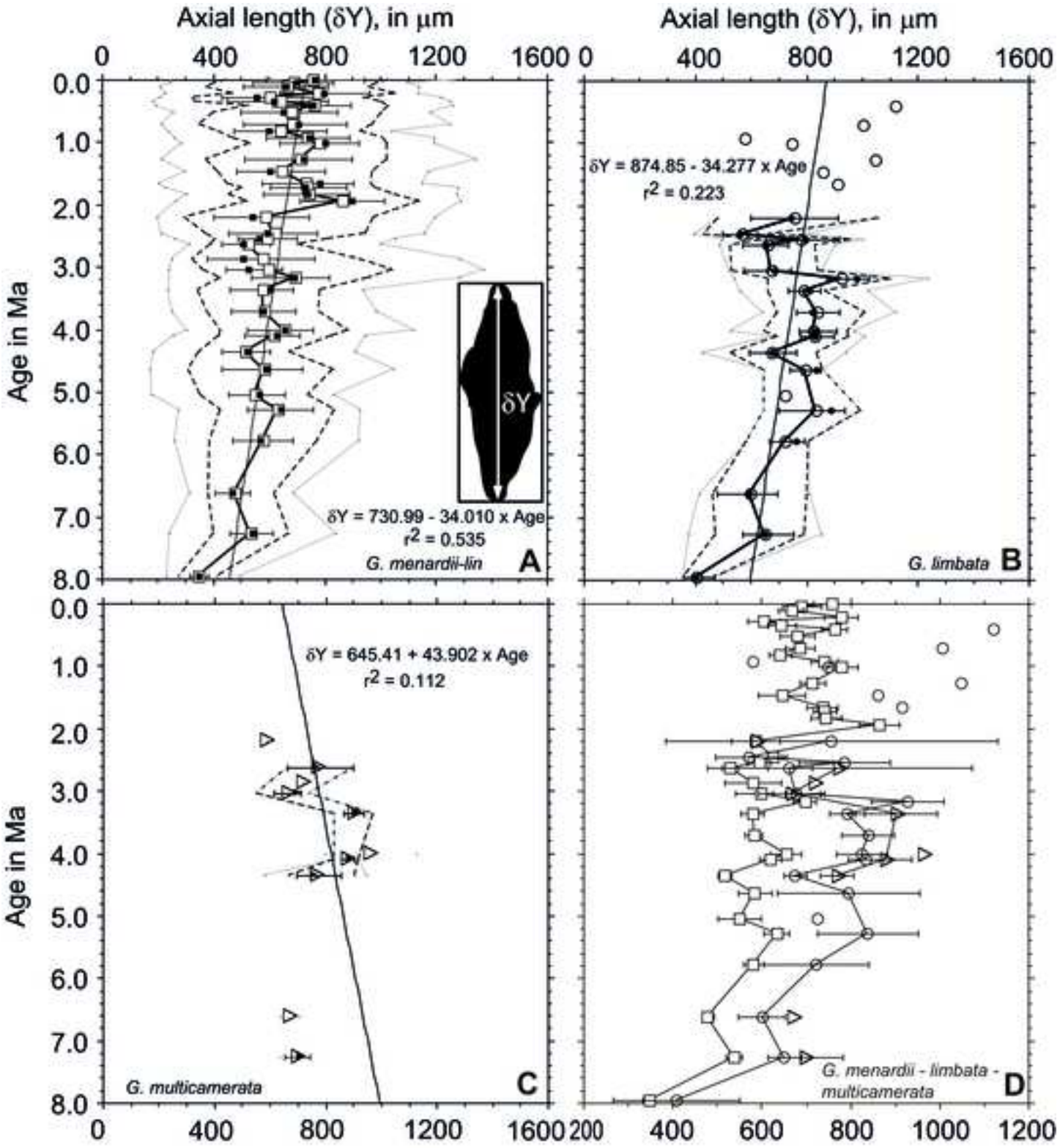

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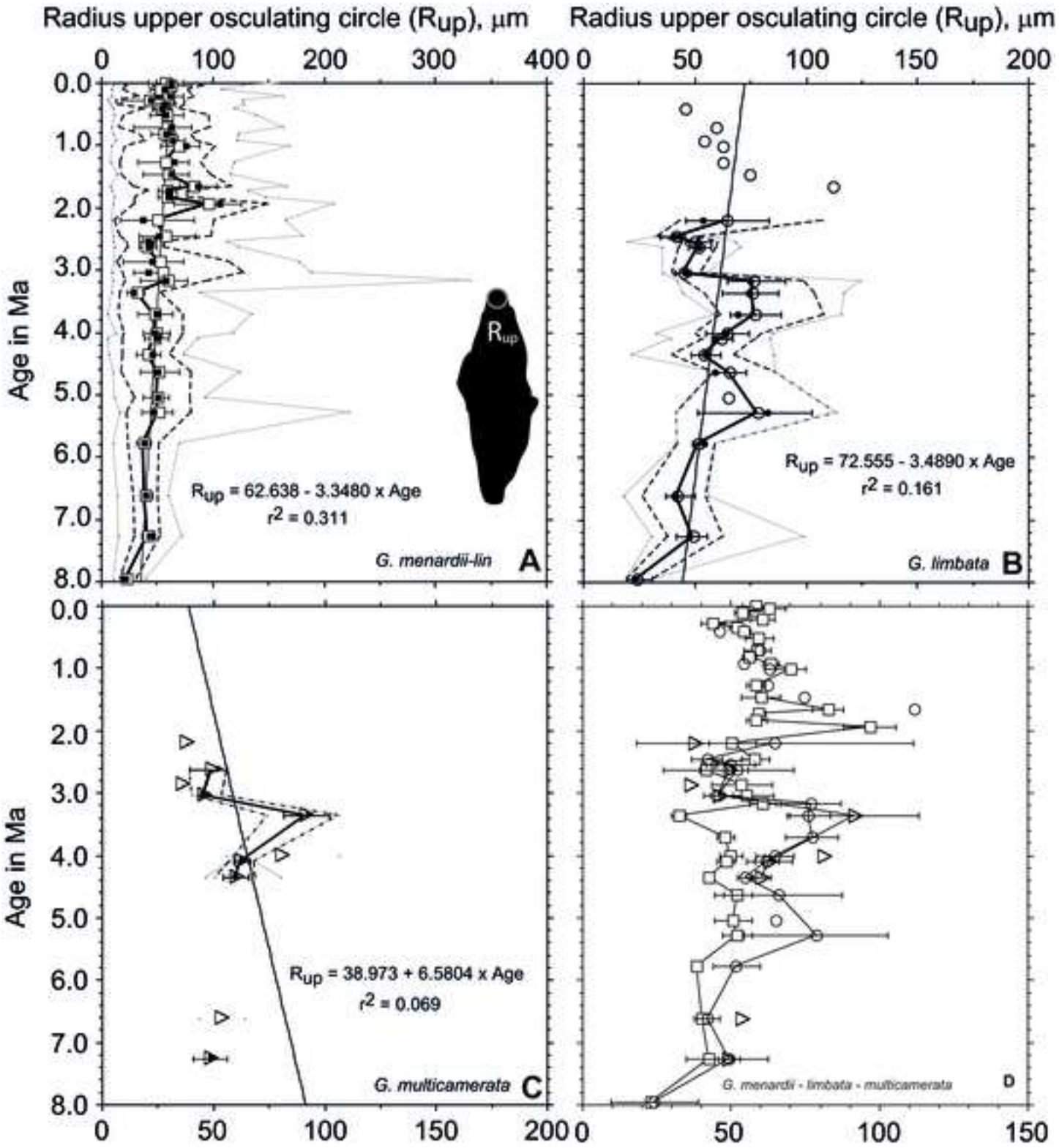

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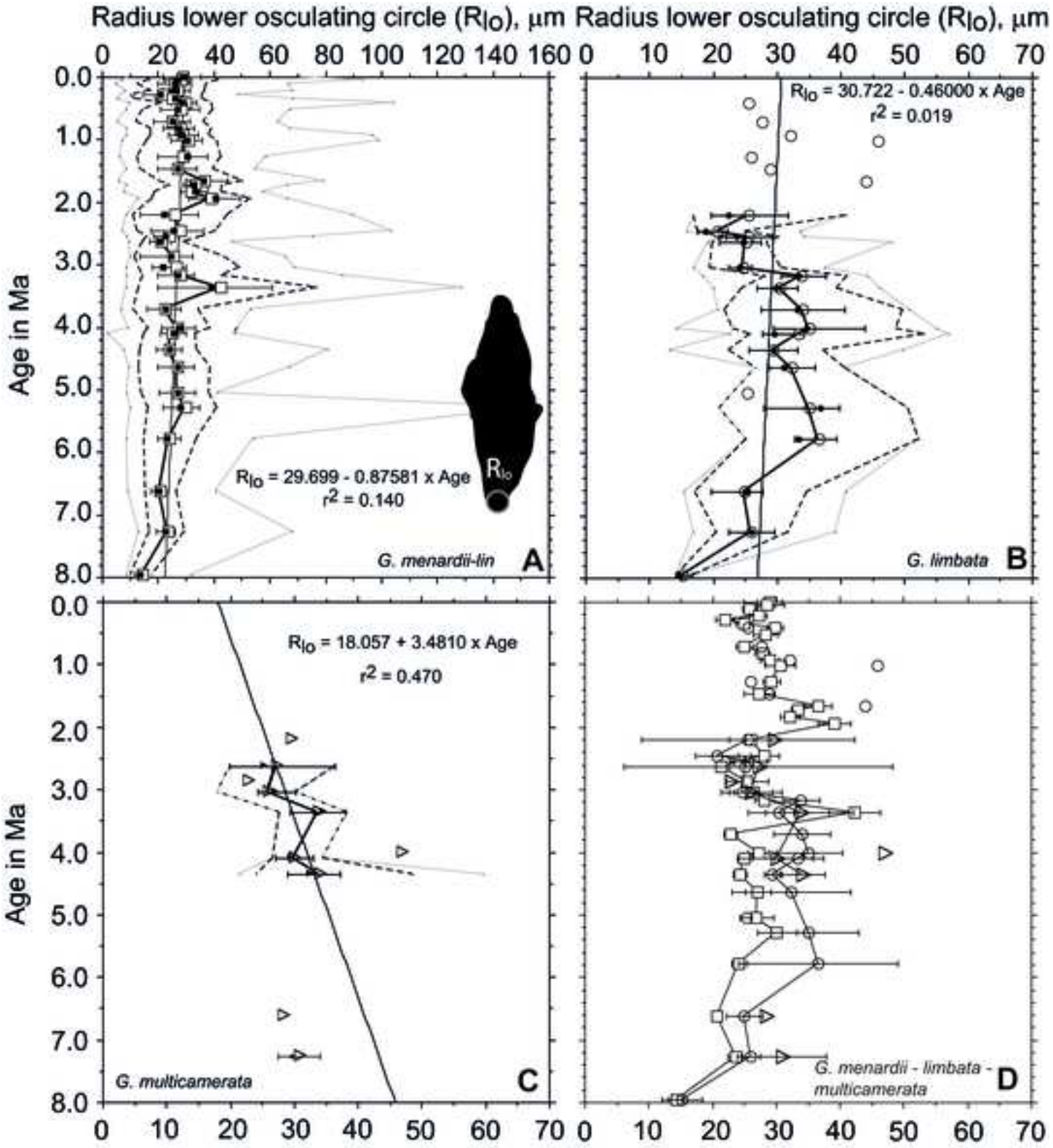

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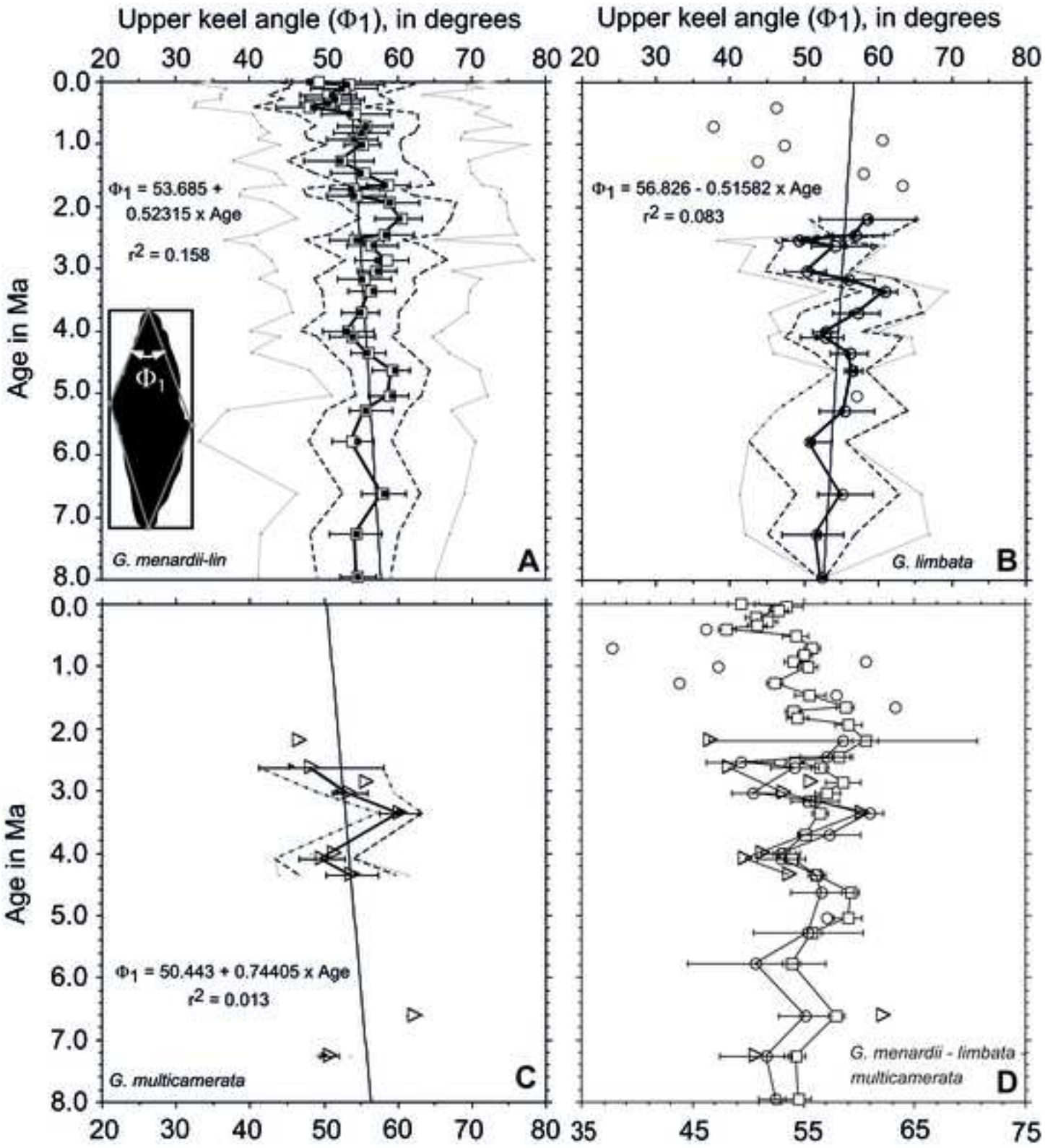

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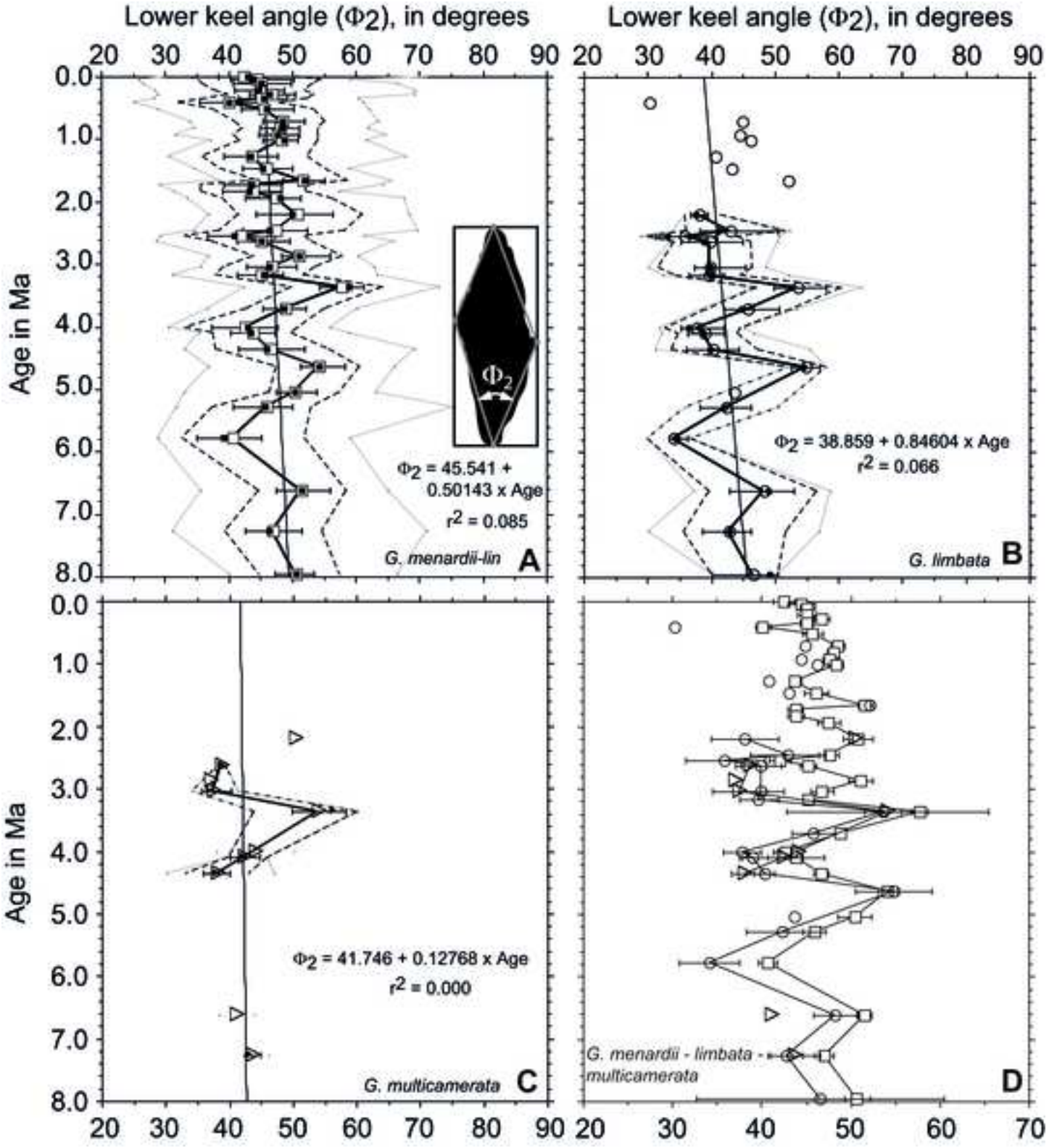

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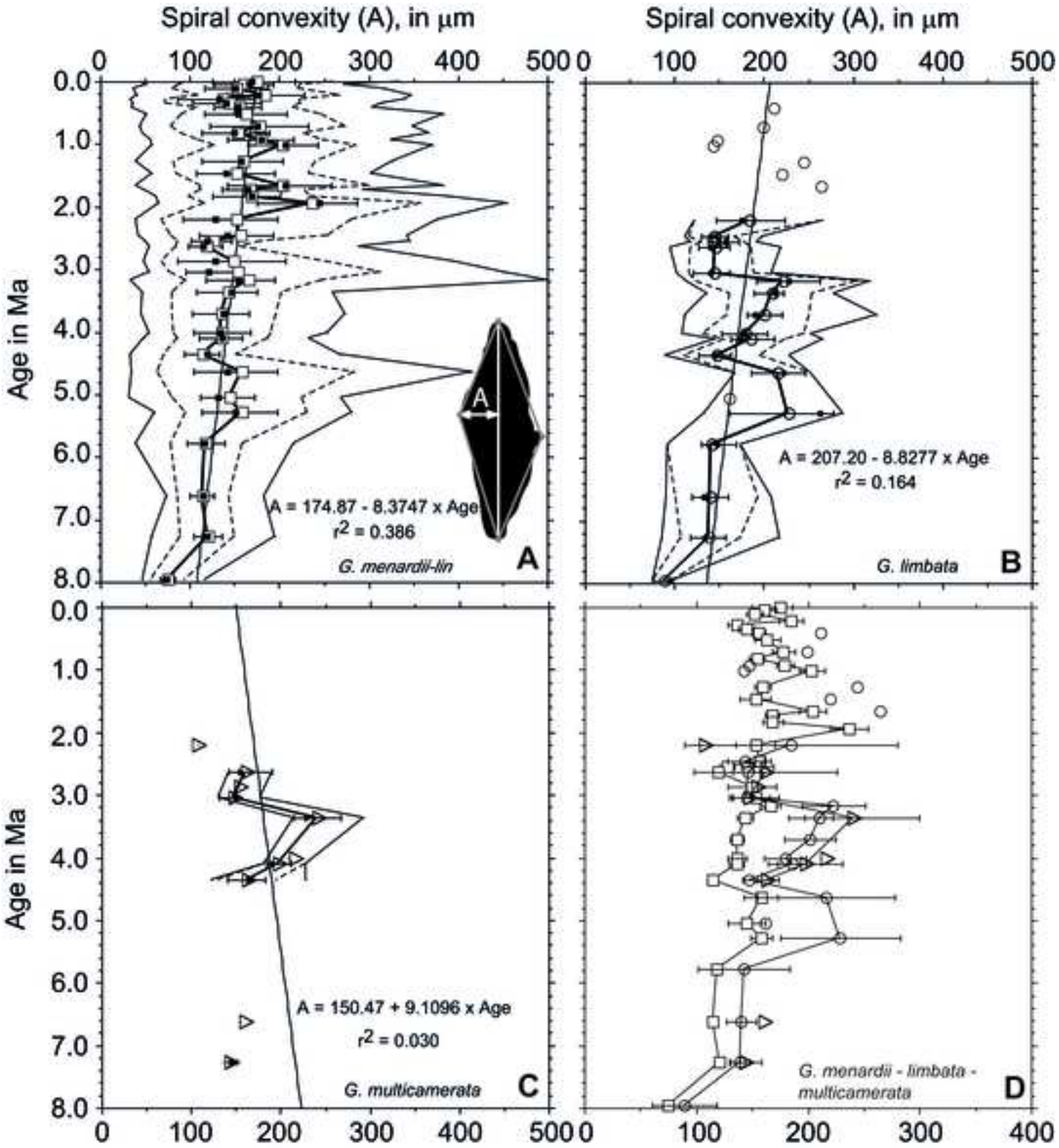

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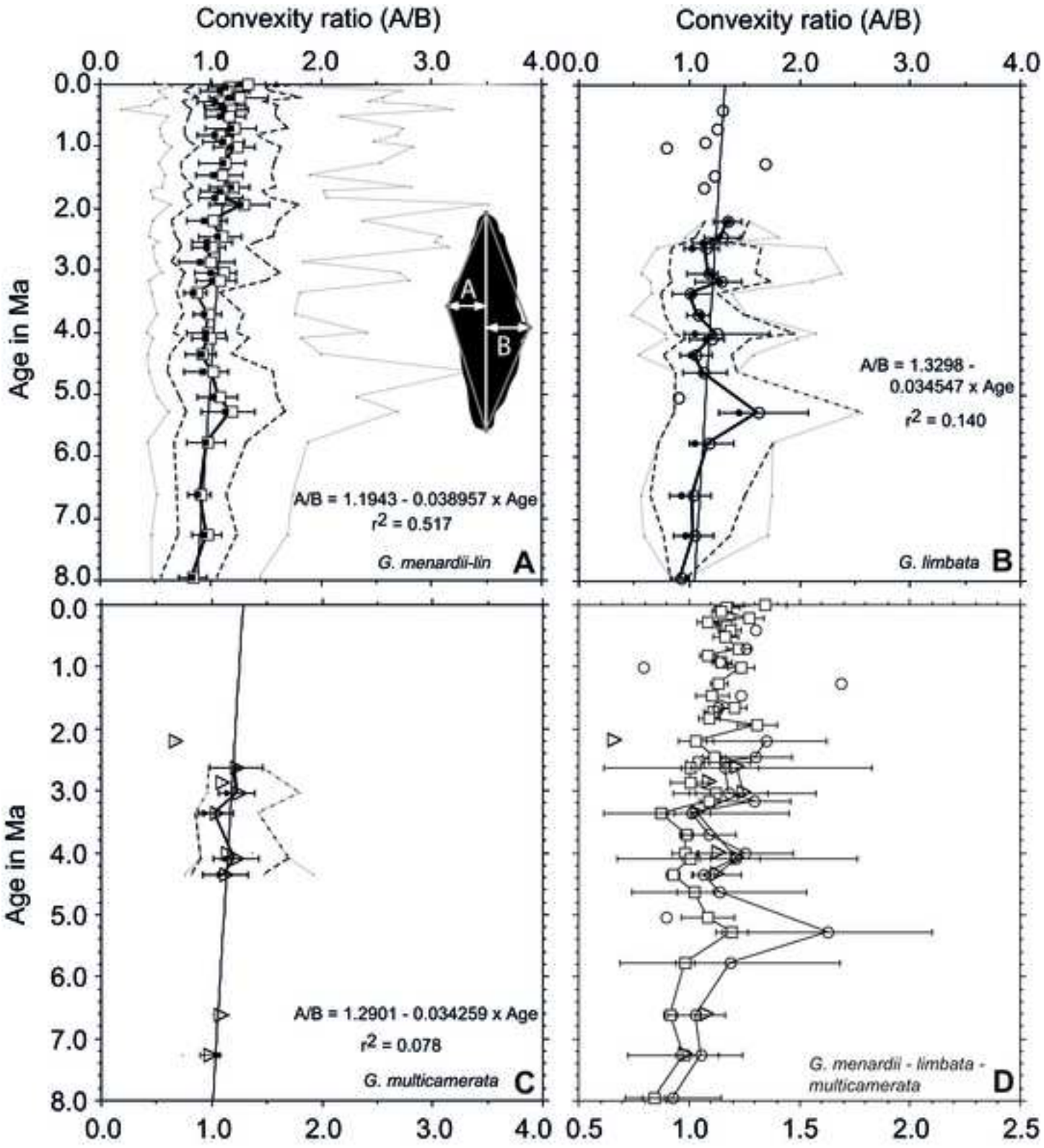

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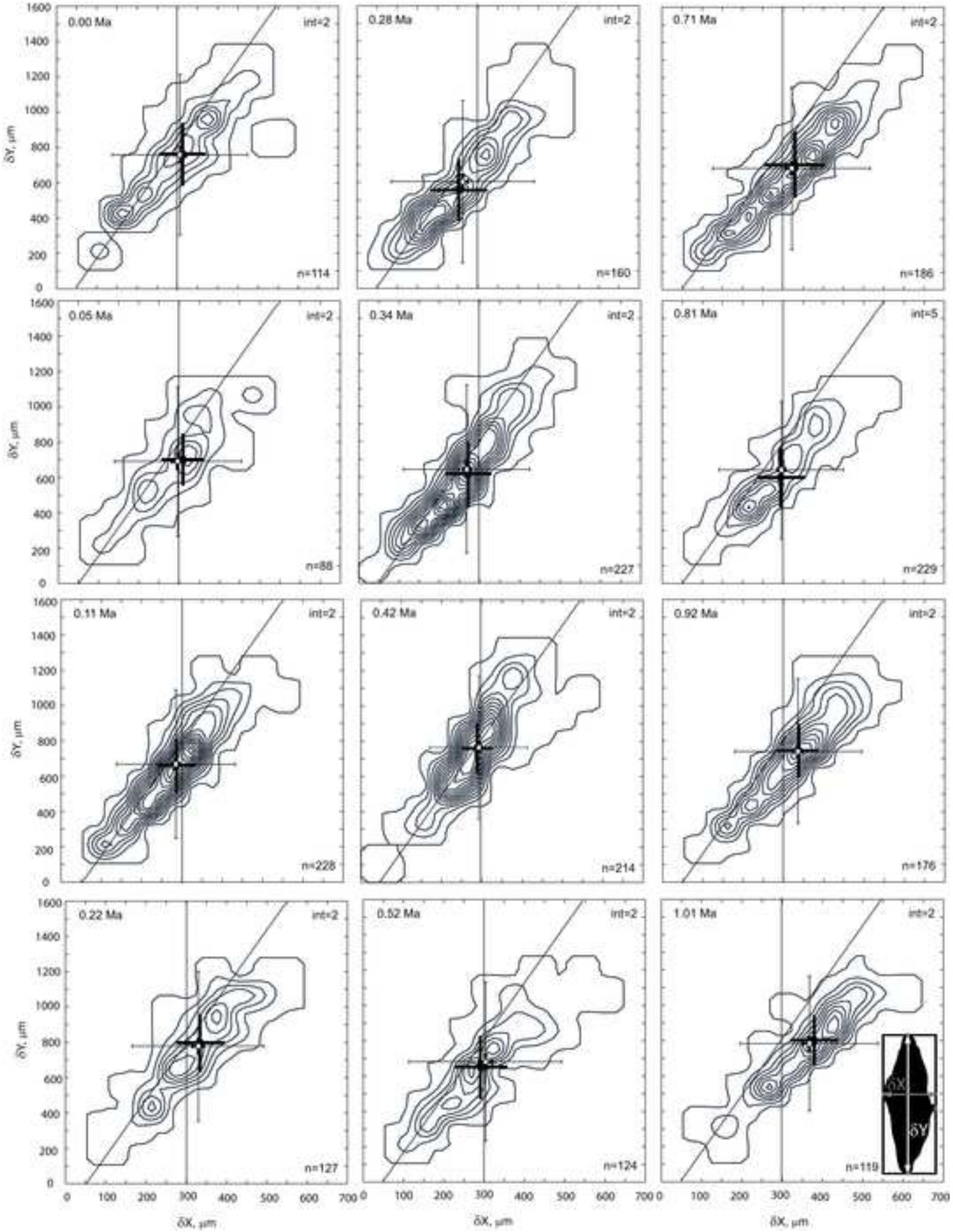

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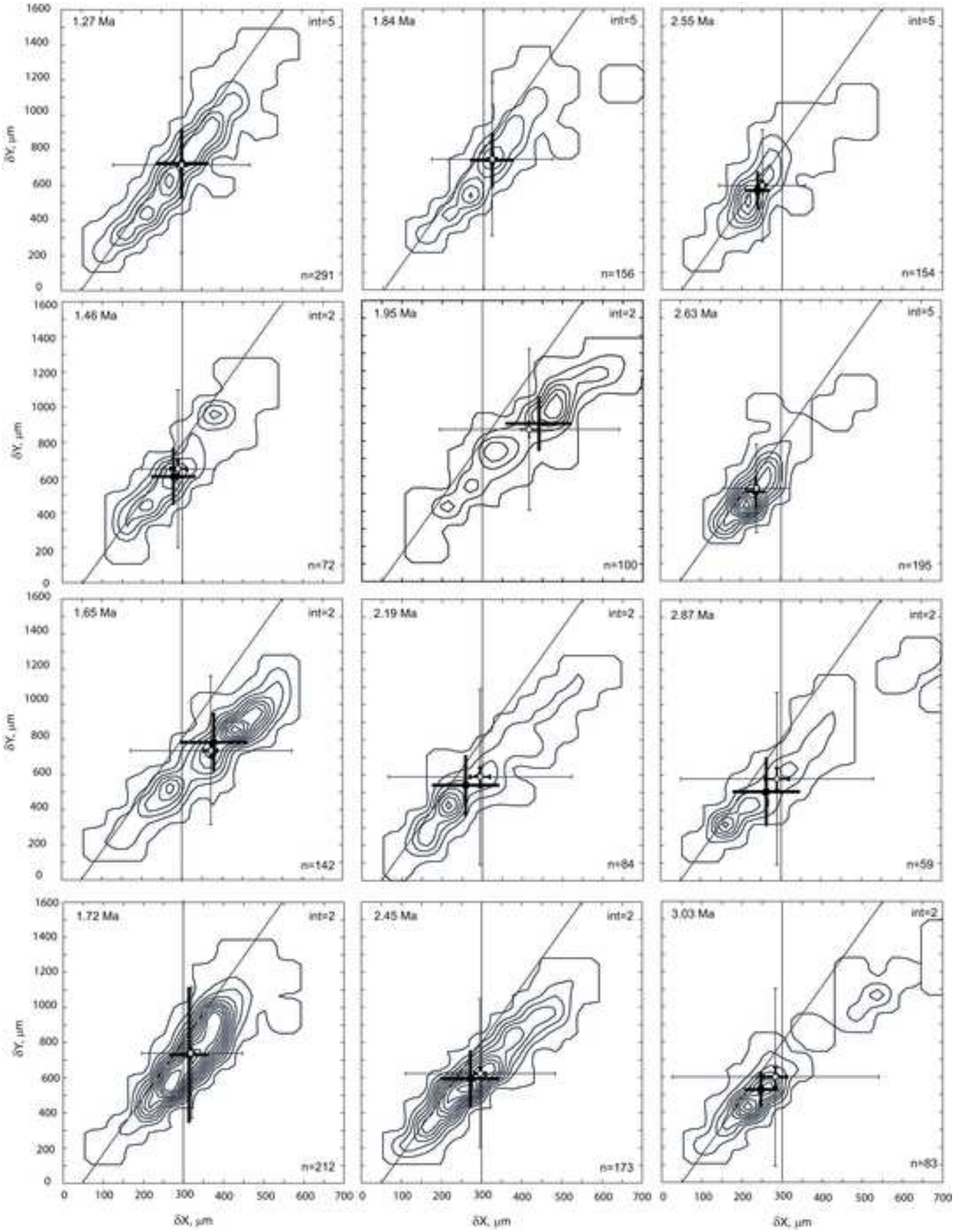

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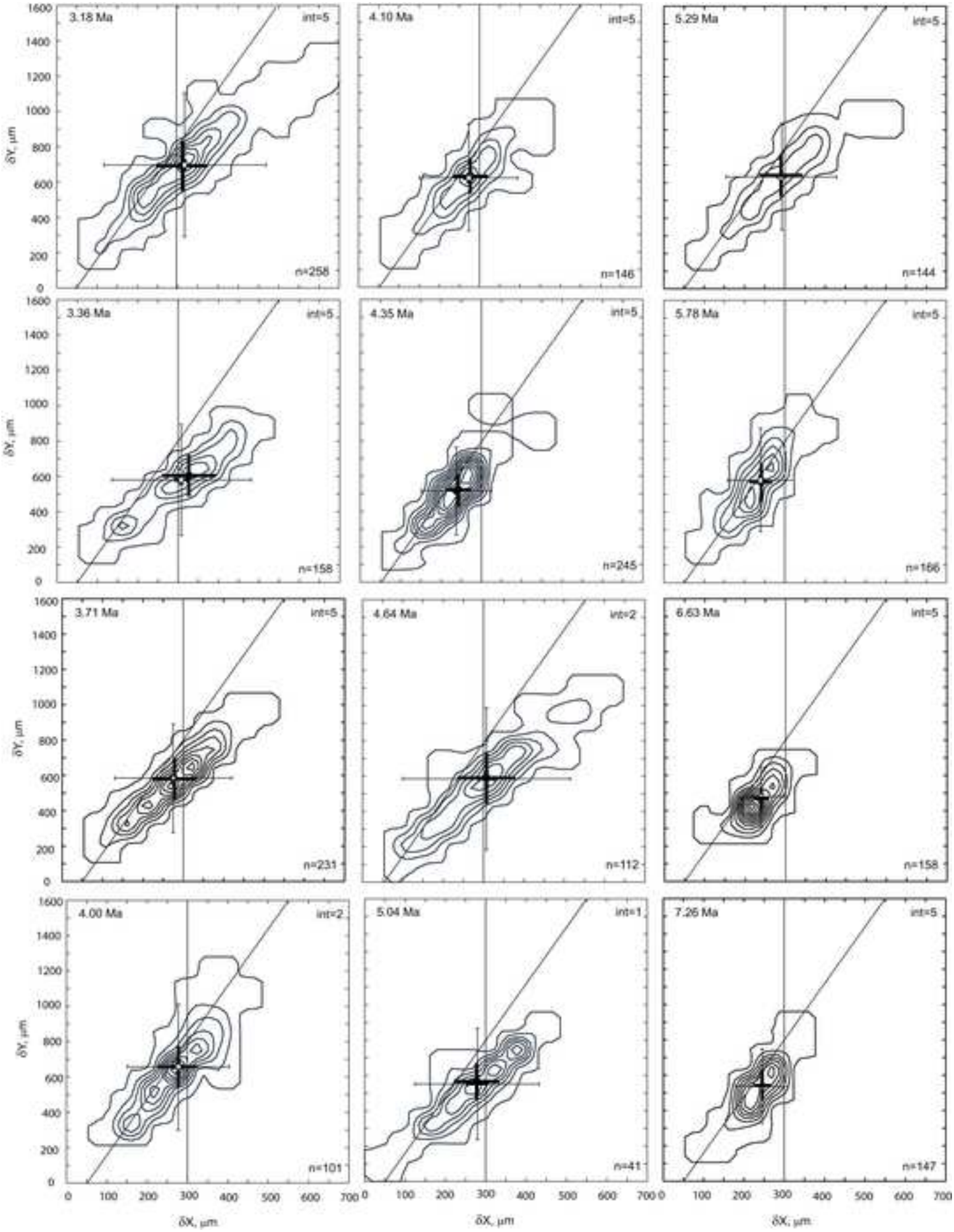

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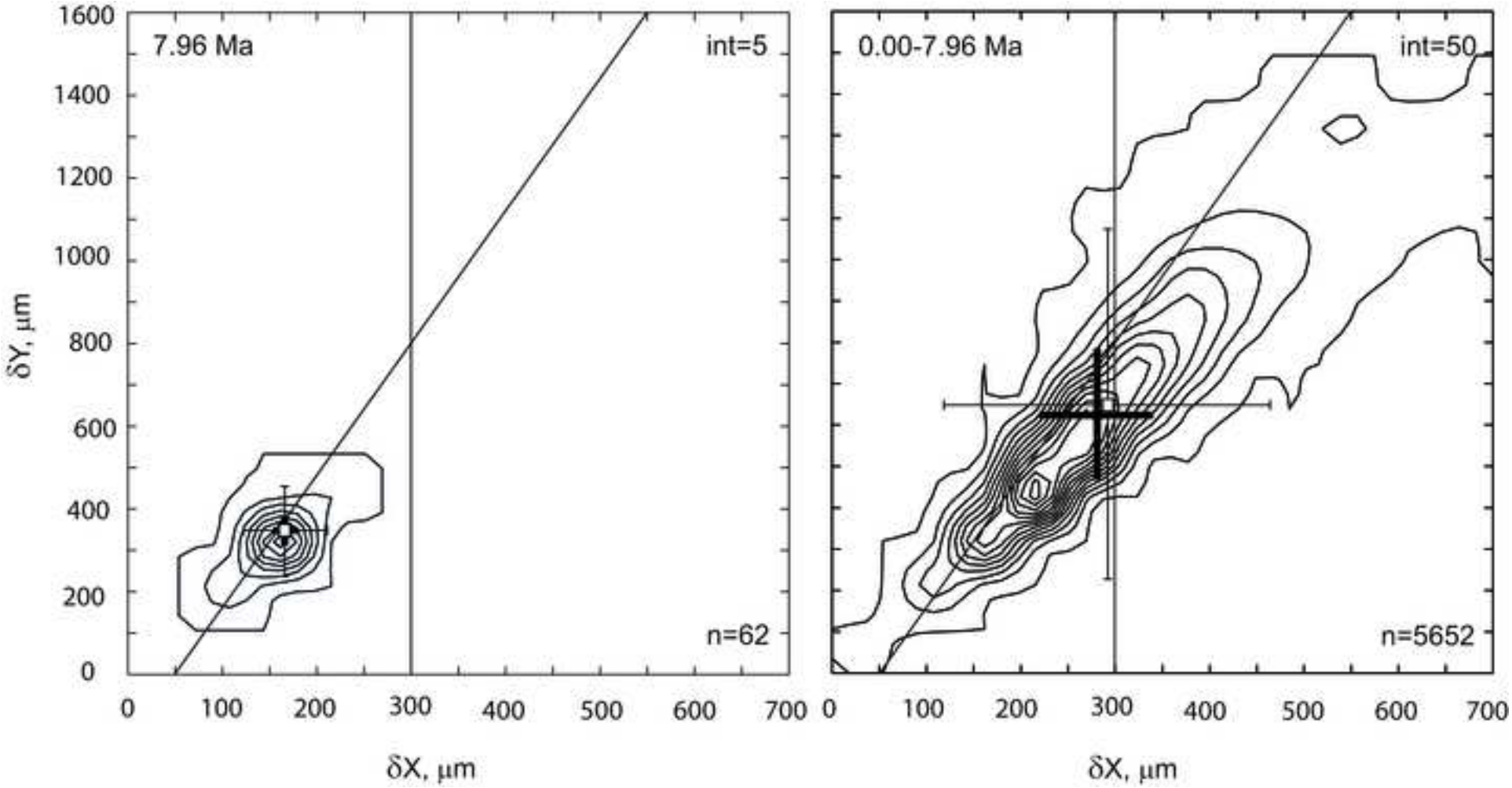

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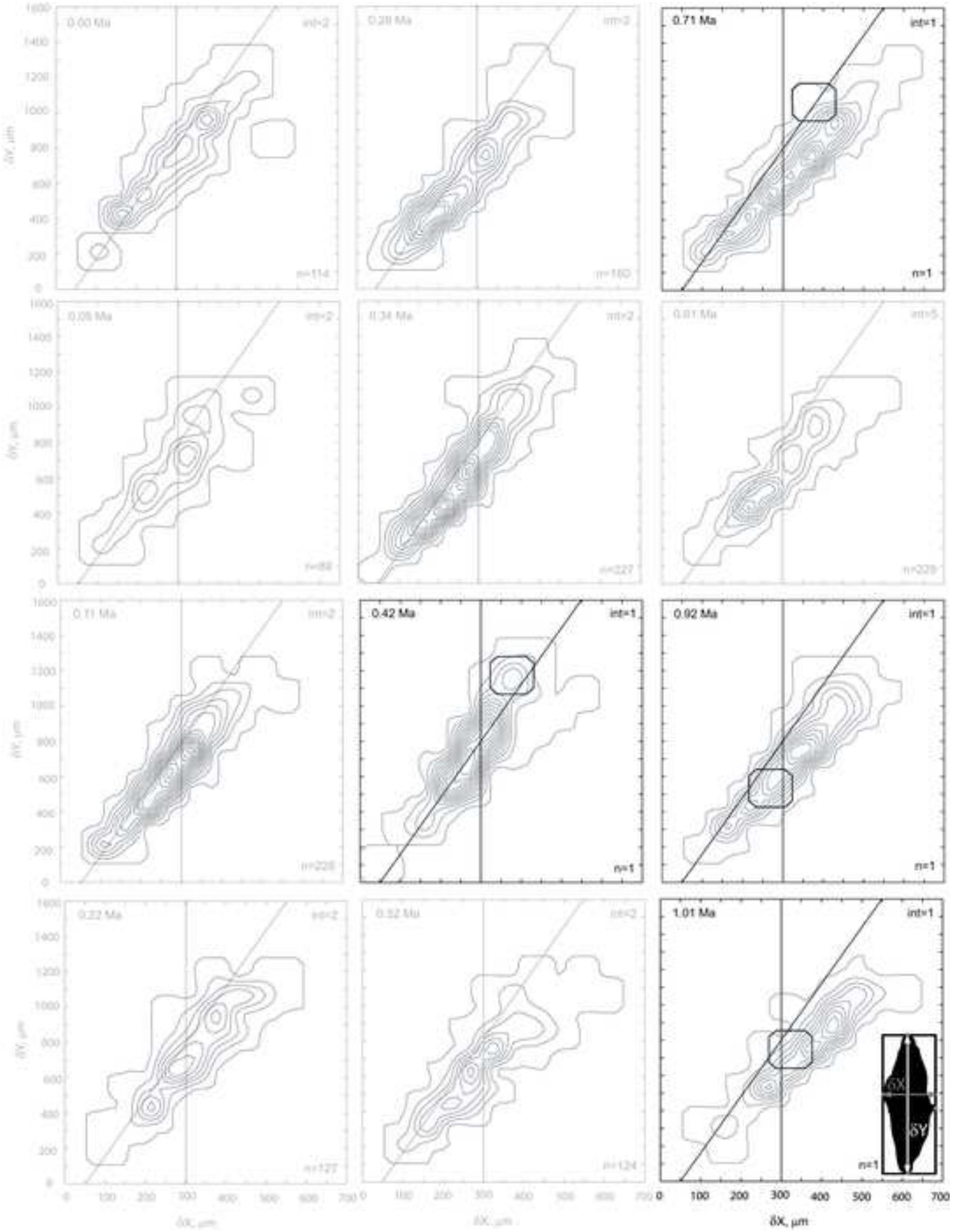

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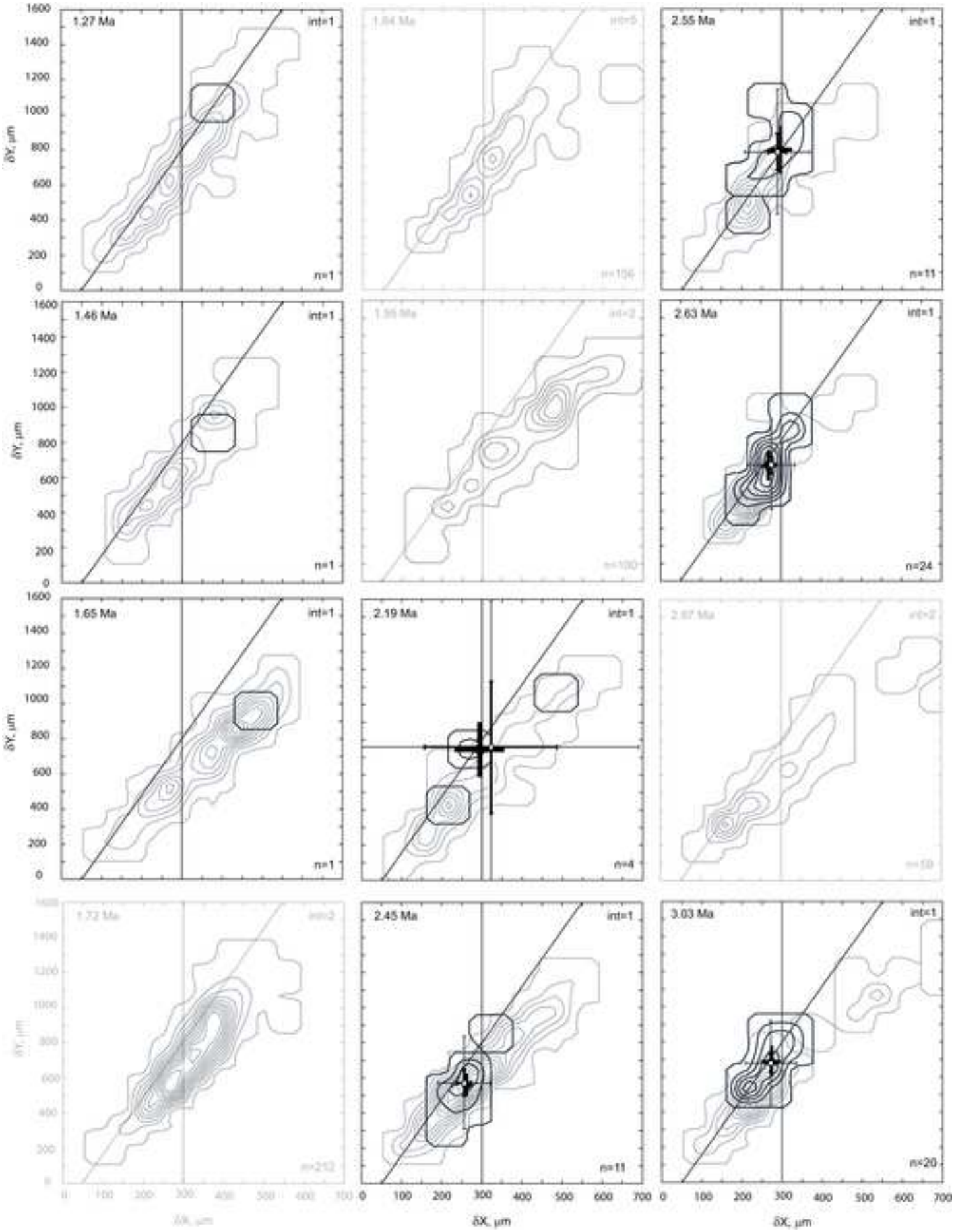

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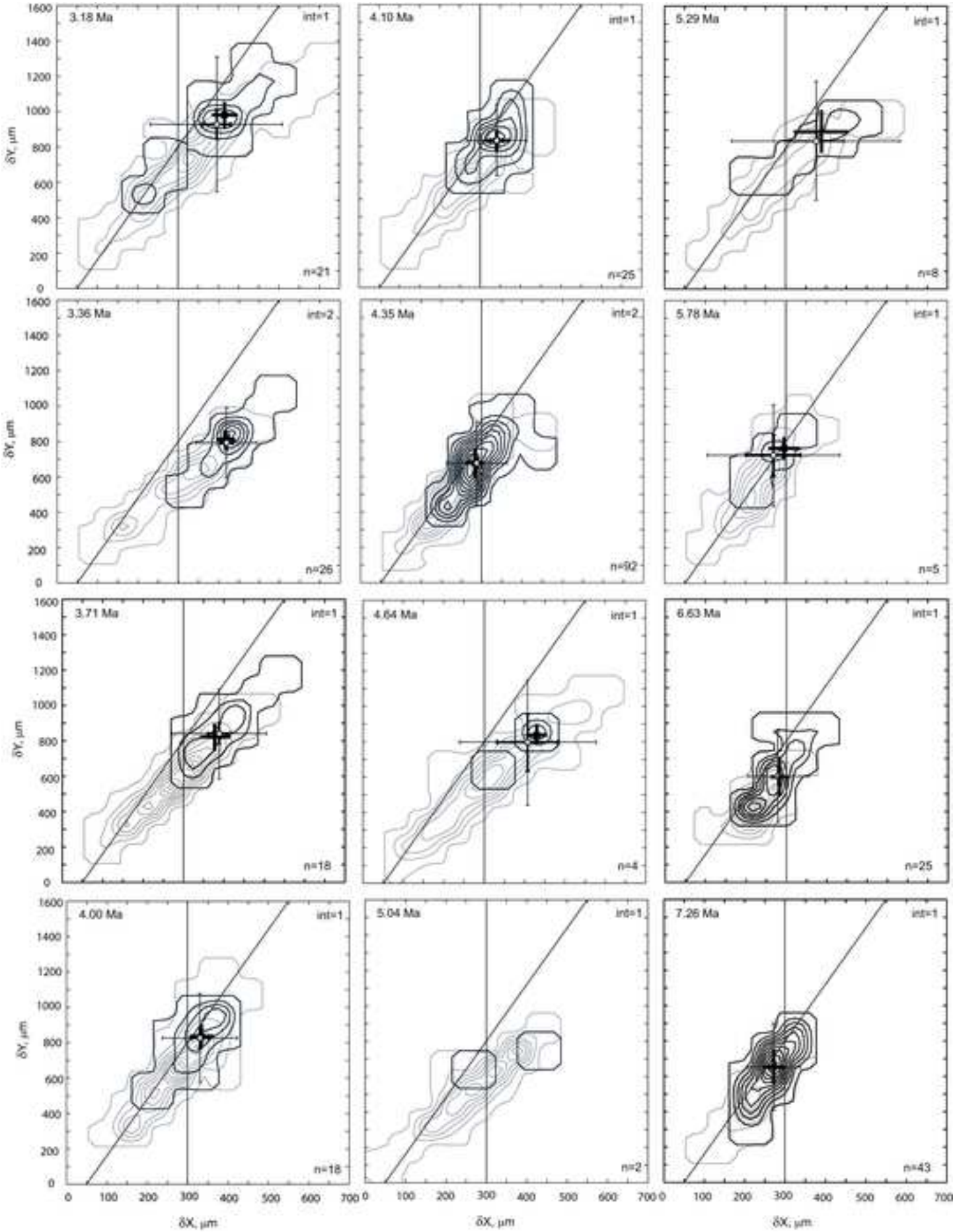

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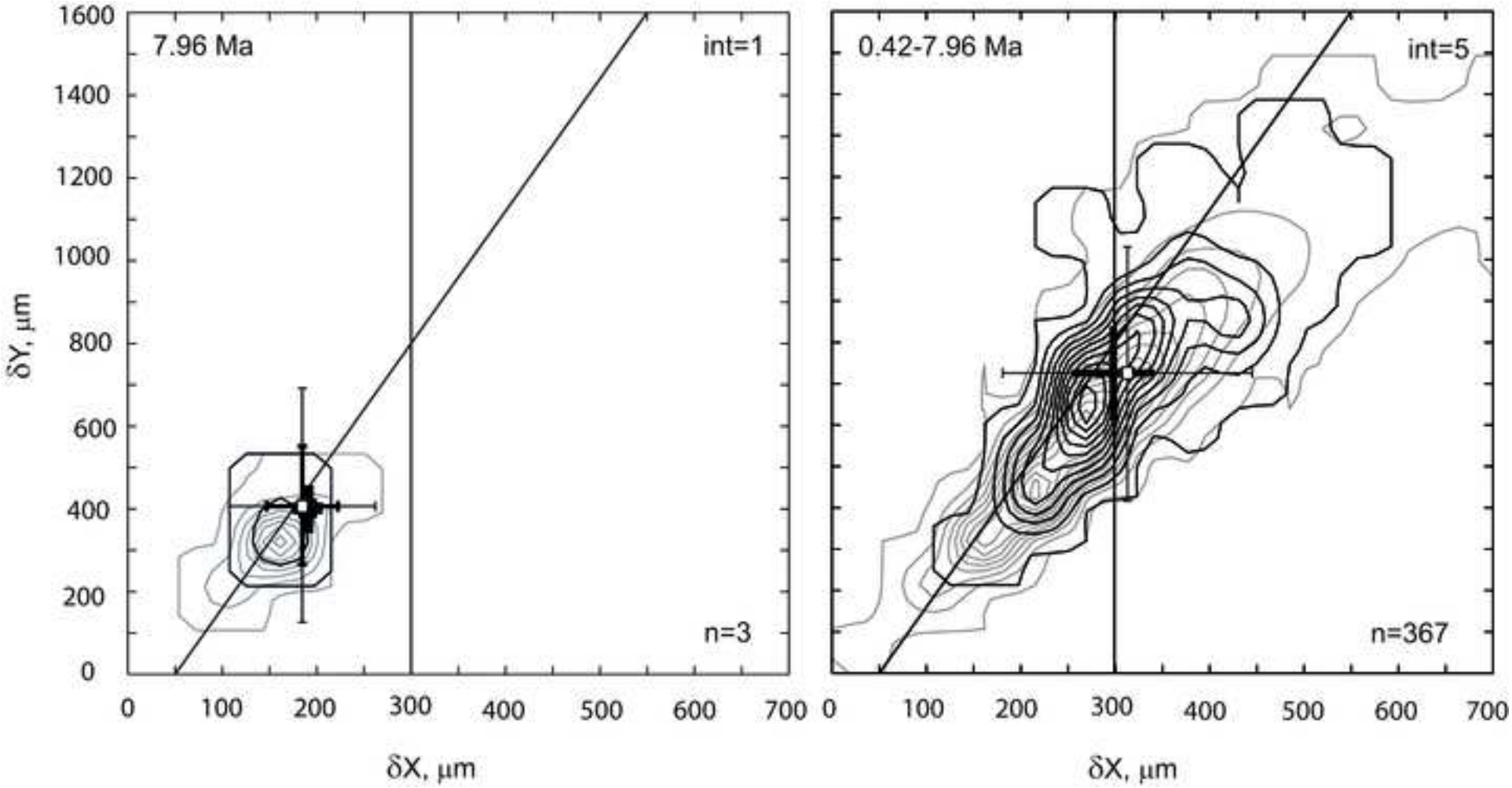

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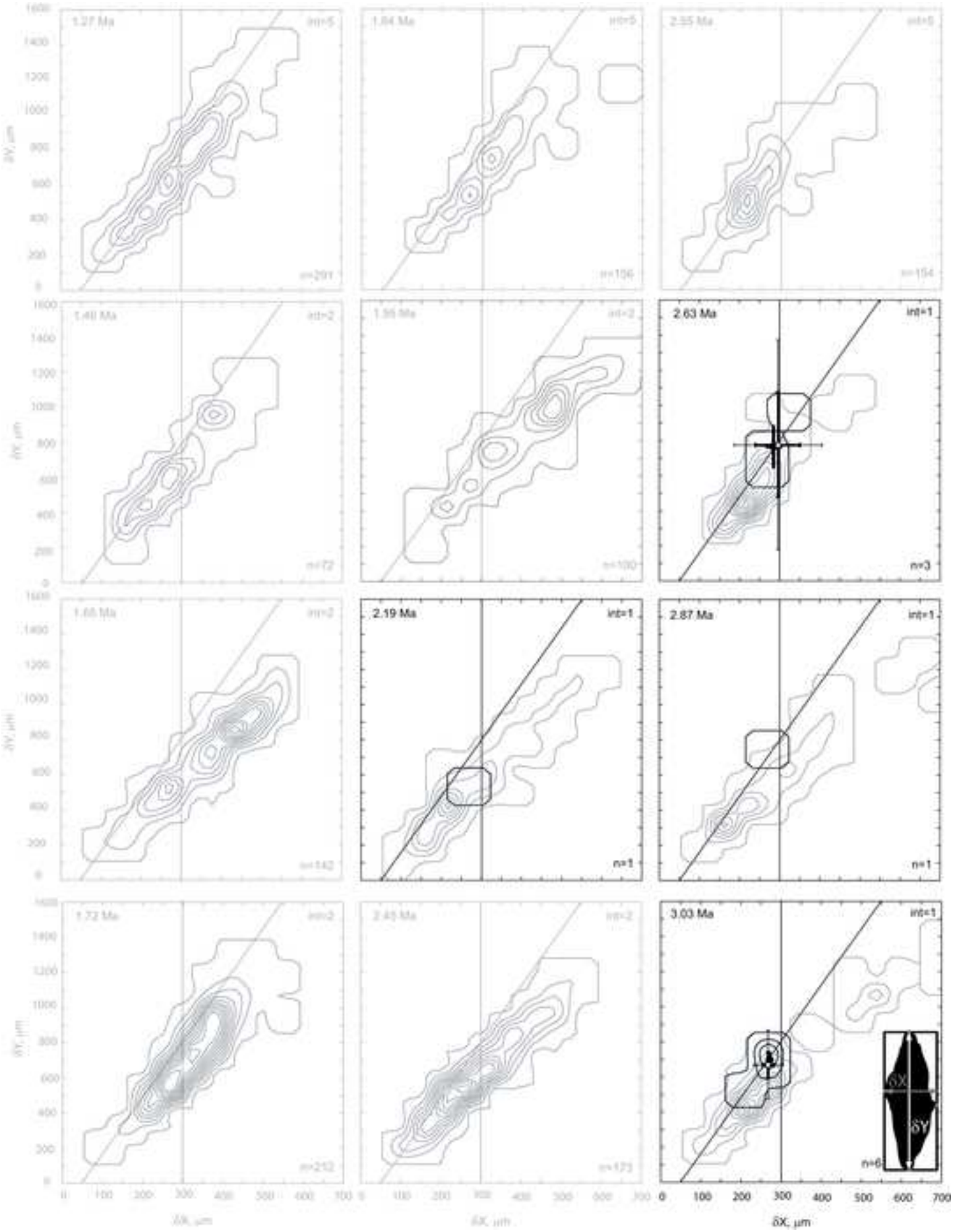

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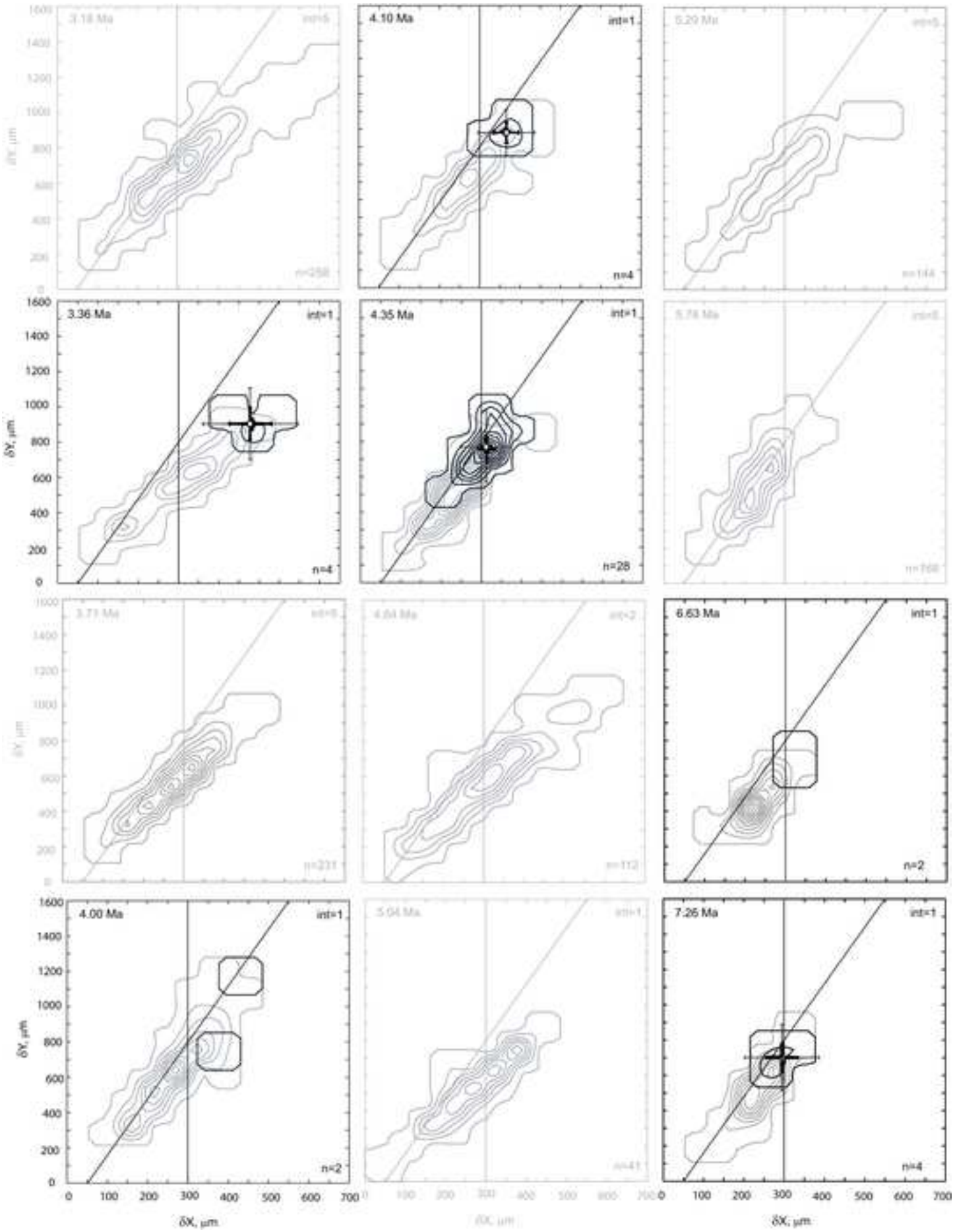

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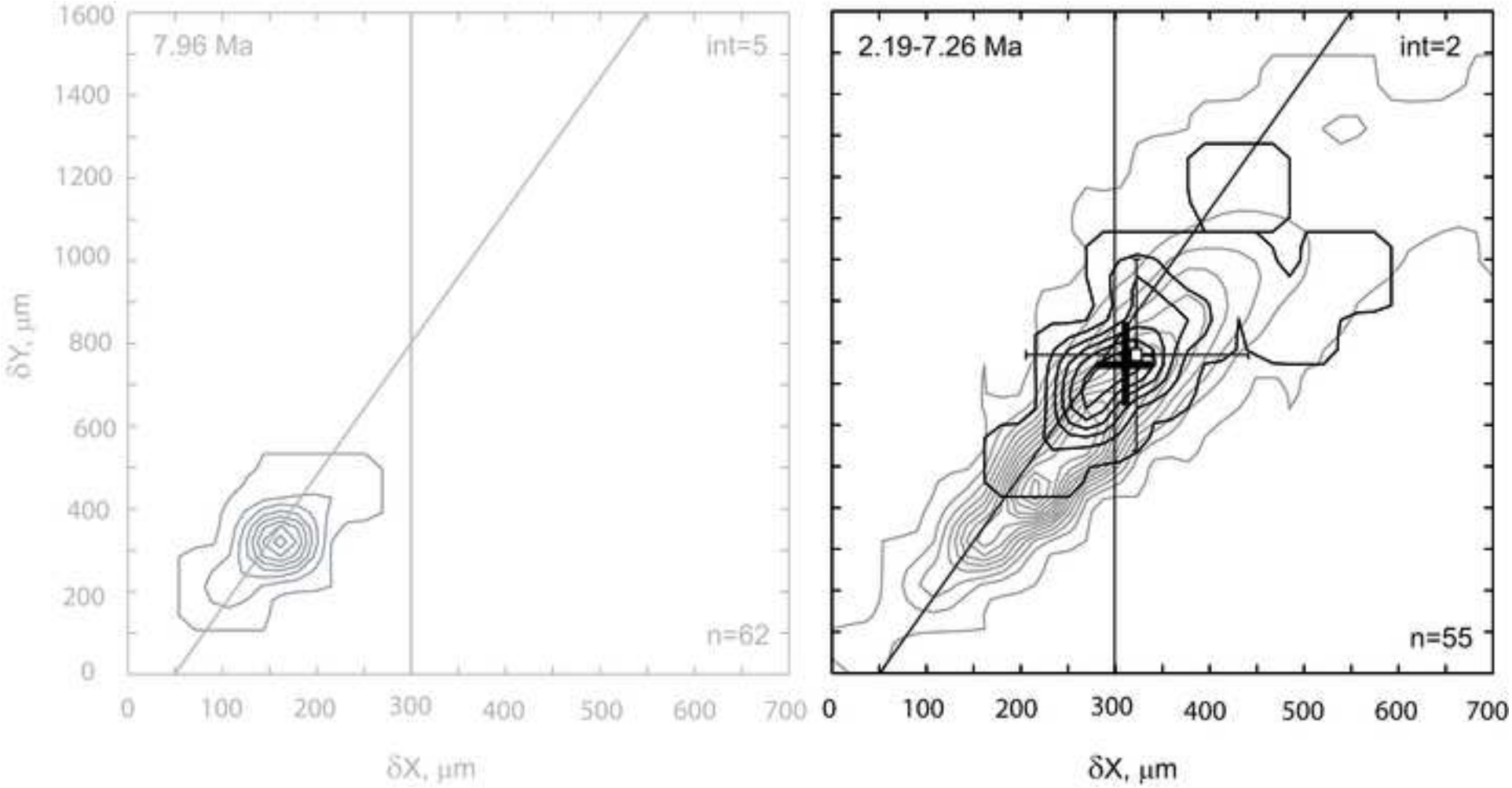

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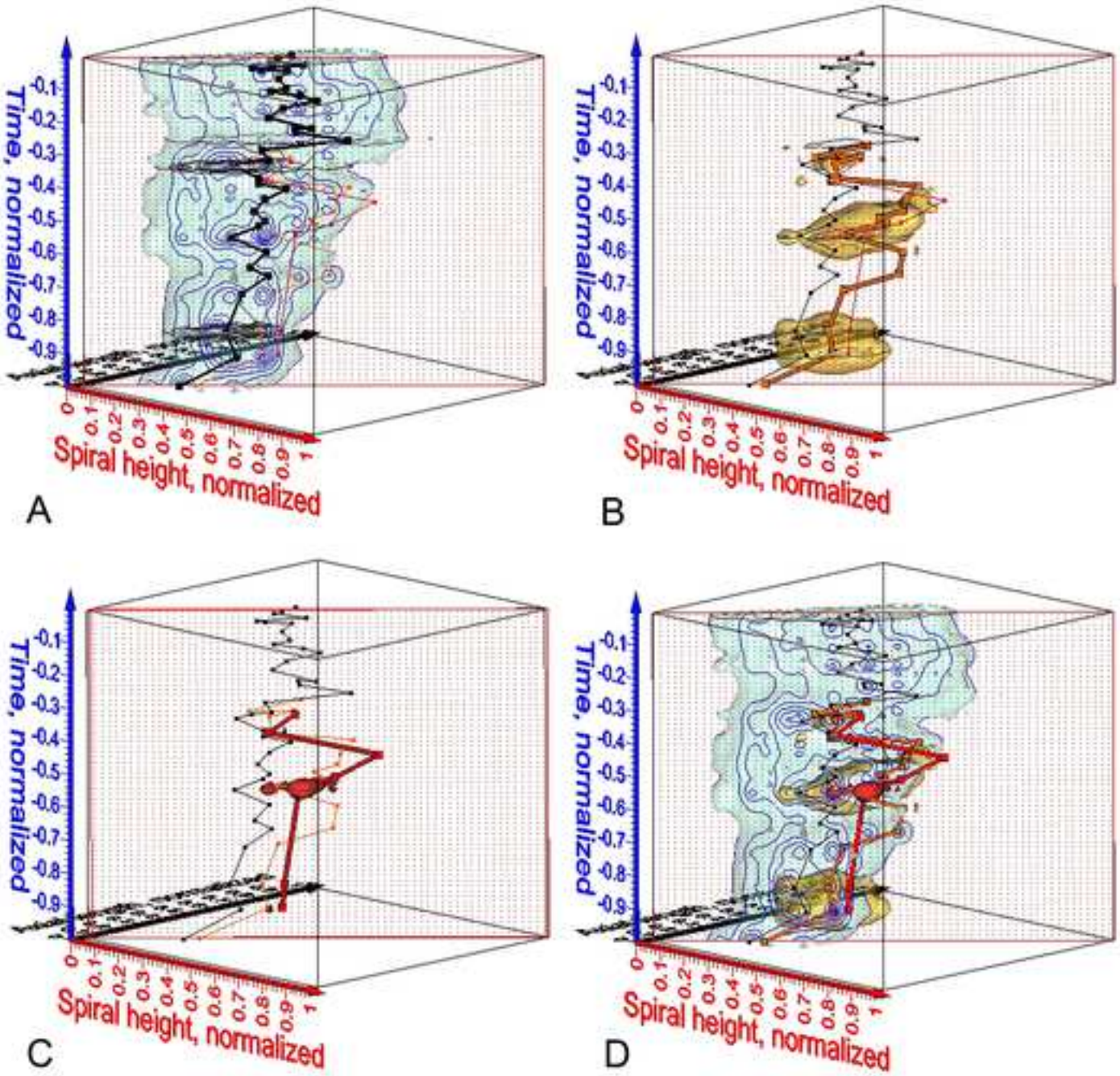

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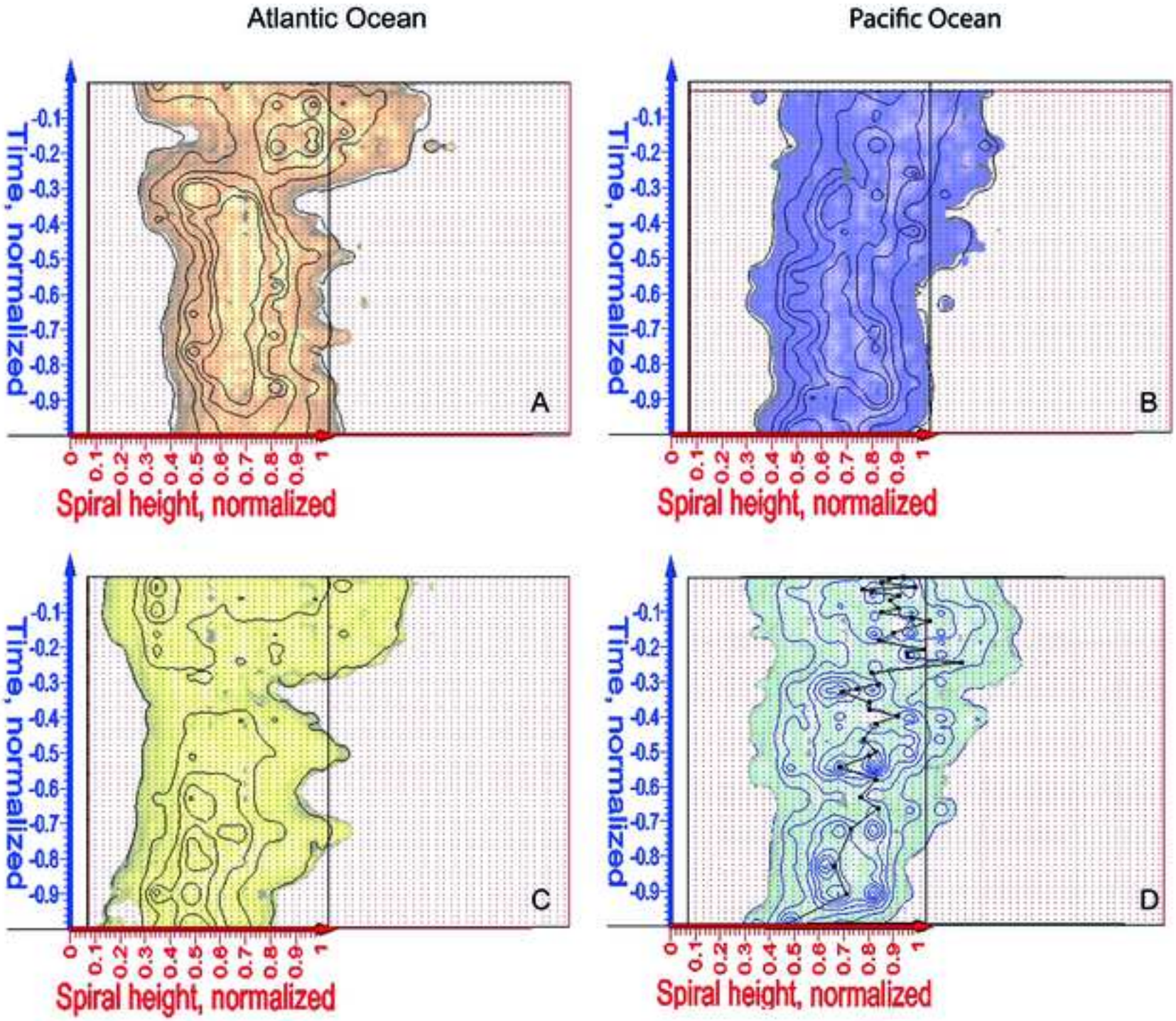

Figure_19.jpg

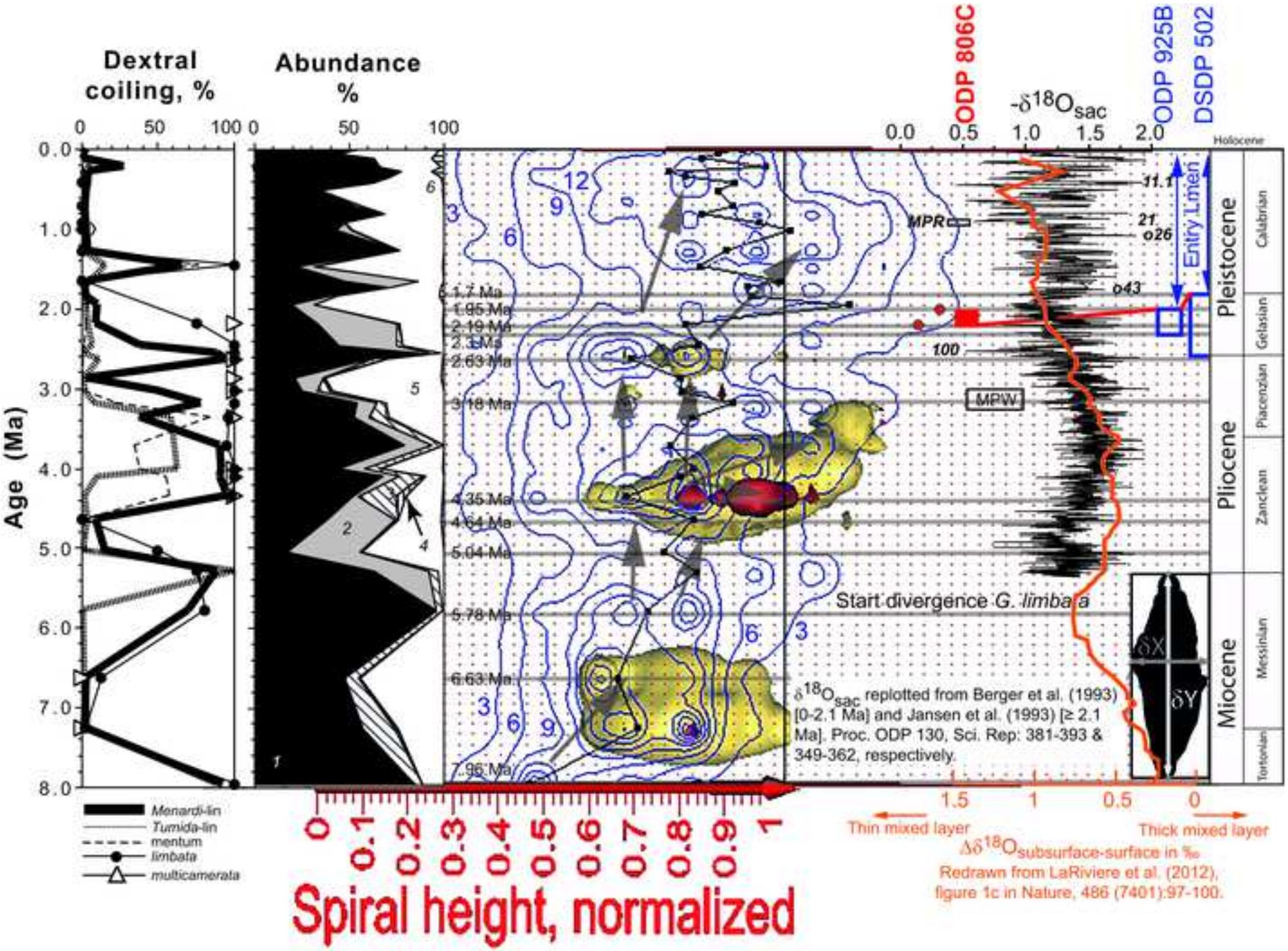

Table captions

Table 1. Sampling intervals at Hole 806C. (*) Between 0 to 102 mbsf the equivalent depths at Hole 806C (mbsf) were correlated from mbsf at 806B to mbsf at 806C by stepwise interpolation using Table 1 in Mayer et al. (1993).

(**) For the interval below 102 mbsf equivalent sample depths at Hole 806C in mbsf were obtained by projecting numerical ages from model "806B fnrdIM loc95 MK_final" onto the age model 806C loc95. See text in section 3.2 and Supplemetal_Info for more details.

Table 2. Control points for age model 806B fnrdIM loc95 MK_final (left pair of values) and for age model 806C loc95 (right pair of values), see Fig. 3.

Hole 806B		Hole 806C	
Depth, mbsf	**Age, Ma**	**Depth, mbsf**	**Age, Ma**
0	0	0	0
1.95122	0.0578489	5.58942	0.612638
2.68293	0.128891	39.0244	1.60165
6.70731	0.37517	57.5203	2.52885
8.35366	0.422531	68.191	3.08517
10.5488	0.536198	77.439	3.29533
16.2195	0.763532	85.2642	3.6044
20.4268	0.919824	94.7154	3.75
22.8049	1.00981	172.053	5.59478
26.0976	1.24188	201.219	6.39835
30.6707	1.46922	286.89	8.30357
55.1829	2.54905	309.248	9.25
59.2073	2.67219	365.142	10.8929
64.02	2.85	411.789	12.0192
77.439	3.21703	519.106	14.8077
94.3338	3.75	574.39	17.3077
110.366	4.19918	576.016	18.2692
136.89	5.09066	602.032	19.1827
164.329	5.6044	679	21.5
180.793	5.89148		
208.333	6.42033		
268.598	7.78022		
290.795	8.30329		
319.817	8.86813		
334.451	9.07967		
339.939	9.6566		
361.789	10.7212		
402.032	11.8544		
465.752	13.6058		
553.438	16.0855		
598.625	19.1776		
739.096	21.5132		

Table 2. Control points for age model 806B fnrdIM loc95 MK_final (left pair of values) and for age model 806C loc95 (right pair of values), see Fig. 3.

Corrected Table 1

Depth 806B, mbsf	Reason for sampling	Reference	Equival. Depth 806C (mbsf)	Age (Ma)	Selected sample 806C
0.15	in stage 1.1	Berger et al. (1993)	0.14224 (*)	0.00445	1H-1,13-15cm
1.75	in stage 3.3	Berger et al.(1993)	1.65948 (*)	0.05188	1H-2,15-17cm
2.45	in stage 5	Berger et al. (1993)	2.32328 (*)	0.106276	1H-2,81-83cm
4.1	in stage 7.1	Berger et al. (1993)	3.88949 (*)	0.215611	1H-3,88-90cm
5.15	in stage 7.3	Berger et al. (1993)	4.92613 (*)	0.279868	1H-4,42-44cm
6.15	in stage 9.1	Berger et al. (1993)	5.91341 (*)	0.341064	2H-1,30-32cm
8.35	in stage 11.1	Berger et al. (1993)	8.08542 (*)	0.422426	2H-2,98-100cm
10.2	in stage 13.1	Berger et al. (1993)	9.91188 (*)	0.518137	2H-3,130-132cm
14.95	in stage 17.1	Berger et al. (1993)	14.6798 (*)	0.712639	2H-7,7-9cm
17.55	in stage 21	Berger et al. (1993)	17.4542 (*)	0.812957	3H-2,84-86cm
20.35	in stage 25 (o24)	Berger et al. (1993)	20.4457 (*)	0.916971	3H-4,84-86cm
22.75	in stage 29 (o26)	Berger et al. (1993)	22.8352 (*)	1.00773	3H-6,23-25cm
26.6	in stage 39 (o32)	Berger et al. (1993)	25.7765 (*)	1.26686	4H-1,117-119cm
30.55	in stage 47 (o36)	Berger et al. (1993)	29.1800 (*)	1.46322	4H-4,7-9cm
34.85	in stage 57 (o41)	Berger et al. (1993)	33.7587 (*)	1.65333	4H-7,15-17cm
36.35	in stage 61 (o44)	Berger et al. (1993)	36.2612 (*)	1.71941	5H-2,65-67cm
39.15	in (o47)	Berger et al. (1993)	38.8713 (*)	1.84276	5H-4,26-28cm
41.55	in (o49)	Berger et al. (1993)	41.4996 (*)	1.94848	5H-5,139-141cm
47.07	at (a)	Jansen et al. (1993)	47.4362 (*)	2.19165	6H-3,83-85cm
52.88	at (3)	Jansen et al. (1993)	52.0133 (*)	2.4476	6H-6,90-92cm
55.27	MIS 100 (cold spell)	Jansen et al. (1993)	55.1851 (*)	2.55172	7H-2,58-60cm
57.97	MIS 105 close 2.59 Ma	Jansen et al. (1993)	57.3849 (*)	2.63433	7H-3,127-129cm
64.67	close 2.88 Ma	Jansen, et al. (1993)	63.6946 (*)	2.86778	8H-1,108-110cm
70.57	warm interval at (7)	Jansen et al. (1993)	70.1692 (*)	3.02915	8H-6,6-8cm
76.17	close 3.2 Ma at (9)	Jansenet al. (1993)	75.2621 (*)	3.18232	9H-3,15-17cm
82.07	3.39 Ma at (10)	Jansen et al. (1993)	82.5908 (*)	3.36312	10H-1,98-100cm
92.25	Peak $\delta^{18}O_{sacculifer}$ at 806B at -1.92	Jansen et al. (1993)	91.8914 (*)	3.706	11H-1,78-79cm
103.27	close 3.99 Ma at (11)	Jansen et al. (1993)	105.212 (**)	4.00037	12H-4,10-12cm
106.67	close 4.14 Ma at (12)	Jansen et al. (1993)	109.205 (**)	4.09563	12H-6,110-112cm
114.82	close 4.35 Ma at (13)	Jansen et al. (1993)	119.822 (**)	4.34888	14H-1,21-23cm
123.37	close 4.62 Ma at (14)	Jansen et al. (1993)	131.869 (**)	4.63625	15H-2,126-128cm
135.37	close 5.0 Ma at (17,18)	Jansen et al. (1993)	148.777 (**)	5.03957	17H-1,67-69cm
Close 154.52	Close lowest pt@806B, and close 806C-18H-2, 30-32cm with many *G. menardii*, see Chaisson et al. (1993)	Jansen et al (1993) Chaisson et al. (1993)	159.39 (**)	5.293	18H-2,28-30cm
174.4	at 5.78 Ma	806B fnrdIM loc95	178.776 (**)	5.78001	20H-2,67-69cm
Not relevant	Close 806C-23H-5,31 FAD *G. multicamerata*	Chaisson et al. (1993)	211.39 (**)	6.625	23H-5,28-30cm
Not relevant	Close 806C-26H-5,30 *G. menardii* abundant	Chaisson et al. (1993)	239.88 (**)	7.258	26H-5,27-29cm
276.2	at 7.96 Ma	806B fnrdIM loc95	271.412 (**)	7.95936	29H-7,30-32cm

Table 1. Sampling intervals at Hole 806C. See table captions and text in section 3.2 for explanations.

Captions to Appendices 1 through 11

Appendix 1. Range Chart of recorded planktic foraminifers from Hole 806C (Excel file).

Appendix 2. Details about image processing and derivation of measurements.

Appendix 3. Summary statistics for morphometric measurements illustrated in Figs. 6 through 14 from *G. menardii*-lineage at Hole 806C.

Appendix 4. Summary statistics for morphometric measurements illustrated in Figs. 6 through 14 from *G. limbata* at Hole 806C.

Appendix 5. Summary statistics for morphometric measurements illustrated in Figs. 6 through 14 from *G. multicamerata* at Hole 806C.

Appendix 6. Summary statistics for measurements of transitional forms between *G. menardii and G. tumida* [*mentum*] at Hole 806C.

Appendix 7. Presence/absence of all encountered menardiform globorotalids at Hole 806C during the present study. Census counts are in absolute specimens per investigated sample split.

Appendix 8. Calculated Means, ±95% confidence intervals about the means, ±95% prediction intervals, medians, . ±quartile ranges from the medians of δX and δY and shown in Figures 6 through 18 for the *G. menardii-lineage* at Hole 806C.

Appendix 9. Calculated Means, ±95% confidence intervals, ±95% prediction intervals, medians, . ±quartile ranges from the medians of δX and δY and shown in Figures 6 through 18 for *G. limbata* at Hole 806C.

Appendix 10. Calculated Means, ±95% confidence intervals, ±95% prediction intervals, medians, . ±quartile ranges from the medians of δX and δY and shown in Figures 6 through 18 for the *G. multicamerata* at Hole 806C.

Appendix 11. Interactive version of the VDD illustrated in Fig. 18D. Please activate for 3D contents after opening the pdf file with Adobe Acrobat Reader by double click on the first option. Use the mouse wheel to zoom in or out. By dragging the cursor with the left mouse button held down it is possible to tilt and rotate the VDD. Use functions in the embedded menu for switching between orthogonal or perspective projection, or for changing model render modes. Use CAD optimized illumination from the lamp symbol for best representation of the VDD. Note, that variation in δX and δY through time is minimal in view parallel to the diagonal plane and maximal perpendicular to it. Also note, that densities of adult specimens (for example larger specimens in *G. multicamerata*) fall outside of the diagonal plane. This interactive VDD was generated from .iv format generated in

Voxler 2 by conversion to pdf with PDF3D ReportGen version 2.17.4 from Visual Technology Services Ltd (https://www.pdf3d.com).

Captions to RawData

RawData_1. Morphometric measurements from individual specimens for *Globorotalia menardii*-lin, from which statistics illustrated in Figs. 6A through 14A were derived. Age in Ma; #pts=number of outline points from Tiff image; Species=species code; #Ch=number of chambers in final whorl; Coil=coiling direction; DX (mu)=spiral height in μm; DY (mu)=axial length in μm; R=DX/DY=Ratio of Spiral height/axial length; Ar (mm2)=Keel view area in mm^2; D10 % (width of keel in μm in lower keel region at 10% of axial height); D90 % (width of keel in μm in upper keel region at 90% of axial height); D9010%=absolute difference between D90 and D10 indicating degree of "twistedness" of test; Dmin%; Dmax%; Phi1(d)=upper keel angle in degrees; Phi2(d)=lower keel angle in degrees; A (mu)=Spiral convexity in μm; B (mu)=umbilical convexity in μm; Ratio A/B; Phi3(d)=Apical angle in degrees; RL (mu)=radius of osculating circle in lower keel region in μm; RU (mu)=radius of osculating circle in upper keel region in μm. See text for explanations; see also code descriptions in Knappertsbusch (2015b). MorphCol 2004-2013. A collection of Fortran 77 programs for geometric morphometry.
http://doi.pangaea.de/10.1594/PANGAEA.848773

RawData_2. Morphometric measurements from individual specimens for *Globorotalia limbata*, from which statistics illustrated in Figs. 6B through 14B were derived. See captions to RawData_1 for explanations.

RawData_3. Morphometric measurements from individual specimens for *Globorotalia multicamerata*, from which statistics illustrated in Figs. 6C through 14C were derived. See captions to RawData_1 for explanations.

RawData_4. Morphometric measurements from individual specimens for transitional forms between *Globorotalia menardii* and *G. tumida*, from which statistics were derived, but which were excluded from analysis in the present study, see taxonomic remarks about transitional forms in section 3 of the text. For explanation of abbreviations see captions to RawData_1.

RawData_5. Gridded files for *Globorotalia menardii-lin*. Bivariate frequency matrices per sample of δX (spiral height, .horizontal rows) versus δY (axial length, vertical columns) at bin-widths of $\Delta\delta$X=50μm and $\Delta\delta$Y=100μm as obtained with program Grid2.out (see text). Numbers indicate absolute local frequencies (F_s) in specimens per grid-cell. The sample age is indicated in the header line of each matrix. Example: In the sample at 0 Ma (0_00 Ma) there are two frequency maxima (Fs=12), e.g., one in the grid-cell at $\Delta\delta$X=150-200μm/$\Delta\delta$Y=400-500μm and one in the grid-cell at $\Delta\delta$X=350-400μm/$\Delta\delta$Y=900-1000μm. From these" gridded files" contour diagrams in Figs. 16, 17, and 18 were generated, see section 2.4.7. in the text.

RawData_6. Gridded files for *Globorotalia limbata*. See captions to RawData_5 for explanations.

RawData_7. Gridded files for *Globorotalia multicamerata*. See captions to RawData_5 for explanations.

RawData_8. Gridded files for *Globorotalia menardii-tumida,* transitional forms. See captions to RawData_5 for explanation of variables (not plotted and excluded from analysis in the present study, see taxonomic remarks about transitional forms in section 3 of the text).

RawData_9. Unscaled input data matrix to Voxler2 for construction of VDD for *Globorotalia menardii-lin* in Fig. 18A. It was created by applying auxiliary program Grid_to_Vox3.out to combined gridded δX,δY matrices, see section 2.4.8. in the text. DX=X-center of grid-cell with bin-width $\Delta\delta$X=50μm (spiral height); DY=Y-center of grid-cell with bin-width $\Delta\delta$Y=100μm (axial length); Z=negative Age in Ma; Fs=local absolute number of specimens per grid-cell (voxel). For best visual representation in Voxler 2, values of DX, DY, and Z were divided by 675, 1550, and 8, respectively, see section 2.4.8. in the text.

RawData_10. Unscaled input data matrix to Voxler2 for construction of VDD for *Globorotalia limbata* in Fig. 18B. See captions to RawData_9 for explanations.

RawData_11. Unscaled input data matrix to Voxler2 for construction of VDD for *Globorotalia multicamerata* in Fig. 18C. See captions to RawData_9 for explanations.

RawData_12. Unscaled input data matrix to Voxler2 for construction of VDD for transitional forms of *Globorotalia menardii-tumida* (VDD not reproduced). See captions to RawData_9 for explanation of variables (not plotted and excluded from analysis in the present study, see taxonomic remarks about transitional forms in section 3 of the text).

RawData_13. .Imaging Log ODP 130, Site 806C, Ontong-Java Plateau (Excel file).

RawData_14. Site 806C data processing overview (Excel file).